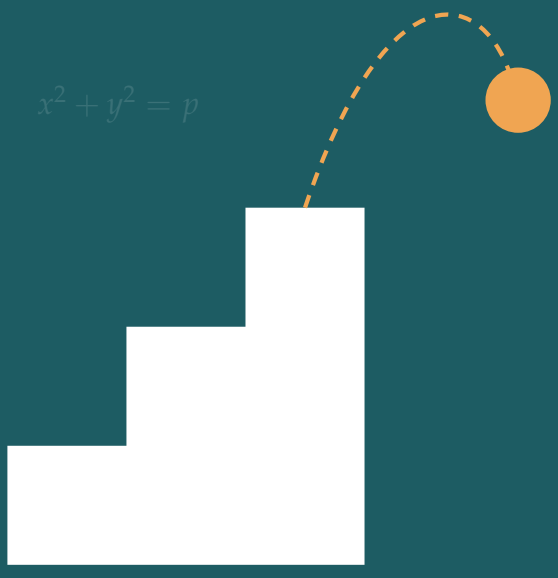
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

2003 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

2003 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

2003 Worked Solutions

Worked solutions to all 42 questions of the 2003 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 2003, Question 1

Pure • Sums of Powers

STEP I 2003, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q1

The key idea

We are told the shape of the answer, a polynomial in n with unknown coefficients, so all that is left is to pin down those coefficients. A polynomial identity that holds for every n certainly holds for a few convenient values of n, and each value gives one linear equation. Starting the sums at a negative index is a deliberate trick: it lets us use small values such as $n=-1$ and $n=-2$, which makes the equations pleasantly symmetric.

Solution

The question uses r both as the summation variable and as a coefficient. To avoid the clash we write the summation variable as k, keeping p,q,r,s and a,b,c,d,e for the coefficients.

Sum of squares

Let $S(n)=\sum_{k=-1}^{n} k^2$, so that

$$S(n)=pn^3+qn^2+rn+s.$$

The sum starting at $k=-1$ makes sense for every $n\geq -1$, and we have

$$S(-1)=1, \qquad S(0)=1+0=1, \qquad S(1)=1+0+1=2, \qquad S(2)=1+0+1+4=6.$$

Substituting $n=-1,0,1,2$ into the given form gives the four equations

$$-p+q-r+s=1, \tag{1}$$
$$s=1, \tag{2}$$
$$p+q+r+s=2, \tag{3}$$
$$8p+4q+2r+s=6. \tag{4}$$

From (2), $s=1$, so (1) and (3) become $-p+q-r=0$ and $p+q+r=1$. Adding these gives $2q=1$, so $q=\frac{1}{2}$, and then $p+r=\frac{1}{2}$. Equation (4) becomes

$$8p+2+2r+1=6 \implies 8p+2r=3 \implies 6p+2(p+r)=3,$$

so $6p = 2$, giving $p = \frac{1}{3}$ and $r = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$. Hence

$$\sum_{k=-1}^{n} k^2 = \tfrac{1}{3}n^3 + \tfrac{1}{2}n^2 + \tfrac{1}{6}n + 1.$$

Ask yourself The question wants the sum from $k = 0$, but we have found the sum from $k = -1$. How do they differ?

They differ only by the $k = -1$ term, which is $(-1)^2 = 1$. So

$$\sum_{k=0}^{n} k^2 = \tfrac{1}{3}n^3 + \tfrac{1}{2}n^2 + \tfrac{1}{6}n = \tfrac{1}{6}n\left(2n^2 + 3n + 1\right) = \tfrac{1}{6}n(n+1)(2n+1),$$

as required.

Sum of cubes

Now let $T(n) = \sum_{k=-2}^{n} k^3 = an^4 + bn^3 + cn^2 + dn + e$. There are five unknowns, so we use the five values $n = -2, -1, 0, 1, 2$:

$$T(-2) = -8, \quad T(-1) = -9, \quad T(0) = -9, \quad T(1) = -8, \quad T(2) = 0.$$

This gives

$$16a - 8b + 4c - 2d + e = -8, \tag{5}$$
$$a - b + c - d + e = -9, \tag{6}$$
$$e = -9, \tag{7}$$
$$a + b + c + d + e = -8, \tag{8}$$
$$16a + 8b + 4c + 2d + e = 0. \tag{9}$$

From (7), $e = -9$. The equations come in pairs at $\pm n$, so it pays to add and subtract each pair: the even powers and the odd powers then separate.

Using $e = -9$, (6) and (8) become $a - b + c - d = 0$ and $a + b + c + d = 1$. Adding and subtracting,

$$a + c = \tfrac{1}{2}, \qquad b + d = \tfrac{1}{2}.$$

Similarly (5) and (9) become $16a - 8b + 4c - 2d = 1$ and $16a + 8b + 4c + 2d = 9$. Adding and subtracting,

$$32a + 8c = 10, \qquad 16b + 4d = 8,$$

that is, $16a + 4c = 5$ and $4b + d = 2$. Substituting $c = \frac{1}{2} - a$ gives $12a + 2 = 5$, so $a = \frac{1}{4}$ and $c = \frac{1}{4}$. Substituting $d = \frac{1}{2} - b$ gives $3b + \frac{1}{2} = 2$, so $b = \frac{1}{2}$ and $d = 0$. Hence

$$\sum_{k=-2}^{n} k^3 = \tfrac{1}{4}n^4 + \tfrac{1}{2}n^3 + \tfrac{1}{4}n^2 - 9.$$

The sum from $k = -2$ contains the extra terms $(-2)^3 + (-1)^3 = -9$ compared with the sum from $k = 0$, so we add 9:

$$\sum_{k=0}^{n} k^3 = \tfrac{1}{4}n^4 + \tfrac{1}{2}n^3 + \tfrac{1}{4}n^2 = \tfrac{1}{4}n^2\left(n^2 + 2n + 1\right) = \tfrac{1}{4}n^2(n+1)^2,$$

as required.

Checks

For $n = 10$ the formulae give $\frac{1}{6} \cdot 10 \cdot 11 \cdot 21 = 385$ and $\frac{1}{4} \cdot 100 \cdot 121 = 3025$, and adding up $0^2 + 1^2 + \cdots + 10^2$ and $0^3 + 1^3 + \cdots + 10^3$ directly gives the same numbers. Notice also that $3025 = 55^2 = (0 + 1 + \cdots + 10)^2$: the sum of the first n cubes is the square of the sum of the first n integers. We also checked by computer that both formulae hold exactly for every n from 0 to 59.

Common slips

- Forgetting the $k = -1$ (or $k = -2$ and $k = -1$) terms when passing from the sum we computed to the sum the question wants. Note that $(-2)^3 + (-1)^3 = -9$, not $+9$.
- Trying to use only four values of n for the cubes: five unknowns need five equations.
- Confusing the coefficient r with the summation variable r in the question; renaming one of them avoids errors.

Beyond STEP

Why may we assume the answer is a polynomial of degree one higher? Because the difference $P(n) - P(n-1)$ of a polynomial of degree $m+1$ has degree m, so one can always solve for a polynomial whose differences are n^m; this is the start of the *calculus of finite differences*, and the coefficients that appear in general are governed by the Bernoulli numbers (Faulhaber's formula). The method used here, fixing a polynomial from its values at a few points, is *polynomial interpolation*, which reappears in numerical analysis as Lagrange interpolation. For more problems in this spirit, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 2003, Question 2

Pure • Algebraic Identities

STEP I 2003, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q2

The key idea

The candidate's answer is right exactly when it satisfies the equation, so substitute it and clear the fractions. Each part then turns into a polynomial condition on the given numbers. In the first part that condition is a quadratic form that can't vanish for real non-zero numbers. In the second it's a cubic expression that factorises beautifully, and the factors are exactly the three equations we're asked about.

Solution

The first question

Throughout, a and b are real and non-zero. The candidate's answer is correct precisely when $x = a + b$ satisfies the equation. For that to make sense at all we need $a + b \neq 0$, and then we need

$$\frac{1}{a+b} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}.$$

Multiplying through by $ab(a+b)$, which is non-zero, this is equivalent to

$$ab = (a+b)^2, \qquad \text{that is,} \qquad a^2 + ab + b^2 = 0.$$

Completing the square,

$$a^2 + ab + b^2 = \left(a + \tfrac{1}{2}b\right)^2 + \tfrac{3}{4}b^2.$$

Both terms are squares of real numbers, so both are ≥ 0, and the second is strictly positive because $b \neq 0$. Hence $a^2 + ab + b^2 > 0$ for every real a and non-zero real b, so the condition can never hold.

(If $a + b = 0$, the candidate's "answer" $x = 0$ doesn't even make sense in the equation, so that case is no better.) Therefore there are no values of a and b for which $x = a + b$ is correct, as required.

The second question

Now a, b, c are non-zero. The answer $x = a+b+c$ is correct if and only if $a+b+c \neq 0$ and

$$\frac{1}{a+b+c} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{bc+ca+ab}{abc}.$$

Multiplying through by $abc(a+b+c) \neq 0$, this is equivalent to

$$(a+b+c)(ab+bc+ca) - abc = 0. \qquad (*)$$

Ask yourself How can we spot that the left-hand side of $(*)$ factorises?

The target result tells us to expect the factors $a+b$, $b+c$ and $c+a$. Test one: put $b = -a$. Then $a+b+c = c$ and $ab+bc+ca = -a^2 - ac + ca = -a^2$, so the left-hand side becomes $c(-a^2) - (-a^2c) = 0$. So $a+b$ is a factor, and by symmetry so are $b+c$ and $c+a$. The left-hand side has degree 3, so it is a constant multiple of $(a+b)(b+c)(c+a)$; comparing the coefficients of a^2b (it's 1 on both sides) shows the constant is 1. Alternatively, expand both:

$$\begin{aligned}(a+b+c)(ab+bc+ca) - abc &= a^2b + a^2c + b^2a + b^2c + c^2a + c^2b + 3abc - abc \\ &= a^2b + a^2c + b^2a + b^2c + c^2a + c^2b + 2abc \\ &= (a+b)(b+c)(c+a).\end{aligned}$$

So $(*)$ is the same as

$$(a+b)(b+c)(c+a) = 0.$$

Now we prove both directions.

If the answer is correct. Then $a+b+c \neq 0$ and $(*)$ holds, so $(a+b)(b+c)(c+a) = 0$. A product of numbers is zero only if one of the factors is zero, so at least one of $a+b=0$, $b+c=0$, $c+a=0$ holds.

Conversely, suppose at least one of these holds; by symmetry say $a+b=0$. Then $a+b+c = c \neq 0$, so the candidate's answer is a legitimate non-zero number, and the factorised form of $(*)$ holds. Since the steps from the equation to $(*)$ were reversible (we only multiplied by the non-zero number $abc(a+b+c)$), $x = a+b+c$ satisfies the equation. You can also see this directly: $\frac{1}{a} + \frac{1}{b} = \frac{1}{a} - \frac{1}{a} = 0$, so the equation reads $\frac{1}{x} = \frac{1}{c}$, whose solution is $x = c = a+b+c$.

Hence the candidate's answer is correct if and only if at least one of $a+b=0$, $b+c=0$, $c+a=0$ holds. ■

Notice that the argument never used that a, b, c are real; the factorisation is an algebraic identity, so the result holds for non-zero complex numbers too. The first part, by contrast, really does need a and b real: over the complex numbers, $a^2+ab+b^2=0$ has solutions such as $b = a\,\mathrm{e}^{2\pi\mathrm{i}/3}$.

Checks

- Take $a = 2$, $b = -2$, $c = 5$: then $\frac{1}{2} - \frac{1}{2} + \frac{1}{5} = \frac{1}{5}$ and $a + b + c = 5$, so the candidate is right.
- Take $a = b = c = 1$: the true answer is $x = \frac{1}{3}$, but the candidate says 3; and indeed $(a+b)(b+c)(c+a) = 8 \neq 0$.
- Expand $(a+b)(b+c)(c+a)$ at $a = 1$, $b = 2$, $c = 3$: it gives $3 \cdot 5 \cdot 4 = 60$, while $(a+b+c)(ab+bc+ca) - abc = 6 \cdot 11 - 6 = 60$.
- We tested both directions exhaustively on all 17 576 triples of non-zero rationals n/d with $|n| \leq 6$ and $d \in \{1, 2, 3\}$, using exact fractions, and found no exceptions.

Common slips

- Writing $a^2 + ab + b^2 = (a+b)^2 - ab$ and stopping: that doesn't show positivity. Completing the square does.
- Proving only one direction in the second part. "If and only if" needs both, and the converse must check that $a + b + c \neq 0$.
- Forgetting that $abc(a+b+c)$ must be non-zero before multiplying through, which is what makes the steps reversible.

Beyond STEP

The expression $(a+b)(b+c)(c+a)$ is a symmetric polynomial, and the identity $(a+b)(b+c)(c+a) = e_1e_2 - e_3$, where $e_1 = a+b+c$, $e_2 = ab+bc+ca$ and $e_3 = abc$, is an instance of the fundamental theorem of symmetric polynomials: every symmetric polynomial is a polynomial in these elementary ones. The same e_1, e_2, e_3 are, up to sign, the coefficients of the cubic with roots a, b, c, which is why they turn up throughout the theory of equations. Allenby's *Rings, Fields and Groups* discusses polynomial rings in this spirit, and Siklos's *Advanced Problems in Mathematics* has more STEP practice in spotting factorisations.

WORKED SOLUTION

STEP I 2003, Question 3

Pure • Half-Angle Trigonometric Equations

STEP I 2003, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q3

The key idea

Each part compares a function of $\frac{1}{2}\theta$ with the same function of θ, so write everything in terms of the half-angle using a double-angle formula. Each equation then becomes a polynomial equation in a single quantity ($\sin\frac{1}{2}\theta$, $\tan\frac{1}{2}\theta$ or $\cos\frac{1}{2}\theta$), which we can factorise or solve as a quadratic. The only care needed is to keep track of which values the half-angle function can actually take, and where the functions are defined.

Solution

Part (i)

Using $\sin\theta = 2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta$, the equation $2\sin\frac{1}{2}\theta = \sin\theta$ is equivalent to

$$2\sin\tfrac{1}{2}\theta - 2\sin\tfrac{1}{2}\theta\cos\tfrac{1}{2}\theta = 0, \qquad \text{that is,} \qquad 2\sin\tfrac{1}{2}\theta\left(1-\cos\tfrac{1}{2}\theta\right) = 0.$$

So the equation holds if and only if $\sin\frac{1}{2}\theta = 0$ or $\cos\frac{1}{2}\theta = 1$.

But if $\cos\frac{1}{2}\theta = 1$ then $\sin^2\frac{1}{2}\theta = 1 - \cos^2\frac{1}{2}\theta = 0$, so $\sin\frac{1}{2}\theta = 0$ anyway. The second possibility is therefore contained in the first, and

$$2\sin\tfrac{1}{2}\theta = \sin\theta \iff \sin\tfrac{1}{2}\theta = 0,$$

as required. (In other words, the solutions are exactly $\theta = 2n\pi$ for integer n.)

Part (ii)

Let $t = \tan\frac{1}{2}\theta$. For the equation to make sense we need both $\tan\frac{1}{2}\theta$ and $\tan\theta$ to be defined, so $\cos\frac{1}{2}\theta \neq 0$ and $\cos\theta \neq 0$. The second condition means $t^2 \neq 1$, since $\cos\theta = \dfrac{1-t^2}{1+t^2}$. With these restrictions, $\tan\theta = \dfrac{2t}{1-t^2}$, and the equation becomes

$$2t = \frac{2t}{1-t^2}.$$

Multiplying by $1 - t^2$, which is non-zero, gives $2t - 2t^3 = 2t$, so $t^3 = 0$ and $t = 0$.

Ask yourself Is it safe to multiply through by $1 - t^2$, and have we lost or gained any solutions?

Multiplying by a non-zero quantity is reversible, so the equation in t is exactly equivalent to $t = 0$ on the set where both tangents are defined. Conversely, $t = 0$ gives $\tan\theta = 0$ and $\tan\frac{1}{2}\theta = 0$, and both sides are 0.

Now $\tan\frac{1}{2}\theta = 0$ exactly when $\frac{1}{2}\theta = n\pi$. So the solutions are

$$\theta = 2n\pi, \qquad n \in \mathbb{Z},$$

the same set as in part (i).

Part (iii)

Let $c = \cos\frac{1}{2}\theta$. Using $\cos\theta = 2\cos^2\frac{1}{2}\theta - 1 = 2c^2 - 1$, the equation $2\cos\frac{1}{2}\theta = \cos\theta$ is equivalent to

$$2c = 2c^2 - 1 \iff 2c^2 - 2c - 1 = 0 \iff c = \frac{2 \pm \sqrt{4+8}}{4} = \frac{1 \pm \sqrt{3}}{2}.$$

Since c is a cosine, $-1 \le c \le 1$. The root $\frac{1}{2}(1+\sqrt{3}) \approx 1.37$ is too big, while $\frac{1}{2}(1-\sqrt{3}) \approx -0.37$ is allowed. So the equation holds if and only if

$$\cos\tfrac{1}{2}\theta = \tfrac{1}{2}(1-\sqrt{3}) = -\cos\phi,$$

where ϕ is as defined in the question. This ϕ exists and is unique because $0 < \frac{1}{2}(\sqrt{3}-1) < 1$ and cos decreases from 1 to 0 on $[0, \frac{1}{2}\pi]$.

Ask yourself How do we turn $\cos\frac{1}{2}\theta = -\cos\phi$ into the given form?

Write $-\cos\phi = \cos(\pi - \phi)$. The general solution of $\cos x = \cos\alpha$ is $x = 2m\pi \pm \alpha$, so

$$\tfrac{1}{2}\theta = 2m\pi \pm (\pi - \phi) \qquad \text{that is,} \qquad \theta = 4m\pi \pm (2\pi - 2\phi), \quad m \in \mathbb{Z}.$$

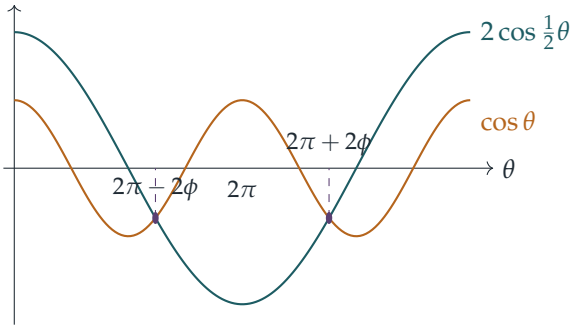
Take the two signs separately.

- With the $+$ sign: $\theta = 4m\pi + 2\pi - 2\phi = (4n+2)\pi - 2\phi$ with $n = m$.
- With the $-$ sign: $\theta = 4m\pi - 2\pi + 2\phi = \big(4(m-1)+2\big)\pi + 2\phi = (4n+2)\pi + 2\phi$ with $n = m - 1$.

As m runs over all integers, so does n in each case. Hence

$$2\cos\tfrac{1}{2}\theta = \cos\theta \iff \theta = (4n+2)\pi \pm 2\phi \text{ for some integer } n.$$

Every step was an equivalence, so both directions are proved. As a direct check of the converse: if $\theta = (4n+2)\pi \pm 2\phi$ then $\cos\frac{1}{2}\theta = \cos\big((2n+1)\pi \pm \phi\big) = -\cos\phi = c$, which satisfies $2c = 2c^2 - 1$.



Over one period $0 \leq \theta \leq 4\pi$ of $2\cos\frac{1}{2}\theta$, the curves meet twice, symmetrically about $\theta = 2\pi$, where $2\cos\frac{1}{2}\theta$ reaches its minimum of -2.

Checks

- Numerically $\cos\phi \approx 0.3660$ gives $\phi \approx 1.1961$, so the first positive solution of (iii) is $2\pi - 2\phi \approx 3.8911$. There $2\cos\frac{1}{2}\theta \approx -0.7321$ and $\cos\theta \approx -0.7321$, as required. A root search of $2\cos\frac{1}{2}\theta - \cos\theta$ on $[-6\pi, 6\pi]$ found exactly the six values $(4n+2)\pi \pm 2\phi$ in that range.
- A similar search for (ii), avoiding the poles, found roots only at multiples of 2π.
- Sanity check for (i) and (ii): $\theta = \pi$ fails (i), since $2\sin\frac{1}{2}\pi = 2 \neq 0 = \sin\pi$, and is outside the domain in (ii), where $\tan\frac{1}{2}\pi$ is undefined.

Common slips

- In (i), cancelling $\sin\frac{1}{2}\theta$ without saying it might be zero, or giving $\cos\frac{1}{2}\theta = 1$ as a separate family of solutions without noticing it is already included.
- In (ii), forgetting that θ must avoid the points where $\tan\theta$ or $\tan\frac{1}{2}\theta$ is undefined before clearing denominators.
- In (iii), keeping the root $\frac{1}{2}(1+\sqrt{3})$, which is bigger than 1, or mishandling the $\pm$ when converting $4m\pi \pm (2\pi - 2\phi)$ to the form $(4n+2)\pi \pm 2\phi$.

Beyond STEP

Each part solves an equation by substituting a single function of the half-angle, which turns a trigonometric equation into an algebraic one. The substitution $t = \tan\frac{1}{2}\theta$ in (ii) is especially powerful: it expresses $\sin\theta$, $\cos\theta$ and $\tan\theta$ as rational functions of t, which is why it is used in first-year calculus to integrate any rational function of $\sin\theta$ and $\cos\theta$. Geometrically it parametrises the unit circle by rational points. Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice with these ideas.

WORKED SOLUTION

STEP I 2003, Question 4

Pure • Trigonometric Inequalities

STEP I 2003, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q4

The key idea

The tempting move is to multiply both sides by $\cos\theta$, but $\cos\theta$ changes sign on $[0, 2\pi)$, and multiplying by a negative number reverses an inequality. So we split into the cases $\cos\theta > 0$ and $\cos\theta < 0$. One case turns out to be trivial, because $1 + \sin\theta$ is never negative. The other reduces to a single sine inequality once we write $\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$.

Solution

Write $f(\theta) = \dfrac{1 + \sin\theta}{\cos\theta}$. Throughout, $0 \le \theta < 2\pi$ and $\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$.

Case 1: $\cos\theta < 0$**, that is,** $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$

Since $\sin\theta \ge -1$, the numerator $1 + \sin\theta$ is at least 0, and the denominator is negative. So $f(\theta) \le 0 < 1$.

Every θ in $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ satisfies the inequality.

Case 2: $\cos\theta > 0$**, that is,** $0 \le \theta < \frac{\pi}{2}$ **or** $\frac{3\pi}{2} < \theta < 2\pi$

Multiplying by the positive number $\cos\theta$ keeps the direction of the inequality, so

$$f(\theta) \le 1 \iff 1 + \sin\theta \le \cos\theta \iff \sin\theta - \cos\theta \le -1.$$

Now $\sin\theta - \cos\theta = \sqrt{2}\left(\sin\theta\cos\frac{\pi}{4} - \cos\theta\sin\frac{\pi}{4}\right) = \sqrt{2}\sin\left(\theta - \frac{\pi}{4}\right)$, so the condition becomes

$$\sin\left(\theta - \tfrac{\pi}{4}\right) \le -\tfrac{1}{\sqrt{2}}.$$

Ask yourself Which values of $\phi = \theta - \frac{\pi}{4}$ do we need to look at?

We need only the two ranges allowed in this case:

- If $0 \le \theta < \frac{\pi}{2}$, then $-\frac{\pi}{4} \le \phi < \frac{\pi}{4}$. On this interval $\sin\phi$ is increasing and $\sin\phi \ge \sin(-\frac{\pi}{4}) = -\frac{1}{\sqrt{2}}$, with equality only at $\phi = -\frac{\pi}{4}$. So the only solution here is $\theta = 0$. (Check: $f(0) = 1/1 = 1$, so equality holds.)
- If $\frac{3\pi}{2} < \theta < 2\pi$, then $\frac{5\pi}{4} < \phi < \frac{7\pi}{4}$. On this open interval $\sin\phi < -\frac{1}{\sqrt{2}}$, since $\sin\phi$ equals $-\frac{1}{\sqrt{2}}$ at both ends and dips to -1 at $\phi = \frac{3\pi}{2}$ in between. So *every* θ in $\left(\frac{3\pi}{2}, 2\pi\right)$ is a solution.

Putting it together

The value $\theta = \frac{3\pi}{2}$ is excluded because $\cos\theta = 0$ there. Combining the cases, the inequality holds exactly when

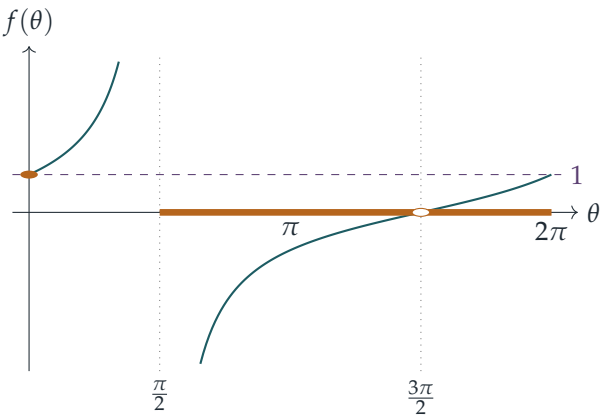
$$\theta = 0 \qquad \text{or} \qquad \frac{\pi}{2} < \theta < \frac{3\pi}{2} \qquad \text{or} \qquad \frac{3\pi}{2} < \theta < 2\pi.$$

A slicker view

Using $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$, $\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$ and $1 = \cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}$,

$$f(\theta) = \frac{\left(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\right)^2}{\left(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\right)\left(\cos\frac{\theta}{2} - \sin\frac{\theta}{2}\right)} = \frac{1 + \tan\frac{\theta}{2}}{1 - \tan\frac{\theta}{2}} = \tan\left(\frac{\theta}{2} + \frac{\pi}{4}\right),$$

cancelling the common factor, which is nonzero because $\cos\theta \neq 0$. (At $\theta = \pi$ the middle expression is undefined, but the first and last are equal there too.) As θ runs over $[0, 2\pi)$, $\psi = \frac{\theta}{2} + \frac{\pi}{4}$ runs over $[\frac{\pi}{4}, \frac{5\pi}{4})$. The graph of $\tan\psi$ shows at once that $\tan\psi \le 1$ only at $\psi = \frac{\pi}{4}$ and for $\frac{\pi}{2} < \psi < \frac{5\pi}{4}$. This is the same answer, once we remove $\psi = \pi$ (that is, $\theta = \frac{3\pi}{2}$).



The solution set (amber) is where the curve lies on or below the dashed line $f = 1$. Note the hole at $\theta = \frac{3\pi}{2}$: there $f(\theta) \to 0$, but f is not defined.

Checks

- $\theta = 0$: $f(0) = 1$, on the boundary. Just above 0 the numerator exceeds 1 while the denominator is below 1, so $f > 1$. This confirms that $\theta = 0$ is an isolated solution.
- $\theta = \frac{7\pi}{4}$: $f = \left(1 - \frac{1}{\sqrt{2}}\right) / \frac{1}{\sqrt{2}} = \sqrt{2} - 1 \approx 0.414 \leq 1$. Good.
- $\theta = \pi$: $f = 1/(-1) = -1 \leq 1$. Good.
- We tested 2×10^5 random θ in $[0, 2\pi)$ and found complete agreement with the stated solution set. Near $\theta = \frac{3\pi}{2}$ we found $f \approx 5 \times 10^{-5}$, consistent with $f \to 0$.

Common slips

- Multiplying through by $\cos\theta$ without splitting into cases. This gives $1 + \sin\theta \leq \cos\theta$ everywhere, which wrongly throws away the whole interval $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$.
- Forgetting the isolated solution $\theta = 0$. The question allows $\theta = 0$, and equality counts.
- Including $\theta = \frac{3\pi}{2}$. The expression is $\frac{0}{0}$ there, so it is not defined, even though its limit is 0.

Beyond STEP

The lesson is that inequalities are only preserved by *increasing* operations: multiplying by a positive number, applying an increasing function, and so on. Keeping track of this is the daily business of real analysis. The removable singularity at $\theta = \frac{3\pi}{2}$, where f has a limit but no value, is another theme there: continuity is about limits agreeing with values. Abbott's *Understanding Analysis* develops both ideas carefully and readably.

WORKED SOLUTION

STEP I 2003, Question 5

Pure • Binomial Series

STEP I 2003, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q5

The key idea

In any binomial expansion, write down the general term and track the power of x it carries; the constant term is the one where that power is zero. In part (ii) the expansion is not a finite one, so first pull out the dominant factor: for large $|x|$ that is x^6, leaving a series in $3/x$; for small $|x|$ it is $3x^5$, leaving a series in $x/3$. Which powers of x can then appear settles everything.

Solution

Part (i)

The general term of $(2x + x^{-2})^6$ is

$$\binom{6}{r}(2x)^{6-r}\left(x^{-2}\right)^r = \binom{6}{r}2^{6-r}x^{6-3r}, \qquad r = 0, 1, \ldots, 6.$$

This is independent of x exactly when $6 - 3r = 0$, that is $r = 2$, and then the term is

$$\binom{6}{2}2^4 = 15 \times 16 = 240,$$

as required.

In the same way, the general term of $(ax^3 + bx^{-2})^{5n}$ is

$$\binom{5n}{r}\left(ax^3\right)^{5n-r}\left(bx^{-2}\right)^r = \binom{5n}{r}a^{5n-r}b^r\,x^{15n-5r}.$$

The power of x is zero when $15n = 5r$, so $r = 3n$ (which lies in $0 \leq r \leq 5n$). The term independent of x is therefore

$$\binom{5n}{3n}a^{2n}b^{3n} = \frac{(5n)!}{(2n)!\,(3n)!}\,a^{2n}b^{3n}.$$

Part (ii)

Large $|x|$. Taking out the factor x^6,

$$f(x) = \left(x^6(1+3/x)\right)^{1/2} = x^3\Big(1+\frac{3}{x}\Big)^{1/2},$$

taking $(x^6)^{1/2} = x^3$ as the question intends (strictly this is $|x|^3$; see the note below). For $|x| > 3$ we have $|3/x| < 1$, so the binomial series for $(1+u)^{1/2}$ with $u = 3/x$ converges:

$$(1+u)^{1/2} = 1 + \tfrac{1}{2}u - \tfrac{1}{8}u^2 + \tfrac{1}{16}u^3 - \tfrac{5}{128}u^4 + \cdots,$$

where the u^3 coefficient is

$$\binom{1/2}{3} = \frac{\frac{1}{2}\cdot(-\frac{1}{2})\cdot(-\frac{3}{2})}{3!} = \frac{3/8}{6} = \frac{1}{16}.$$

Multiplying the series by x^3, the term $\binom{1/2}{k}3^k x^{-k}$ becomes $\binom{1/2}{k}3^k x^{3-k}$, so only $k = 3$ gives a constant:

$$f(x) = x^3 + \tfrac{3}{2}x^2 - \tfrac{9}{8}x + \frac{1}{16}\cdot 27 - \frac{405}{128x} + \cdots,$$

and the term independent of x is $\frac{27}{16}$, as required.

? **Ask yourself** What happens when $x < -3$?

The square root means the non-negative root, and $(x^6)^{1/2} = |x|^3$, which equals $-x^3$ when $x < 0$. So for $x < -3$ every term of the expansion changes sign, and the constant term is $-\frac{27}{16}$. The stated result $\frac{27}{16}$ is exactly right for $x > 3$; for $x < -3$ it holds if we read $(x^6)^{1/2}$ as x^3, which is evidently what the question intends.

Small $|x|$. Now the dominant term inside the bracket is $3x^5$, so we take that out instead:

$$x^6 + 3x^5 = 3x^5\Big(1+\frac{x}{3}\Big).$$

For $f(x)$ to be real we need $x^5(x+3) \geq 0$; with $|x| < 3$ the factor $x+3$ is positive, so we need $x \geq 0$. (For $-3 < x < 0$, f isn't defined as a real function at all.) For $0 < x < 3$,

$$f(x) = \sqrt{3}\,x^{5/2}\Big(1+\frac{x}{3}\Big)^{1/2} = \sqrt{3}\,x^{5/2}\sum_{k\geq 0}\binom{1/2}{k}\frac{x^k}{3^k},$$

and the series converges since $|x/3| < 1$. Every term is a constant multiple of

$$x^{5/2+k}, \qquad k = 0, 1, 2, \ldots,$$

and $\frac{5}{2} + k$ is never an integer, so in particular never 0. Hence the expansion of $f(x)$ in powers of x for $|x| < 3$ has no term independent of x. (At $x = 0$ itself, $f(0) = 0$.)

Checks

- For large positive x, $f(x) - \left(x^3 + \frac{3}{2}x^2 - \frac{9}{8}x\right)$ should approach $\frac{27}{16} = 1.6875$. At $x = 1000$ a calculator gives 1.68434, and the next term of the series, $-\frac{405}{128x} \approx -0.00316$, accounts for the gap.
- Near 0, $f(x) \approx \sqrt{3}\,x^{5/2}$: at $x = 0.01$, $f(0.01) = \sqrt{10^{-12} + 3 \times 10^{-10}} \approx 1.734 \times 10^{-5}$, close to $\sqrt{3} \times 10^{-5}$, and certainly not tending to a non-zero constant.
- For the general formula, $n = 1$, $a = b = 1$ gives $\binom{5}{3} = 10$, the constant term of $(x^3 + x^{-2})^5$; expanding directly confirms it.

Common slips

- Writing the general term with the powers the wrong way round, e.g. $(2x)^r(x^{-2})^{6-r}$, and then mismatching r with the binomial coefficient. Either labelling works if you are consistent.
- Getting the sign of $\binom{1/2}{3}$ wrong: two negative factors $(-\frac{1}{2})(-\frac{3}{2})$ make it positive.
- Expanding $(x^6 + 3x^5)^{1/2}$ for small x by taking out x^6: that gives a series in $3/x$, which diverges when $|x| < 3$.

Beyond STEP

The two expansions of f are valid on different regions, and which factor you pull out decides which region you get. This is the idea of expanding a function about different points, and it leads to Laurent series and to Puiseux series (expansions in fractional powers, like $x^{5/2+k}$ here) in complex analysis. The branch of the square root that caused the sign issue for $x < -3$ is also a central theme there. Needham's *Visual Complex Analysis* explains both pictorially.

WORKED SOLUTION

STEP I 2003, Question 6

Pure • Integration by Partial Fractions

STEP I 2003, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q6

The key idea

Both quadratics factorise, so the natural tool is partial fractions. The "different cases" come from the moment the two linear factors coincide: then partial fractions break down and we integrate a perfect square instead. Look hard at the two quadratics before starting part (ii): a simple shift of variable turns one into the other.

Solution

Part (i)

The quadratic factorises, since $2 \times a = 2a$ and $2 + a = a + 2$:

$$x^2 + (a+2)x + 2a = (x+2)(x+a).$$

Because $a > 0$, both factors are positive for $0 \le x \le 1$, so the integrand is continuous on the whole interval and the integral exists for every $a > 0$. The two factors are equal precisely when $a = 2$, and that is the dividing line.

Case $a \neq 2$. The factors are distinct, and

$$\frac{1}{(x+2)(x+a)} = \frac{1}{a-2}\left(\frac{1}{x+2} - \frac{1}{x+a}\right),$$

as we can check by recombining: the numerator becomes $(x+a) - (x+2) = a - 2$. Since both $x+2$ and $x+a$ are positive on $[0,1]$, we need no modulus signs in the logarithms:

$$\begin{aligned}\int_0^1 \frac{\mathrm{d}x}{(x+2)(x+a)} &= \frac{1}{a-2}\Big[\ln(x+2) - \ln(x+a)\Big]_0^1 \\ &= \frac{1}{a-2}\Big(\ln 3 - \ln(1+a) - \ln 2 + \ln a\Big) \\ &= \frac{1}{a-2}\ln\left(\frac{3a}{2(a+1)}\right).\end{aligned}$$

Ask yourself When $0 < a < 2$ the factor $\frac{1}{a-2}$ is negative. Is the answer still positive, as it must be for a positive integrand?

Yes. We have $\dfrac{3a}{2(a+1)} < 1 \iff 3a < 2a+2 \iff a < 2$, so for $0 < a < 2$ the logarithm is negative too, and the quotient is positive. Likewise for $a > 2$ both are positive. If you prefer every piece positive, write the answer for $0 < a < 2$ as $\dfrac{1}{2-a}\ln\left(\dfrac{2(a+1)}{3a}\right)$; it is the same number.

Case $a = 2$. Now the integrand is $(x+2)^{-2}$, and

$$\int_0^1 \frac{\mathrm{d}x}{(x+2)^2} = \left[-\frac{1}{x+2}\right]_0^1 = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}.$$

Part (ii)

Here the quadratic is $u^2 + au + a - 1 = (u+1)(u+a-1)$, but before repeating the work, substitute $u = x+1$:

$$(x+1)^2 + a(x+1) + a - 1 = x^2 + (a+2)x + 2a.$$

This is exactly the quadratic of part (i). Also $\mathrm{d}u = \mathrm{d}x$, and $u = 1, 2$ correspond to $x = 0, 1$. So

$$\int_1^2 \frac{\mathrm{d}u}{u^2 + au + a - 1} = \int_0^1 \frac{\mathrm{d}x}{x^2 + (a+2)x + 2a},$$

and the answers are the same as in part (i):

$$\int_1^2 \frac{\mathrm{d}u}{u^2 + au + a - 1} = \begin{cases} \dfrac{1}{a-2}\ln\left(\dfrac{3a}{2(a+1)}\right), & a \neq 2, \\ \dfrac{1}{6}, & a = 2. \end{cases}$$

Directly: on $[1,2]$ we have $u + 1 > 0$ and $u + a - 1 \geq a > 0$, so the integrand is again continuous, and the factors coincide exactly when $a - 1 = 1$, that is $a = 2$. Partial fractions give $\frac{1}{a-2}\left[\ln\frac{u+1}{u+a-1}\right]_1^2 = \frac{1}{a-2}\left(\ln\frac{3}{a+1} - \ln\frac{2}{a}\right)$, which is the same expression.

Checks

- **Continuity at $a = 2$.** As $a \to 2$, the logarithm tends to $\ln 1 = 0$, so the formula is of the form $0/0$. Its limit is the derivative of $\ln 3a - \ln(2a+2)$ at $a = 2$, namely $\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, which agrees with the special case. The two cases join up smoothly, as they should.
- **A special value.** With $a = 1$ the integrand of (i) is $\frac{1}{(x+1)(x+2)}$ and the formula gives $-\ln\frac{3}{4} = \ln\frac{4}{3} \approx 0.2877$.
- **Numerically.** Simpson's rule on both integrals, for $a = 1$, $a = 2$, a close to 0 and 2, and 200 random values of a up to 20, agrees with the formula to better than 10^{-10}.

Common slips

- Forgetting the case $a = 2$, where the partial-fraction coefficient $\frac{1}{a-2}$ is undefined.
- Getting the sign of the partial fractions the wrong way round. Recombine to check.
- Treating $0 < a < 2$ and $a > 2$ as genuinely different answers. One formula covers both, provided you keep track of signs.

Beyond STEP

The behaviour at $a = 2$ is a small example of a removable singularity: a formula that looks undefined at a point, but whose limit there is exactly the value of the integral. In analysis you learn when an integral depending on a parameter is continuous (or differentiable) in that parameter, which explains why the two cases had to agree. Bartle and Sherbert, *Introduction to Real Analysis*, treats limits, continuity and the Riemann integral carefully.

WORKED SOLUTION

STEP I 2003, Question 7

Pure • Digits and Divisibility

STEP I 2003, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q7

The key idea

Writing $N = 100a + 10b + c$ and setting $S = N$ gives one equation in three digits. Gather it as

$$99a + (10b - b^2) = c^3 - c,$$

and the two facts from the first paragraph slot straight in: the b-term is trapped between 0 and 25, and both $99a$ and $c^3 - c$ are multiples of 3. The first fact means $c^3 - c$ must sit just above a multiple of 99; the second cuts down the possible values of b. After that there are only a handful of cases to check by hand.

Solution

The two preliminary facts

Completing the square,

$$10k - k^2 = 25 - (5 - k)^2.$$

Since $(5 - k)^2 \geq 0$, we get $10k - k^2 \leq 25$. Since $0 \leq k \leq 9$, we have $-4 \leq 5 - k \leq 5$, so $(5 - k)^2 \leq 25$ and hence $10k - k^2 \geq 0$. (Equality at 0 happens for $k = 0$ and at 25 for $k = 5$.)

Next, $k - 1$, k, $k + 1$ are three consecutive integers, so exactly one of them is a multiple of 3. Hence their product $k(k - 1)(k + 1)$ is divisible by 3. (This holds for every integer k, not just $0 \leq k \leq 9$.)

Finding the numbers

Let $N = 100a + 10b + c$ with digits $1 \leq a \leq 9$ (as $N \geq 100$) and $0 \leq b, c \leq 9$. Then $S = a + b^2 + c^3$, and

$$\begin{aligned} S = N &\iff a + b^2 + c^3 = 100a + 10b + c \\ &\iff 99a + (10b - b^2) = c^3 - c = (c - 1)c(c + 1). \qquad (*) \end{aligned}$$

Ask yourself Why put $10b - b^2$ on the same side as $99a$?

Because that is the combination we have just learned to control. With $t = 10b - b^2$, the first fact says $0 \le t \le 25$, and $99a$ is a multiple of 99 that is much bigger than 25. So $(*)$ says that $c^3 - c$ is a multiple of 99 plus a small "remainder" t, which pins things down very quickly.

Restricting b. By the second fact, 3 divides $c^3 - c = (c-1)c(c+1)$, and 3 divides $99a$. So by $(*)$, 3 divides $t = b(10-b)$. Since $10 - b \equiv 1 - b \pmod 3$, this needs $b \equiv 0$ or $b \equiv 1 \pmod 3$. The possibilities are

b	0	1	3	4	6	7	9
$t = 10b - b^2$	0	9	21	24	24	21	9

so $t \in \{0, 9, 21, 24\}$. (The values $b = 2, 5, 8$ give $t = 16, 25, 16$, none divisible by 3.)

Restricting c. Since $a \ge 1$ and $t \ge 0$, $(*)$ gives $c^3 - c \ge 99$. The values of $c^3 - c$ are

c	0	1	2	3	4	5	6	7	8	9
$c^3 - c$	0	0	6	24	60	120	210	336	504	720

so $c \ge 5$.

The cases. For each $c \ge 5$ we need a multiple $99a$ (with $1 \le a \le 9$) lying in the range $[c^3 - c - 25,\ c^3 - c]$, and then $t = c^3 - c - 99a$ must be one of $0, 9, 21, 24$.

- $c = 5$: $c^3 - c = 120$. The only multiple of 99 in $[95, 120]$ is 99, giving $a = 1$ and $t = 21$, so $b = 3$ or $b = 7$. This gives $N = 135$ and $N = 175$.
- $c = 6$: $c^3 - c = 210$. The only multiple of 99 in $[185, 210]$ is 198, giving $a = 2$ and $t = 12$. But 12 is not one of $0, 9, 21, 24$ (indeed $10b - b^2 = 12$ has no integer root). No solution.
- $c = 7$: $c^3 - c = 336$. The multiples nearby are 297 and 396, neither in $[311, 336]$. No solution.
- $c = 8$: $c^3 - c = 504$. The only multiple of 99 in $[479, 504]$ is 495, giving $a = 5$ and $t = 9$, so $b = 1$ or $b = 9$. This gives $N = 518$ and $N = 598$.
- $c = 9$: $c^3 - c = 720$. The multiples nearby are 693 and 792, neither in $[695, 720]$. No solution.

Every step above is reversible, so these four do satisfy $S = N$; checking directly,

$$135 : 1 + 3^2 + 5^3 = 1 + 9 + 125 = 135, \qquad 175 : 1 + 7^2 + 5^3 = 1 + 49 + 125 = 175,$$
$$518 : 5 + 1^2 + 8^3 = 5 + 1 + 512 = 518, \qquad 598 : 5 + 9^2 + 8^3 = 5 + 81 + 512 = 598.$$

So the numbers with $S = N$ are exactly

$$N = 135, \quad 175, \quad 518, \quad 598.$$

Checks

A brute-force search over all 900 three-digit numbers, computing $a + b^2 + c^3$ for each, finds exactly 135, 175, 518 and 598, agreeing with the case analysis. A neat structural check: in each pair the two values of b add to 10 ($3 + 7$ and $1 + 9$). That is forced, because $t = 10b - b^2$ is unchanged when b is replaced by $10 - b$.

Common slips

- Forgetting that $a \geq 1$, and so admitting "solutions" such as 000 or 001 that are not three-digit numbers.
- Stopping after finding one b for a given a and c: since $t(b) = t(10 - b)$, solutions come in pairs.
- Using only the divisibility by 3 and not the bound $0 \leq t \leq 25$: it is the bound that leaves at most one possible a for each c.

Beyond STEP

Numbers equal to a sum built from powers of their own digits are a classic recreation (the Armstrong or "narcissistic" numbers, such as $153 = 1^3 + 5^3 + 3^3$, are the best-known family). The serious idea underneath is the one used here: bound the size of each side to make the search finite, then use congruences to prune it. The same pairing of size estimates with modular arithmetic drives the solution of many Diophantine equations. Davenport's *The Higher Arithmetic* is an excellent introduction to congruences and their uses.

WORKED SOLUTION

STEP I 2003, Question 8

Pure • Logistic Differential Equation

STEP I 2003, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q8

The key idea

Translate "the rate at which A converts into B" into a derivative: it is the rate at which the volume of B grows. Because $x = 1 - y$, this gives a differential equation in y alone with separable variables, and partial fractions finish the integration. The final part is about whether the formula for $x = 1 - y$ can ever reach zero, and the answer rests on the fact that an exponential is never infinite.

Solution

Setting up the equation

The volume of B is yV, and V is fixed. A turns into B at rate $kVxy$, so the volume of B increases at that rate:

$$\frac{\mathrm{d}}{\mathrm{d}t}(yV) = kVxy \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}t} = kxy = ky(1-y),$$

dividing by V and using $x = 1 - y$. (Equivalently, the volume of A falls at this rate: $V\,\mathrm{d}x/\mathrm{d}t = -kVxy$, which says the same thing because $\mathrm{d}x/\mathrm{d}t = -\mathrm{d}y/\mathrm{d}t$.)

Solving it

At the start $x > 0$ and $y > 0$, so $0 < y < 1$. While $0 < y < 1$ we can divide by $y(1-y)$ and separate the variables:

$$\int \frac{\mathrm{d}y}{y(1-y)} = \int k\,\mathrm{d}t.$$

By partial fractions, $\dfrac{1}{y(1-y)} = \dfrac{1}{y} + \dfrac{1}{1-y}$, so

$$\ln y - \ln(1-y) = kt + c,$$

$$\ln \frac{y}{1-y} = kt + c,$$

where we've used that y and $1-y$ are both positive, so no modulus signs are needed. Exponentiating and writing $D = \mathrm{e}^c$,

$$\frac{y}{1-y} = D\mathrm{e}^{kt}.$$

Then $y = D\mathrm{e}^{kt} - yD\mathrm{e}^{kt}$, so $y\left(1 + D\mathrm{e}^{kt}\right) = D\mathrm{e}^{kt}$ and

$$y = \frac{D\mathrm{e}^{kt}}{1 + D\mathrm{e}^{kt}},$$

as required. Putting $t = 0$ shows $D = \dfrac{y_0}{1-y_0} = \dfrac{y_0}{x_0}$, where x_0 and y_0 are the starting values; in particular $D > 0$.

Ask yourself We assumed $0 < y < 1$ to do the integration. How do we know this stays true for all time?

Suppose not, and let T be the first time at which y reaches 0 or 1. On $0 \le t < T$ the working above is valid, so y is given by the formula there. Since y is continuous, $y(T)$ is the limit of the formula as $t \to T$, namely $D\mathrm{e}^{kT}/(1 + D\mathrm{e}^{kT})$. With $D > 0$ this lies strictly between 0 and 1, which is a contradiction. So $0 < y < 1$ for all t, and the formula holds for all t.

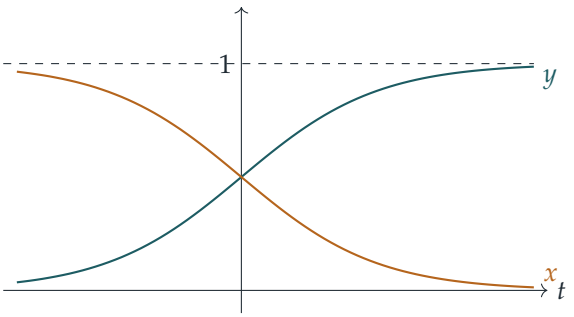
Does A ever completely convert?

No. From the formula,

$$x = 1 - y = \frac{1}{1 + D\mathrm{e}^{kt}},$$

and since $D\mathrm{e}^{kt}$ is finite for every t, $x > 0$ at every time: some of A is always left. However, as $t \to \infty$, $D\mathrm{e}^{kt} \to \infty$ (because $k > 0$ and $D > 0$), so $x \to 0$ and $y \to 1$. The conversion gets ever closer to complete but never finishes in finite time.

This makes physical sense: the reaction rate $kVxy$ is proportional to the amount of A left, so the reaction slows down as A runs out.



The sketch (with $D = k = 1$) shows the S-shaped logistic curve for y approaching 1 and the curve for x approaching 0, neither ever reaching its limit.

Checks

- Differentiate the answer: with $u = De^{kt}$, $y = u/(1+u)$ and $\mathrm{d}y/\mathrm{d}t = ku/(1+u)^2 = ky(1-y)$, as it should.
- With $D = k = 1$ (so we start with equal volumes) the formula gives $y = \frac{3}{4}$ at $t = \ln 3$. We also stepped the equation forward numerically for 200 random choices of k and starting value, and it matched the formula to within 10^{-12}, with x staying positive throughout.

Common slips

- Treating "$y = 1$ in the limit" as "$y = 1$ at some time". A limit as $t \to \infty$ is not attained at any finite time.
- Forgetting that the step $\ln(y/(1-y))$ needs $0 < y < 1$, or not saying why that condition persists.

Beyond STEP

The equation $\dot{y} = ky(1-y)$ is the logistic equation, the standard model for growth that is limited by a resource. Here $y = 0$ and $y = 1$ are equilibrium solutions, and the uniqueness theorem for differential equations says no other solution can ever cross or touch them, which is a slicker way to settle the final part. Boyce and DiPrima, *Elementary Differential Equations*, discusses the logistic equation and this kind of stability analysis in detail.

WORKED SOLUTION

STEP I 2003, Question 9

Mechanics • Projectile Through a Point

STEP I 2003, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

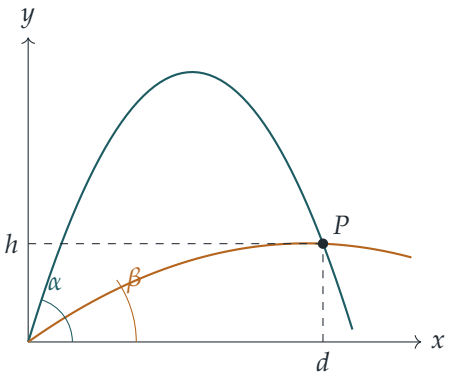
step.maths.org/questions/03-s1-q9

The key idea

Write down the horizontal and vertical displacements, eliminate the time, and replace $\sec^2\theta$ by $1+\tan^2\theta$: the trajectory equation becomes a quadratic in $T=\tan\theta$. After that everything is about this quadratic. Two angles means two distinct real roots, so look at the discriminant. And $\alpha+\beta$ is controlled by the sum and product of the roots, through the formula for $\tan(\alpha+\beta)$.

Solution

We model the particle as moving freely under gravity alone (no air resistance), with the origin at the point of projection, x horizontal towards P and y vertically upwards. So P is the point (d,h) with $d>0$. We take $h>0$, meaning that P is above the point of projection; we come back to $h\leq 0$ at the end.



The quadratic in T

The horizontal velocity is constant, $V\cos\theta$, and the vertical motion has constant acceleration $-g$, so at time t

$$x = Vt\cos\theta, \qquad y = Vt\sin\theta - \tfrac{1}{2}gt^2.$$

The particle reaches $x = d$ at time $t = d/(V\cos\theta)$ (here $\cos\theta > 0$, as the particle moves towards P). Substituting this into y and setting $y = h$,

$$h = d\tan\theta - \frac{gd^2}{2V^2\cos^2\theta} = dT - \frac{gd^2}{2V^2}(1+T^2),$$

using $\sec^2\theta = 1 + \tan^2\theta$. Now multiply through by $\dfrac{2V^2}{gd^2} = \dfrac{2k}{d}$:

$$\frac{2kh}{d} = 2kT - (1+T^2), \qquad \text{that is,} \qquad T^2 - 2kT + \frac{2kh}{d} + 1 = 0,$$

as required.

Two distinct angles

For a given V (and so a given k), the possible angles of projection are the $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ whose tangent is a root of this quadratic. Since tan is strictly increasing on that interval, distinct real roots T give distinct angles. So we need the quadratic to have two distinct real roots, which happens exactly when its discriminant is positive:

$$4k^2 - 4\left(\frac{2kh}{d} + 1\right) > 0 \iff k^2d^2 - 2h(kd) - d^2 > 0 \iff (kd-h)^2 > h^2 + d^2,$$

multiplying by $d^2 > 0$ and then completing the square in kd.

Ask yourself Why does the given condition imply $(kd-h)^2 > h^2 + d^2$, when squaring inequalities can go wrong?

If $kd > h + \sqrt{h^2+d^2}$, then $kd - h > \sqrt{h^2+d^2} > 0$. Both sides are positive, so squaring keeps the inequality: $(kd-h)^2 > h^2 + d^2$. So the discriminant is positive and there are two distinct roots T, hence two distinct angles of projection.

(The converse holds too. The other way to have $(kd-h)^2 > h^2+d^2$ is $kd < h - \sqrt{h^2+d^2}$, but the right-hand side is negative since $\sqrt{h^2+d^2} > |h|$, while $kd = V^2/g > 0$. So two angles are possible *exactly* when $kd > h + \sqrt{h^2+d^2}$.)

Both angles are genuinely *above* the horizontal. The roots have sum $2k > 0$ and product $1 + 2kh/d > 0$ (as $h > 0$), so both are positive.

The sum of the angles

Let $\tan\alpha$ and $\tan\beta$ be the two roots. From the quadratic,

$$\tan\alpha + \tan\beta = 2k, \qquad \tan\alpha\tan\beta = 1 + \frac{2kh}{d}.$$

The product is not 1 (since $kh/d \neq 0$), so $\alpha + \beta \neq \frac{\pi}{2}$ and the compound-angle formula applies:

$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{2k}{-2kh/d} = -\frac{d}{h}.$$

Note that k has cancelled: the sum of the angles doesn't depend on the speed.

Ask yourself $\tan(\alpha+\beta) = -d/h$ has infinitely many solutions. Which one is $\alpha+\beta$?

We showed $\alpha, \beta \in (0, \frac{\pi}{2})$, so $0 < \alpha+\beta < \pi$. Moreover $\tan(\alpha+\beta) < 0$, so $\alpha+\beta$ lies in $(\frac{\pi}{2}, \pi)$. The one angle in $(\frac{\pi}{2}, \pi)$ with tangent $-d/h$ is $\pi - \arctan(d/h)$, because $\arctan(d/h) \in (0, \frac{\pi}{2})$ and $\tan(\pi - u) = -\tan u$. Therefore

$$\alpha + \beta = \pi - \arctan\left(\frac{d}{h}\right),$$

as required.

There is a nice way to see this. $\arctan(d/h)$ is the angle between OP and the upward vertical, so $\pi - \arctan(d/h) = \frac{\pi}{2} + \gamma$, where γ is the angle of elevation of P. Then $\frac{1}{2}(\alpha+\beta) = \frac{\pi}{4} + \frac{\gamma}{2}$: the two directions of projection are symmetric about the line bisecting the angle between OP and the vertical.

Checks

- If $h = 0$ (the target is on the same level) the formula gives $\alpha + \beta = \pi - \frac{\pi}{2} = \frac{\pi}{2}$: the familiar complementary pair of angles with equal range.
- Take $d = 3$, $h = 4$, $k = 4$. Then $kd = 12 > 4 + 5$, and the roots are $T = 4 \pm \sqrt{13/3}$, giving $\alpha \approx 1.4078$ and $\beta \approx 1.0903$. So $\alpha + \beta \approx 2.4981$, which equals $\pi - \arctan(0.75) \approx 2.4981$.
- A numerical simulation of the flight at each of the two angles, for thousands of random d, h and k, passes through (d, h) and confirms both the condition and the sum of the angles.

Common slips

- Forgetting that $\sec^2\theta = 1 + \tan^2\theta$, and so never reaching a quadratic in T.
- Writing $\alpha + \beta = \arctan(-d/h) = -\arctan(d/h)$, which is negative. The inverse tangent only returns angles in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so you must use the range of $\alpha+\beta$ to pick the correct branch.
- Taking $h < 0$ (a target below the point of projection) without care. Then one root may be negative, and the correct general statement is $\alpha + \beta = \frac{\pi}{2} + \arctan(h/d)$. This agrees with the question's form only when $h > 0$.

Beyond STEP

The boundary case $kd = h + \sqrt{h^2 + d^2}$, where the two angles merge, traces out the *envelope* of all trajectories at speed V: the "parabola of safety" $y = \dfrac{V^2}{2g} - \dfrac{gx^2}{2V^2}$. Points inside it can be hit in two ways, points on it in one way, and points outside not at all. Envelopes of families of curves, found by solving an equation and its derivative with respect to the parameter at the same time, come up again in differential equations and optics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops projectile and constrained motion with this kind of care.

WORKED SOLUTION

STEP I 2003, Question 10

Mechanics • Centre of Mass of a Lamina

STEP I 2003, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q10

The key idea

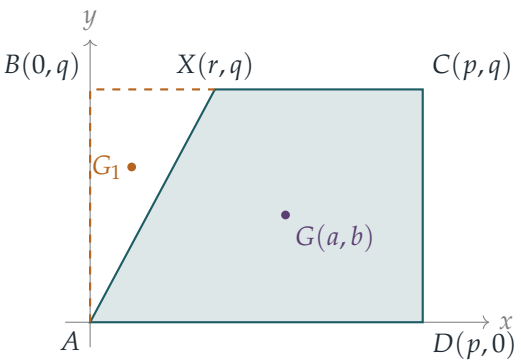
A shape with a piece removed is best handled by *subtraction*: the rectangle's moment about each axis equals the trapezium's moment plus the triangle's moment, with masses proportional to areas. For the suspension, a lamina hanging freely from a point comes to rest with its centre of gravity directly below that point, so the line AG is vertical. The 45° condition then turns into a very simple relation between a and b.

Solution

We model the lamina as uniform and thin, so the mass of any piece is proportional to its area and the centre of gravity of a triangle is at its centroid (the mean of its vertices).

The centre of gravity

With the given axes, $A = (0,0)$, $D = (p,0)$, $C = (p,q)$, and so $B = (0,q)$. Since X lies on BC with $BX = r$, we have $X = (r,q)$, where $0 \le r \le p$.



The three pieces are as follows.

- The rectangle $ABCD$: area pq, centre of gravity $(\frac{p}{2}, \frac{q}{2})$.
- The triangle ABX (removed): area $\frac{1}{2}qr$, centre of gravity G_1 at the mean of $(0,0)$, $(0,q)$ and (r,q), which is $(\frac{r}{3}, \frac{2q}{3})$.
- The trapezium $AXCD$ that remains: area $pq - \frac{1}{2}qr = \frac{1}{2}q(2p-r)$, centre of gravity $G = (a,b)$.

The rectangle's moment about each axis is the sum of the moments of its two parts. Taking moments about the y-axis (that is, comparing x-coordinates),

$$pq \cdot \frac{p}{2} = \frac{qr}{2} \cdot \frac{r}{3} + \frac{q(2p-r)}{2}\, a.$$

Dividing by $\frac{q}{2}$ gives $p^2 = \frac{r^2}{3} + (2p-r)a$, and since $2p - r \geq p > 0$ we may divide by it:

$$a = \frac{3p^2 - r^2}{3(2p-r)}.$$

Similarly, about the x-axis,

$$pq \cdot \frac{q}{2} = \frac{qr}{2} \cdot \frac{2q}{3} + \frac{q(2p-r)}{2}\, b,$$

so $pq = \frac{2qr}{3} + (2p-r)b$ and

$$b = \frac{q(3p-2r)}{3(2p-r)}.$$

As a quick check, $r = 0$ (nothing cut off) gives $a = \frac{p}{2}$, $b = \frac{q}{2}$, the centre of the rectangle.

Suspending from A

When the lamina hangs freely from A and is in equilibrium, only two forces act on it: its weight, through G, and the reaction at A. Two forces in equilibrium must act along the same line, so G lies vertically below A.

Ask yourself How does "AD is at 45° below the horizontal" become an equation in a and b?

Since AG is vertical and AD makes 45° with the horizontal, AD also makes 45° with AG; that is, the angle GAD is 45°. In our axes AD is the positive x-axis and $G = (a,b)$ with $a, b > 0$ (it lies inside the trapezium), so $\tan \angle GAD = b/a$. Hence $b/a = 1$, that is,

$$a = b.$$

Using the expressions above and cancelling the common positive denominator $3(2p-r)$,

$$\begin{aligned} & 3p^2 - r^2 = q(3p-2r) \\ \iff \quad & r^2 - 2qr + 3pq - 3p^2 = 0. \end{aligned}$$

By the quadratic formula,

$$r = q \pm \sqrt{q^2 - (3pq - 3p^2)} = q \pm \sqrt{q^2 - 3pq + 3p^2}.$$

The quantity under the root is $(q - \frac{3}{2}p)^2 + \frac{3}{4}p^2 > 0$, so both roots are real and distinct. Write $s = \sqrt{q^2 - 3pq + 3p^2} > 0$.

Choosing the sign. Since X lies on the side BC, which has length p, we need $0 \le r \le p$. We show that the root $q + s$ is always bigger than p.

- If $q \ge p$, then $q + s > q \ge p$, because $s > 0$.
- If $q < p$, then $p - q > 0$ and

$$s^2 - (p - q)^2 = (q^2 - 3pq + 3p^2) - (p^2 - 2pq + q^2) = 2p^2 - pq = p(2p - q) > 0,$$

since $q < p < 2p$. So $s > p - q$ and hence $q + s > p$.

Either way the plus sign would put X beyond C, off the lamina, so it is impossible. Therefore

$$r = q - \sqrt{q^2 - 3pq + 3p^2},$$

as required.

For completeness: this root really does lie in $[0, p]$ exactly when $p \le q \le 2p$. Indeed $q - s \ge 0 \iff q^2 \ge s^2 \iff 3p(q - p) \ge 0 \iff q \ge p$, and (when $q > p$) $q - s \le p \iff (q - p)^2 \le s^2 \iff p(2p - q) \ge 0 \iff q \le 2p$. So the situation described can only arise for such shapes, which matches intuition: the uncut square ($q = p$, $r = 0$) hangs from a corner with its diagonal vertical, and at the other extreme ($q = 2p$, $r = p$) the shape is the right-angled triangle ACD, whose centroid $(\frac{2p}{3}, \frac{q}{3})$ again has $a = b$.

Checks

- Setting $r = 0$ recovers the rectangle's centre $(\frac{p}{2}, \frac{q}{2})$, and $r = p$ gives $(\frac{2p}{3}, \frac{q}{3})$, the centroid of triangle ACD.
- With $p = 1$, $q = 1.5$ the formula gives $r = 1.5 - \sqrt{0.75} \approx 0.634$, and then $a = b \approx 0.634$, so AG is indeed at 45° to AD. The other root, ≈ 2.37, exceeds $p = 1$.
- We also checked the centre-of-gravity formulae against the general polygon centroid formula for thousands of random p, q, r.

Common slips

- Using the triangle's centroid as $(\frac{r}{3}, \frac{q}{3})$: two of its vertices are at height q, so its centroid is at height $\frac{2q}{3}$.

- Adding the triangle's moment instead of subtracting it: the trapezium is the rectangle *minus* the triangle.
- Rejecting the plus sign with a vague "r must be small". The real reason is $r \leq p$, and that needs a short argument, split by whether $q \geq p$.

Beyond STEP

Centres of mass by subtraction are a discrete version of computing $\iint x \, \mathrm{d}A \big/ \iint \mathrm{d}A$, and at university you meet the same idea with double integrals, for laminas of varying density and for solids. The hanging condition is the simplest case of a general principle: a rigid body pivoted at a point is in stable equilibrium when its centre of mass is directly below the pivot, where its potential energy is least. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats centres of mass and rigid-body equilibrium carefully.

WORKED SOLUTION

STEP I 2003, Question 11

Mechanics • Collisions on an Inclined Plane

STEP I 2003, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q11

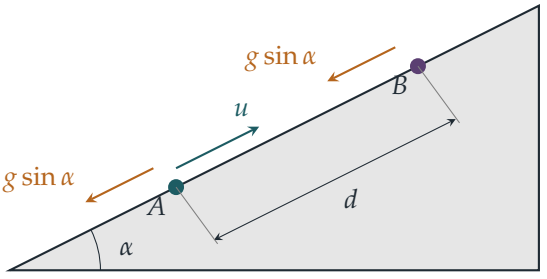
The key idea

On a smooth slope every particle has the same acceleration, $g\sin\alpha$ down the line of greatest slope, whatever its velocity. So gravity cannot change the *relative* motion of the two particles: seen from one of them, the other moves at constant velocity. That makes the collision time easy to find, and it tells us the relative velocity at impact without any messy kinematics. The energy loss in a collision depends only on that relative velocity, which is why α, g and d all disappear from the answer.

Solution

The model

We treat both particles as point masses moving along the line of greatest slope through A and B. The plane is smooth, so the only forces on each particle are its weight and the normal reaction. Resolving perpendicular to the plane, the reaction balances $mg\cos\alpha$. Resolving along the plane, each particle has acceleration $g\sin\alpha$ down the slope. Take x as the distance up the slope from A.



The time of collision

Let $a = g\sin\alpha$. With up the slope as positive, the particle from A (call it P) and the particle from B (call it Q) have positions

$$x_P = ut - \tfrac{1}{2}at^2,$$
$$x_Q = d - \tfrac{1}{2}at^2.$$

The gap between them is

$$x_Q - x_P = d - ut,$$

because the $\frac{1}{2}at^2$ terms cancel. The gap shrinks steadily from d and is zero exactly when

$$t = \frac{d}{u},$$

which is what we had to show. The cancellation is the key idea at work: both particles fall down the slope in the same way, so only the initial relative speed u matters.

The loss of kinetic energy

At $t = d/u$ the velocities up the slope are

$$v_P = u - \frac{ad}{u}, \qquad v_Q = -\frac{ad}{u}.$$

Write $w = ad/u$ for short. Then $v_P = u - w$ and $v_Q = -w$, and P approaches Q with relative speed $v_P - v_Q = u$.

> **Ask yourself** Why can we use conservation of momentum when gravity is acting?

The collision is instantaneous. Over the tiny time of impact, the finite forces of gravity and the normal reaction produce a negligible impulse compared with the large impulse between the particles. So momentum along the slope is conserved in the collision.

Let the velocities up the slope after the collision be v'_P and v'_Q. Since the masses are equal, conservation of momentum gives

$$v'_P + v'_Q = v_P + v_Q = u - 2w. \tag{1}$$

Newton's experimental law gives the speed of separation as e times the speed of approach:

$$v'_Q - v'_P = e(v_P - v_Q) = eu. \tag{2}$$

Rather than solving for v'_P and v'_Q separately, we use the identity

$$p^2 + q^2 = \tfrac{1}{2}\left[(p+q)^2 + (p-q)^2\right].$$

Applying it before and after the collision, the total kinetic energies are

$$\text{KE before} = \tfrac{1}{2}m\left(v_P^2 + v_Q^2\right) = \tfrac{1}{4}m\left[(u-2w)^2 + u^2\right],$$
$$\text{KE after} = \tfrac{1}{2}m\left(v_P'^2 + v_Q'^2\right) = \tfrac{1}{4}m\left[(u-2w)^2 + e^2u^2\right],$$

using (1) and (2) for the sum and difference after the collision. The sum term is the same in both, so

$$\text{loss of KE} = \tfrac{1}{4}mu^2 - \tfrac{1}{4}me^2u^2 = \tfrac{1}{4}mu^2(1-e^2),$$

as required.

Ask yourself What is the $(u-2w)^2$ term physically, and why does it not change?

Half of $v_P + v_Q$ is the velocity of the centre of mass. The term $\frac{1}{4}m(u-2w)^2$ is the kinetic energy of the total mass $2m$ moving with that velocity, and momentum conservation keeps it fixed. Only the kinetic energy of the motion *relative* to the centre of mass can be lost, and that depends only on the relative speed, u before and eu after.

Checks

- If $e = 1$ (perfectly elastic) the loss is zero, as it should be. If $e = 0$ the particles move together afterwards and the loss is $\frac{1}{4}mu^2$, the full kinetic energy of the relative motion.
- If $\alpha = 0$ the plane is horizontal, Q stays at rest, and P reaches it at speed u after time d/u, which agrees.
- Numerically, with $\alpha = 30^\circ$, $g = 9.8$, $d = 2$, $u = 4$, $e = \frac{1}{2}$ and $m = 1$: at $t = 0.5$ the velocities are 1.55 and -2.45. Afterwards they are -1.45 and 0.55, and the kinetic energy falls from 4.2025 to 1.2025, a loss of 3, which equals $\frac{1}{4} \cdot 16 \cdot \frac{3}{4}$. A random test of 1000 cases, stepping the motion numerically, agreed too.

Common slips

- Taking Q's velocity at impact to be zero. It has been sliding down the slope for time d/u, just as P has been slowing down.
- Getting a sign wrong in Newton's law. Write both velocities in the same positive direction, then separation speed $= e \times$ approach speed.
- Solving fully for v_P' and v_Q' and then expanding four squares. It works, but the sum-and-difference identity avoids almost all the algebra.

Beyond STEP

Both parts are examples of changing to the centre-of-mass frame. A uniform gravitational field gives every particle the same acceleration, so it drops out of relative motion completely. Also, the kinetic energy of any system splits into the kinetic energy of the centre of mass plus the kinetic energy relative to it (König's theorem). In two-body

problems the second part is $\frac{1}{2}\mu v_{\text{rel}}^2$ with the reduced mass $\mu = m_1 m_2/(m_1 + m_2)$, here $\frac{1}{2}m$. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

WORKED SOLUTION

STEP I 2003, Question 12

Probability and Statistics • Expectation of Products

STEP I 2003, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q12

The key idea

An expected value is a sum of (value $\times$ probability) over all equally likely outcomes, so both parts come down to listing the ordered pairs of numbers and adding up their products. The double sum $\sum_r \sum_s rs$ factorises as $\left(\sum r\right)^2$, and that is the whole of part (i). In part (ii) the only change is that the pairs with $r = s$ are no longer possible, so we take the same double sum and remove its diagonal.

Solution

Let X be the number on the first ball and Y the number on the second.

Part (i)

Because the first ball is replaced, the second draw is made from the full bag again and does not depend on the first. So for any $r, s \in \{1, \ldots, n\}$,

$$\mathrm{P}(X = r,\ Y = s) = \mathrm{P}(X = r)\,\mathrm{P}(Y = s) = \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n^2}.$$

There are n^2 such ordered pairs (r, s), each giving product rs, so by the definition of expectation

$$\mathrm{E}(XY) = \sum_{r=1}^{n} \sum_{s=1}^{n} rs\,\mathrm{P}(X = r,\ Y = s) = \frac{1}{n^2} \sum_{r=1}^{n} \sum_{s=1}^{n} rs,$$

which is the expression we were asked to explain.

To simplify it, notice that r does not depend on s, so it comes outside the inner sum:

$$\sum_{r=1}^{n} \sum_{s=1}^{n} rs = \sum_{r=1}^{n} r\left(\sum_{s=1}^{n} s\right) = \left(\sum_{r=1}^{n} r\right)^2 = \tfrac{1}{4}n^2(n+1)^2.$$

Therefore

$$\mathrm{E}(XY) = \frac{(n+1)^2}{4}.$$

(Equivalently, X and Y are independent, so $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y) = \left(\frac{n+1}{2}\right)^2$.)

Part (ii)

We need $n \geq 2$ for two balls to be drawn. Now the first ball is not replaced, so the possible outcomes are the ordered pairs (r, s) with $r \neq s$. For such a pair,

$$\mathrm{P}(X = r,\ Y = s) = \mathrm{P}(X = r)\,\mathrm{P}(Y = s \mid X = r) = \frac{1}{n} \cdot \frac{1}{n-1},$$

since after ball r is removed the second ball is equally likely to be any of the remaining $n - 1$. There are $n(n-1)$ such pairs, all equally likely, and the pairs with $r = s$ have probability 0. Hence

$$\mathrm{E}(XY) = \frac{1}{n(n-1)} \sum_{r \neq s} rs.$$

? **Ask yourself** How do we add up rs over only the pairs with $r \neq s$?

Take the sum over *all* pairs, which we found in part (i), and subtract the diagonal terms $r = s$, which contribute $r \cdot r = r^2$:

$$\sum_{r \neq s} rs = \sum_{r=1}^{n} \sum_{s=1}^{n} rs - \sum_{r=1}^{n} r^2 = \tfrac{1}{4}n^2(n+1)^2 - \tfrac{1}{6}n(n+1)(2n+1).$$

Taking out the common factor $\frac{1}{12}n(n+1)$,

$$\begin{aligned} \sum_{r \neq s} rs &= \tfrac{1}{12}n(n+1)\left[3n(n+1) - 2(2n+1)\right] \\ &= \tfrac{1}{12}n(n+1)\left(3n^2 - n - 2\right) \\ &= \tfrac{1}{12}n(n+1)(n-1)(3n+2). \end{aligned}$$

Dividing by $n(n-1)$, which is non-zero because $n \geq 2$,

$$\mathrm{E}(XY) = \frac{(n+1)(3n+2)}{12},$$

as required.

Checks

Take $n = 3$. With replacement, the nine products have mean $\frac{1}{9}(1+2+3)^2 = 4$, and $\frac{(3+1)^2}{4} = 4$. Without replacement, the unordered pairs $\{1,2\}, \{1,3\}, \{2,3\}$ give products $2, 3, 6$ with mean $\frac{11}{3}$, and $\frac{4 \cdot 11}{12} = \frac{11}{3}$. For $n = 2$ the only product without replacement is 2, and the formula gives $\frac{3 \cdot 8}{12} = 2$. We also confirmed both formulae by exact enumeration for every n up to 12.

Comparing the two answers is a good sanity check too:

$$\frac{(n+1)(3n+2)}{12} - \frac{(n+1)^2}{4} = -\frac{n+1}{12} < 0.$$

Without replacement the answer is smaller, which makes sense: we have lost the large diagonal products such as $n \cdot n$.

Common slips

- Dividing by n^2 in part (ii). There are only $n(n-1)$ possible ordered pairs, and each has probability $\frac{1}{n(n-1)}$.
- Using unordered pairs for the sum but ordered pairs for the count (or the other way round). Summing over unordered pairs halves both the sum and the count, so the ratio is the same, but mixing the two is out by a factor of 2.
- Writing $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y)$ in part (ii). That needs independence, and X and Y are not independent when the ball is not replaced.

Beyond STEP

The difference we found is exactly the covariance: $\mathrm{Cov}(X, Y) = \mathrm{E}(XY) - \mathrm{E}(X)\,\mathrm{E}(Y) = -\frac{n+1}{12}$, and it is negative because a large first number makes a large second number slightly less likely. Covariances like this are what you need to find the variance of a sum when sampling without replacement, which leads to the variance of the hypergeometric distribution and the "finite population correction" used in surveys. Blitzstein and Hwang, *Introduction to Probability*, develops covariance and indicator methods of exactly this kind.

WORKED SOLUTION

STEP I 2003, Question 13

Probability and Statistics • Geometric Distribution Expectation

STEP I 2003, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q13

The key idea

For the first part, count each team's *misses* instead of its goals: after five shots each, the teams are level exactly when they have missed the same number of times. For the expectation, notice that the sudden-death stage happens in rounds of two shots, and every round is an independent trial that ends the shoot-out with the same probability. So the number of rounds is geometric, and the expected number of shots splits into "the ten we always take" plus "the extra ones, weighted by the chance we need them".

Solution

We assume, as the model intends, that every shot is independent of every other shot, and that Team A always scores with probability p_A and Team B with probability p_B. Write $q_A = 1 - p_A$ and $q_B = 1 - p_B$.

The probability α of a draw after ten shots

Let X be the number of shots Team A misses in its first five, and Y the number Team B misses. Each team's five shots are independent trials with the same miss probability, so

$$X \sim \mathrm{B}(5, q_A), \qquad Y \sim \mathrm{B}(5, q_B),$$

and X and Y are independent because the two teams' shots are.

Team A scores $5 - X$ goals and Team B scores $5 - Y$, so the scores are level exactly

when $X = Y$. The events $\{X = Y = i\}$ for $i = 0, 1, \ldots, 5$ are disjoint, so

$$\begin{aligned}\alpha = \mathrm{P}(X = Y) &= \sum_{i=0}^{5} \mathrm{P}(X = i)\,\mathrm{P}(Y = i) \\ &= \sum_{i=0}^{5} \binom{5}{i} q_A^i p_A^{5-i} \binom{5}{i} q_B^i p_B^{5-i} \\ &= \sum_{i=0}^{5} \binom{5}{i}^2 (1-p_A)^i (1-p_B)^i \, p_A^{5-i} p_B^{5-i},\end{aligned}$$

which is the required expression. (Counting goals instead, with $5 - i$ goals each, gives the same sum, since $\binom{5}{i} = \binom{5}{5-i}$.)

The expected number of shots

If the scores are still level after ten shots, the sudden-death stage runs in *rounds*: in each round A takes one shot and B takes one shot. A round settles the match when exactly one of the two shots goes in, and that happens with probability

$$p_A q_B + q_A p_B = p_A(1 - p_B) + p_B(1 - p_A) = p_A + p_B - 2p_A p_B = \beta.$$

Otherwise (both score or both miss) another round is played. We need $\beta > 0$, that is, we exclude the trivial cases $p_A = p_B = 0$ and $p_A = p_B = 1$ in which no round can ever be decisive.

Ask yourself Why can't the shoot-out stop halfway through a round?

The rules say the teams must have taken the same number of shots when one has scored and the other hasn't. Even if A misses, B still has to take its shot (B might miss too). So the sudden-death shots always come in pairs, and the total is $10 + 2R$, where R is the number of rounds played ($R = 0$ if the match is settled in the first ten shots).

Given that sudden death is reached, the rounds are independent trials, each decisive with probability β, so R has a geometric distribution:

$$\mathrm{P}(R = k \mid \text{level after ten}) = (1 - \beta)^{k-1}\beta, \qquad k = 1, 2, 3, \ldots$$

Its mean is

$$\mathrm{E}(R \mid \text{level after ten}) = \sum_{k=1}^{\infty} k\beta(1 - \beta)^{k-1} = \beta \cdot \frac{1}{\big(1 - (1 - \beta)\big)^2} = \frac{1}{\beta},$$

using $\sum_{k \geq 1} k x^{k-1} = (1 - x)^{-2}$ for $|x| < 1$, which applies because $0 < \beta \leq 1$.

If the match is settled in the first ten shots, $R = 0$. Conditioning on whether the scores are level after ten shots,

$$\mathrm{E}(R) = \alpha \cdot \frac{1}{\beta} + (1 - \alpha) \cdot 0 = \frac{\alpha}{\beta}.$$

Hence the expected number of shots is

$$\mathrm{E}(10 + 2R) = 10 + \frac{2\alpha}{\beta},$$

as required.

Checks

- Equal teams with $p_A = p_B = \frac{1}{2}$: by Vandermonde's identity $\sum_i \binom{5}{i}^2 = \binom{10}{5} = 252$, so $\alpha = 252/2^{10} = \frac{63}{256}$, and $\beta = \frac{1}{2}$. The expected number of shots is $10 + \frac{63}{64} = \frac{703}{64} \approx 10.98$.
- If one team is perfect and the other hopeless ($p_A = 1$, $p_B = 0$), then $\alpha = 0$ (only the $i = 0$ term could survive, and it contains $p_B^5 = 0$), so exactly 10 shots are taken, as common sense says.
- We checked the formula for α against a direct enumeration of all 2^{10} outcomes of the first ten shots, and $10 + 2\alpha/\beta$ against a simulation of 2×10^5 shoot-outs; for $p_A = \frac{3}{4}$, $p_B = \frac{2}{3}$ the formula gives 11.264 and the simulation 11.25.

Common slips

- Taking β to be the probability that A wins a round, $p_A(1 - p_B)$: the shoot-out ends when *either* team wins a round.
- Writing the answer as $10 + 2/\beta$, forgetting that sudden death happens only with probability α.
- Counting one shot per step of sudden death instead of two.

Beyond STEP

The step $\mathrm{E}(R) = \alpha\,\mathrm{E}(R \mid \text{level}) + (1 - \alpha)\,\mathrm{E}(R \mid \text{not level})$ is the law of total expectation, one of the most useful tools in first-year probability. The sudden-death stage is also a simple Markov chain with an absorbing state, and the "expected time to absorption" calculation here generalises to random walks and gambler's-ruin problems. Grimmett and Welsh, *Probability: An Introduction*, covers both conditional expectation and these chains.

WORKED SOLUTION

STEP I 2003, Question 14

Probability and Statistics • Poisson Distribution

STEP I 2003, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q14

The key idea

The number of friends Jane sees is not the number who call, but that number capped at 2. A capped variable only cares about three cases: nobody calls, exactly one calls, or two or more call. So we never need to sum the whole Poisson series. We just need $P(0)$ and $P(1)$, and "two or more" is whatever probability is left over. For the new friend, we condition on whether they call.

Solution

The original friends

Let N be the number of friends who call on a given day, so $N \sim \mathrm{Po}(2)$ and

$$P(N = k) = \frac{\mathrm{e}^{-2}2^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

Jane sees $S = \min(N, 2)$ friends. So $S = 0$ when $N = 0$, $S = 1$ when $N = 1$, and $S = 2$ when $N \geq 2$. Now

$$P(N = 0) = \mathrm{e}^{-2}, \qquad P(N = 1) = 2\mathrm{e}^{-2}, \qquad P(N \geq 2) = 1 - 3\mathrm{e}^{-2},$$

and therefore

$$\mathrm{E}(S) = 0 \cdot \mathrm{e}^{-2} + 1 \cdot 2\mathrm{e}^{-2} + 2\left(1 - 3\mathrm{e}^{-2}\right) = 2 - 4\mathrm{e}^{-2},$$

as required.

A neater way to write the same calculation, which we use again below, is this. For any random variable X taking values in $\{0, 1, 2, \ldots\}$,

$$\mathrm{E}\big(\min(X, 2)\big) = 2 - 2P(X = 0) - P(X = 1),$$

because $\min(X, 2)$ falls short of 2 by 2 when $X = 0$, by 1 when $X = 1$, and by nothing otherwise. With $X = N$ this gives $2 - 2\mathrm{e}^{-2} - 2\mathrm{e}^{-2}$ again.

With the new friend

Let $B = 1$ if the new friend calls and $B = 0$ otherwise, so $P(B = 1) = p$. The question says B is independent of N. The total number of callers is now $M = N + B$, and Jane sees $\min(M, 2)$ friends.

? **Ask yourself** Do we need the full distribution of M?

No. By the identity above, we only need $P(M = 0)$ and $P(M = 1)$. Using independence to multiply probabilities:

$$P(M = 0) = P(N = 0)\,P(B = 0) = (1 - p)\mathrm{e}^{-2},$$
$$P(M = 1) = P(N = 1)\,P(B = 0) + P(N = 0)\,P(B = 1) = 2(1 - p)\mathrm{e}^{-2} + p\,\mathrm{e}^{-2}.$$

Hence

$$\begin{aligned}\mathrm{E}\big(\min(M, 2)\big) &= 2 - 2(1 - p)\mathrm{e}^{-2} - 2(1 - p)\mathrm{e}^{-2} - p\,\mathrm{e}^{-2} \\ &= 2 - (4 - 3p)\mathrm{e}^{-2}.\end{aligned}$$

Alternatively, condition on B. If the new friend doesn't call (probability $1 - p$), the average is $2 - 4\mathrm{e}^{-2}$, as before. If they do call (probability p), Jane sees $\min(N + 1, 2)$, which is 1 if $N = 0$ and 2 otherwise, with mean $\mathrm{e}^{-2} + 2(1 - \mathrm{e}^{-2}) = 2 - \mathrm{e}^{-2}$. So the average is

$$(1 - p)(2 - 4\mathrm{e}^{-2}) + p(2 - \mathrm{e}^{-2}) = 2 - (4 - 3p)\mathrm{e}^{-2}.$$

So Jane now sees, on average, $2 - (4 - 3p)\mathrm{e}^{-2}$ friends a day, an increase of $3p\,\mathrm{e}^{-2}$.

Checks

- When $p = 0$, the new friend never calls and we recover $2 - 4\mathrm{e}^{-2} \approx 1.4587$.
- When $p = 1$, we get $2 - \mathrm{e}^{-2} \approx 1.8647$, which is the mean of $\min(N + 1, 2)$, as it should be.
- The answer is increasing in p and always below 2, which makes sense.
- Summing $\min(k + b, 2)$ against the exact distribution, and running 4×10^5 simulated days with $p = 0.3$, both give about 1.580, in agreement with $2 - 3.1\mathrm{e}^{-2}$.

Common slips

- Answering $\mathrm{E}(N) = 2$, or $2 + p$, by forgetting the cap at two friends.
- Treating $N + B$ as Poisson. A Bernoulli variable is not Poisson, so the sum is not Poisson either.
- Missing one of the two ways to get exactly one caller (one old friend and no new friend, or no old friends and the new friend).

Beyond STEP

The identity $\mathrm{E}(\min(X,2)) = P(X \geq 1) + P(X \geq 2)$ is a special case of the tail-sum formula $\mathrm{E}(X) = \sum_{k \geq 1} P(X \geq k)$ for non-negative integer random variables. It is the discrete version of integrating the survival function. Capped means like this one appear in queueing and inventory models, where they give the expected number of customers served when capacity is limited. Grimmett and Welsh, *Probability: An Introduction*, covers the tail-sum formula and conditional expectation, which are the two tools used here.

WORKED SOLUTION

STEP II 2003, Question 1

Pure • Simultaneous Linear Equations

STEP II 2003, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q1

The key idea

The unknown x only ever appears as ax, and y appears with coefficient -1 in every equation. So subtracting consecutive equations kills y completely, and the two differences we get have the same left-hand side. That makes the system consistent only for one value of b, and once it is consistent there is one free parameter. The rest of the question is about choosing that parameter well.

Solution

Number the equations (1), (2), (3). Subtracting (1) from (2), and (2) from (3), gives

$$ax - 2z = 4, \tag{A}$$
$$ax - 2z = b - 7. \tag{B}$$

Any solution of the system satisfies both (A) and (B), so a solution can exist only if $4 = b - 7$, that is $b = 11$. This holds for every value of a.

Part (i)

With $a = 0$ the equations become $-y - z = 3$, $-y - 3z = 7$, $-y - 5z = b$, and x does not appear at all.

If there is a solution, then $b = 11$ by the remark above.

Conversely, suppose $b = 11$. Equation (A) now reads $-2z = 4$, so $z = -2$, and then (1) gives $y = -3 - z = -1$. Checking all three equations with $y = -1$, $z = -2$:

$$1 + 2 = 3, \qquad 1 + 6 = 7, \qquad 1 + 10 = 11 = b,$$

so (with any x at all) we have a solution. Hence the equations have a solution if and only if $b = 11$.

Part (ii)

Now $a \neq 0$ and $b = 11$. Put $z = \lambda$. Equation (A) gives $ax = 4 + 2\lambda$, and since $a \neq 0$ we can divide:

$$x = \frac{4 + 2\lambda}{a}.$$

Then (1) gives $y = ax - z - 3 = 4 + 2\lambda - \lambda - 3 = 1 + \lambda$.

We found these values using only (1) and (2), so we must check they satisfy (3) too:

$$3ax - y - 5z = 3(4 + 2\lambda) - (1 + \lambda) - 5\lambda = 12 + 6\lambda - 1 - 6\lambda = 11 = b.$$

So for every number λ,

$$x = \frac{4 + 2\lambda}{a}, \qquad y = 1 + \lambda, \qquad z = \lambda$$

is a solution with $z = \lambda$, as required. (In fact these are *all* the solutions: any solution has some value λ of z, and the working above then forces x and y.)

Ask yourself Why do we have to check equation (3) separately?

We used (1) and (2) (in the form (1) and (A)) to find x and y. Three equations in three unknowns do not automatically fit together, and (3) is exactly where b comes in. The check is where $b = 11$ gets used.

Part (iii)

With $a = 2$, part (ii) gives every solution as $(x, y, z) = (2 + \lambda,\ 1 + \lambda,\ \lambda)$. Then

$$\begin{aligned} x^2 + y^2 + z^2 &= (2 + \lambda)^2 + (1 + \lambda)^2 + \lambda^2 \\ &= 3\lambda^2 + 6\lambda + 5 \\ &= 3(\lambda + 1)^2 + 2, \end{aligned}$$

which is least, with value 2, exactly when $\lambda = -1$. So the required solution is

$$x = 1, \qquad y = 0, \qquad z = -1.$$

Geometrically, the solutions form a line with direction $(1, 1, 1)$, and we are finding the point of the line closest to the origin. Indeed $(1, 0, -1) \cdot (1, 1, 1) = 0$, so the position vector is perpendicular to the line, as it should be.

Part (iv)

By the remark at the start, b must be 11. If $a = 0$, part (i) forces $y = -1$, $z = -2$, so $y^2 + z^2 = 5$, which is too big. So $a \neq 0$, and by part (ii) the solutions are $x = (4 + 2\lambda)/a$, $y = 1 + \lambda$, $z = \lambda$.

Ask yourself How can x be huge while y and z stay small?

The only place a appears is in $x = (4+2\lambda)/a$. Choosing λ fixes y and z, and then a small enough a makes x as large as we like.

We need $y^2+z^2 = (1+\lambda)^2+\lambda^2 = 2\lambda^2+2\lambda+1 < 1$, that is $2\lambda(\lambda+1) < 0$, that is $-1 < \lambda < 0$. Take $\lambda = -\frac{1}{2}$: then $y = \frac{1}{2}$, $z = -\frac{1}{2}$ and $y^2+z^2 = \frac{1}{2} < 1$, while $x = 3/a$. Choosing

$$a = 10^{-6}$$

gives $x = 3\times 10^6 > 10^6$. So $a = 10^{-6}$ (with $b = 11$) works.

More generally, for $-1 < \lambda < 0$ we have $2 < 4+2\lambda < 4$, so a suitable solution exists exactly when $0 < a < 4\times 10^{-6}$. Negative a cannot work, since then $x < 0$.

Checks

- Substitute $a = 2$, $(x,y,z) = (1,0,-1)$ into the original equations: $2-0+1 = 3$, $4-0+3 = 7$, $6-0+5 = 11$.
- For part (iv), with $a = 10^{-6}$ and $(x,y,z) = (3\times 10^6, \frac{1}{2}, -\frac{1}{2})$: $3-\frac{1}{2}+\frac{1}{2} = 3$, $6-\frac{1}{2}+\frac{3}{2} = 7$, $9-\frac{1}{2}+\frac{5}{2} = 11$.
- We checked with exact fractions that the part (ii) formula satisfies all three equations for hundreds of random a and λ, and a grid search over λ confirms the minimum value 2 at $\lambda = -1$ in part (iii).

Common slips

- Finding x and y from two of the equations and forgetting to check the third.
- In part (iv), dismissing it as impossible because y and z "can't both be small". They can: $y - z = 1$ only forces $y^2+z^2 \geq \frac{1}{2}$.

Beyond STEP

This is a first look at rank and consistency in linear algebra. The coefficient matrix here is singular (its rows satisfy $r_1 - 2r_2 + r_3 = 0$), so the system has either no solutions or a whole line of them, depending on whether the right-hand side satisfies the same relation. Part (iv) also shows why nearly singular systems are dangerous in numerical work: a tiny coefficient a produces an enormous solution. Strang's *Introduction to Linear Algebra* covers all of this, including least-norm solutions like the one in part (iii).

WORKED SOLUTION

STEP II 2003, Question 2

Pure • Inverse Trigonometric Functions

STEP II 2003, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q2

The key idea

The expression $a\cos\theta + b\sin\theta$ can always be written as a single cosine, $R\cos(\theta - \alpha)$. Doing that turns the equation into a statement about arccos, while α itself is an arctan. So once we spot a nice solution θ, the identity for π is just "$\theta = (\theta - \alpha) + \alpha$" rewritten. The only real care needed is in checking that each angle lies in the range of the inverse function we use.

Solution

The first identity

Try the obvious angles in $(\frac{1}{4}\pi, \frac{1}{2}\pi)$. With $\theta = \frac{1}{3}\pi$,

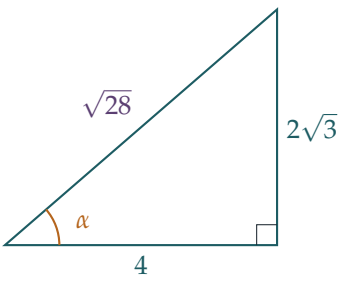
$$4\cos\tfrac{\pi}{3} + 2\sqrt{3}\,\sin\tfrac{\pi}{3} = 4\cdot\tfrac{1}{2} + 2\sqrt{3}\cdot\tfrac{\sqrt{3}}{2} = 2 + 3 = 5,$$

so $\theta = \frac{1}{3}\pi$ works, and $\frac{1}{4}\pi < \frac{1}{3}\pi < \frac{1}{2}\pi$.

Now write the left-hand side in the form $R\cos(\theta - \alpha) = R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$. Matching coefficients,

$$R\cos\alpha = 4, \qquad R\sin\alpha = 2\sqrt{3},$$

so $R^2 = 16 + 12 = 28$, giving $R = \sqrt{28}$, and $\tan\alpha = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2}$. Since $\cos\alpha$ and $\sin\alpha$ are both positive we may take α acute, so $\alpha = \arctan(\sqrt{3}/2)$.



The equation therefore reads

$$\sqrt{28}\cos(\theta-\alpha)=5, \qquad \text{that is,} \qquad \cos(\theta-\alpha)=\frac{5}{\sqrt{28}},$$

and it is satisfied by $\theta=\frac{1}{3}\pi$.

Ask yourself Can we immediately conclude that $\frac{1}{3}\pi-\alpha=\arccos(5/\sqrt{28})$?

Not quite: $\cos x=c$ has many solutions, and $\arccos c$ is only the one in $[0,\pi]$. So we must check that $\frac{1}{3}\pi-\alpha$ lies in $[0,\pi]$. Since tan is increasing on $(0,\frac{1}{2}\pi)$ and $\tan\alpha=\frac{\sqrt{3}}{2}<\sqrt{3}=\tan\frac{\pi}{3}$, we have $0<\alpha<\frac{1}{3}\pi$. Hence

$$0<\tfrac{1}{3}\pi-\alpha<\tfrac{1}{3}\pi\leq\pi,$$

and so $\frac{1}{3}\pi-\alpha=\arccos(5/\sqrt{28})$. Rearranging and multiplying by 3,

$$\pi=3\arccos(5/\sqrt{28})+3\alpha=3\arccos(5/\sqrt{28})+3\arctan(\sqrt{3}/2),$$

as required.

The second identity

Copy the idea: we want $\frac{1}{4}\pi=\arcsin(7\sqrt{2}/10)-\arctan(3/4)$, so let $\beta=\arctan(3/4)$ and look at $\sin(\frac{1}{4}\pi+\beta)$.

Since $0<\frac{3}{4}$, β is acute, and from the 3–4–5 right-angled triangle, $\sin\beta=\frac{3}{5}$ and $\cos\beta=\frac{4}{5}$. Then, by the addition formula,

$$\begin{aligned}\sin\left(\tfrac{1}{4}\pi+\beta\right)&=\sin\tfrac{\pi}{4}\cos\beta+\cos\tfrac{\pi}{4}\sin\beta\\&=\frac{1}{\sqrt{2}}\left(\frac{4}{5}+\frac{3}{5}\right)=\frac{7}{5\sqrt{2}}=\frac{7\sqrt{2}}{10}.\end{aligned}$$

Ask yourself Is $\frac{1}{4}\pi+\beta$ really the angle that arcsin returns?

The range of arcsin is $[-\frac{1}{2}\pi,\frac{1}{2}\pi]$, so we need $\frac{1}{4}\pi+\beta$ in there. Because $\tan\beta=\frac{3}{4}<1=\tan\frac{\pi}{4}$ and tan is increasing on $(0,\frac{1}{2}\pi)$, we have $0<\beta<\frac{1}{4}\pi$, so

$$\tfrac{1}{4}\pi<\tfrac{1}{4}\pi+\beta<\tfrac{1}{2}\pi.$$

Hence $\arcsin(7\sqrt{2}/10)=\frac{1}{4}\pi+\arctan(3/4)$, and multiplying by 4 and rearranging,

$$\pi=4\arcsin(7\sqrt{2}/10)-4\arctan(3/4),$$

which is the result.

Checks

Numerically, $\arctan(\sqrt{3}/2) \approx 0.713724$ and $\arccos(5/\sqrt{28}) \approx 0.333473$; their sum is $1.047198 \approx \pi/3$, and three times it is 3.141593. For the second identity, $\arcsin(7\sqrt{2}/10) \approx 1.428899$ and $\arctan(3/4) \approx 0.643501$; the difference is $0.785398 \approx \pi/4$. A quick scan also shows that $\frac{1}{3}\pi$ is the only solution of the equation in $(\frac{1}{4}\pi, \frac{1}{2}\pi)$: the other solution near there, $\alpha - \arccos(5/\sqrt{28}) \approx 0.380$, is less than $\frac{1}{4}\pi$.

Common slips

- Writing $\arccos(\cos x) = x$ without checking $x \in [0, \pi]$ (or $\arcsin(\sin x) = x$ without $x \in [-\frac{1}{2}\pi, \frac{1}{2}\pi]$). This is exactly where the marks for rigour are.
- Getting $\tan\alpha$ upside down: from $R\cos\alpha = 4$, $R\sin\alpha = 2\sqrt{3}$ we get $\tan\alpha = 2\sqrt{3}/4$, not $4/(2\sqrt{3})$.

Beyond STEP

Identities expressing π as combinations of inverse tangents are the basis of Machin-type formulae, such as $\frac{1}{4}\pi = 4\arctan\frac{1}{5} - \arctan\frac{1}{239}$, which were used to compute π to many digits because the series for $\arctan x$ converges fast for small x. The clean way to see them is through complex numbers: $\arctan(b/a)$ is the argument of $a + b\mathrm{i}$, so adding inverse tangents corresponds to multiplying complex numbers. Needham's *Visual Complex Analysis* develops this geometric viewpoint beautifully.

WORKED SOLUTION

STEP II 2003, Question 3

Pure • Irrational Numbers and Induction

STEP II 2003, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q3

The key idea

Irrationality is a "not" property, so the natural tool is the contrapositive: instead of showing that an irrational number has an irrational cube root, show that a rational cube root forces the number itself to be rational. That single fact then drives the induction, because each u_{n+1} is the cube root of u_n. For the last part, we notice that $u_n \to 1$ and slide the whole sequence along by an integer, which keeps every term irrational.

Solution

Cube roots of irrational numbers

Let x be a real number and let $y = \sqrt[3]{x}$ be its real cube root, so $y^3 = x$. We prove the contrapositive: if y is rational, then x is rational.

Suppose $y = \dfrac{p}{q}$ with p, q integers and $q \neq 0$. Then

$$x = y^3 = \frac{p^3}{q^3},$$

and p^3 and q^3 are integers with $q^3 \neq 0$. So x is a ratio of two integers, that is, rational.

So a rational cube root can only come from a rational number. Equivalently, if x is irrational then $\sqrt[3]{x}$ cannot be rational, so it is irrational. This is what we had to prove.

? **Ask yourself** Why not attack the statement directly?

The direct route would start from "x is not a ratio of integers", and there is nothing concrete to calculate with. The contrapositive starts from "$y = p/q$", which hands us integers to work with, and the proof is then one line. Whenever the hypothesis says that something is *not* rational, it is worth trying the contrapositive or a proof by contradiction.

The induction

Since $3^{n+1} = 3 \cdot 3^n$, the laws of indices give

$$u_{n+1} = 5^{1/3^{n+1}} = \left(5^{1/3^n}\right)^{1/3} = \sqrt[3]{u_n}.$$

So each term is the cube root of the one before.

Base case. $u_1 = 5^{1/3} = \sqrt[3]{5}$, which we are told is irrational.

Inductive step. Suppose u_k is irrational for some positive integer k. Then $u_{k+1} = \sqrt[3]{u_k}$ is the cube root of an irrational number, so by the first part it is irrational.

Hence, by induction, u_n is irrational for every positive integer n.

A sequence converging to m

First we find the limit of u_n. Taking logarithms,

$$\ln u_n = \frac{\ln 5}{3^n} \to 0 \quad \text{as } n \to \infty,$$

and since the exponential function is continuous, $u_n = \mathrm{e}^{\ln u_n} \to \mathrm{e}^0 = 1$. Also $\ln u_n > 0$, so $u_n > 1$ for every n; the terms decrease towards 1:

$$u_1 \approx 1.710, \quad u_2 \approx 1.196, \quad u_3 \approx 1.061, \quad u_4 \approx 1.020, \ \ldots$$

Ask yourself How do we move the limit from 1 to m without spoiling irrationality?

Adding an integer to an irrational number gives an irrational number. Indeed, if a is an integer and $u + a = r$ were rational, then $u = r - a$ would be rational too, since a difference of rationals is rational.

So, for a given integer m, define

$$a_n = u_n + m - 1 = 5^{1/3^n} + m - 1.$$

Since $m - 1$ is an integer and u_n is irrational, each a_n is irrational, and

$$a_n \to 1 + m - 1 = m \quad \text{as } n \to \infty.$$

So (a_n) is an infinite sequence of irrational numbers converging to m. This works for every integer m, including $m = 0$.

(Another natural choice, $m\,u_n$, is also irrational and tends to m, but only when $m \neq 0$: for $m = 0$ every term is 0, which is rational. That is why the shift by $m - 1$ is the safer answer.)

Checks

- Numerically, $u_2^3 \approx 1.19581^3 \approx 1.70998 \approx u_1$, confirming $u_{n+1} = \sqrt[3]{u_n}$.

- With $m = 4$: $a_1 \approx 4.710$, $a_2 \approx 4.196$, $a_3 \approx 4.061$, and a_{20} agrees with 4 to more than eight decimal places.
- The difference $a_n - m = u_n - 1$ is positive and roughly $\ln 5/3^n$, so it shrinks by a factor of about 3 at each step.

Common slips

- Proving the converse ("the cube of an irrational is irrational"), which is false: $\left(\sqrt[3]{2}\right)^3 = 2$.
- Using $m\,u_n$ without noticing that it fails for $m = 0$.
- Forgetting to justify that $u_n \to 1$; "the power tends to zero" needs a word about continuity, or the logarithm argument above.

Beyond STEP

This question shows that the irrational numbers are *dense* in the real line: every real number, not just every integer, is a limit of irrationals (try $q_n + \sqrt{2}/n$ with rationals $q_n \to x$). The rationals are dense too, yet in a precise sense there are far more irrationals than rationals: the rationals are countable and the reals are not. Abbott's *Understanding Analysis* covers density, countability and limits of sequences beautifully in its opening chapters.

WORKED SOLUTION

STEP II 2003, Question 4

Pure • Circle Segment Areas

STEP II 2003, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q4

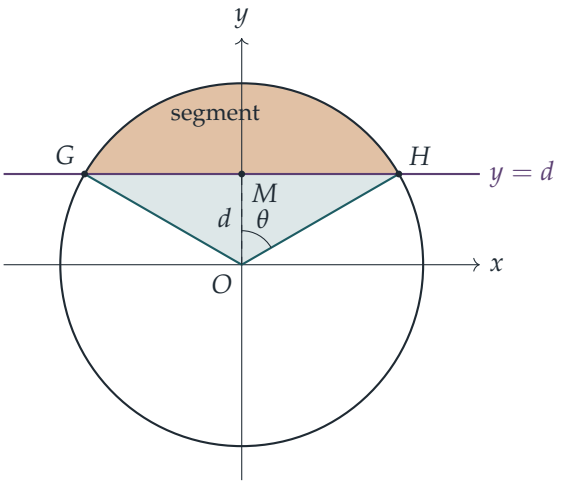
The key idea

A minor segment is a sector with a triangle removed, so the first result is just "sector minus triangle". For the rest, notice that in $(*)$ the number d is really the *perpendicular distance from the centre of the circle to the line*. Sliding or rotating the picture changes no areas, so in every case we keep $(*)$ and replace d by that distance. All the work is in computing the distance.

Solution

The basic result

Because the line meets the circle at two points, $0 < d < R$. Putting $y = d$ into $x^2 + y^2 = R^2$ gives $x = \pm\sqrt{R^2 - d^2}$, so $G = \left(-\sqrt{R^2 - d^2}, d\right)$ and $H = \left(\sqrt{R^2 - d^2}, d\right)$. Let $M = (0, d)$ be the midpoint of GH, and let $\theta = \angle MOH$.



In the right-angled triangle OMH the hypotenuse is $OH = R$ and the adjacent side is $OM = d$, so $\cos\theta = d/R$. Since $0 < d/R < 1$, we have $\theta = \arccos(d/R)$ with $0 < \theta < \frac{\pi}{2}$, and by symmetry $\angle GOH = 2\theta < \pi$.

The sector OGH with angle 2θ has area

$$\tfrac{1}{2}R^2(2\theta) = R^2 \arccos\left(\frac{d}{R}\right).$$

Triangle OGH has base $GH = 2\sqrt{R^2 - d^2}$ and height $OM = d$, so its area is $d\sqrt{R^2 - d^2}$. Because $2\theta < \pi$, the centre O lies on the opposite side of GH from the minor arc, so the sector is exactly the triangle together with the minor segment. Hence

$$\text{area of minor segment} = R^2 \arccos\left(\frac{d}{R}\right) - d\sqrt{R^2 - d^2},$$

which is $(*)$.

Ask yourself What does d actually stand for in $(*)$, as far as the geometry is concerned?

Nothing in the argument used the axes: it used only the radius R and the distance $OM = d$ from the centre to the chord. Translating or rotating the whole figure preserves both, and preserves area. So for any circle of radius R and any line at perpendicular distance p from the centre, with $0 \le p < R$, the minor segment has area

$$R^2 \arccos\left(\frac{p}{R}\right) - p\sqrt{R^2 - p^2}. \qquad (\dagger)$$

(If $p = 0$ the line is a diameter, the two pieces are equal halves, and $(\dagger)$ gives $\frac{1}{2}\pi R^2$ correctly.) In each part we find p and substitute it into $(\dagger)$.

Part (i)

The centre is (a, b) and the line $y = c$ is horizontal, so the distance from the centre to the line is $p = |c - b|$. Replace d by $|c - b|$:

$$R^2 \arccos\left(\frac{|c-b|}{R}\right) - |c-b|\sqrt{R^2 - (c-b)^2}.$$

The modulus matters: c may be below b.

Part (ii)

Here the centre is O. We need the distance from O to $y = mx + c$. Let the line make angle α with the x-axis, so $\tan\alpha = m$, and let it cross the y-axis at $C = (0, c)$. Drop the perpendicular from O to the line, meeting it at F. The angle between OC (vertical) and OF (perpendicular to the line) equals the angle between the horizontal and the line, namely α. So in the right-angled triangle OFC,

$$p = OF = |c|\,|\cos\alpha| = \frac{|c|}{\sqrt{1 + \tan^2\alpha}} = \frac{|c|}{\sqrt{1 + m^2}}.$$

(This is the usual formula $|Ax_0 + By_0 + C|/\sqrt{A^2+B^2}$ for the line $mx - y + c = 0$ and the point $(0,0)$.) Then

$$\sqrt{R^2 - p^2} = \sqrt{R^2 - \frac{c^2}{1+m^2}} = \frac{\sqrt{R^2(1+m^2) - c^2}}{\sqrt{1+m^2}},$$

so the area of the minor segment is

$$R^2 \arccos\left(\frac{|c|}{R\sqrt{1+m^2}}\right) - \frac{|c|\sqrt{R^2(1+m^2)-c^2}}{1+m^2}.$$

The line meets the circle exactly when $c^2 < R^2(1+m^2)$, so the square root is real.

Part (iii)

Shift the origin to the centre by writing $X = x - a$, $Y = y - b$. The circle becomes $X^2 + Y^2 = R^2$ and the line becomes

$$Y + b = m(X + a) + c \quad \Longleftrightarrow \quad Y = mX + k, \qquad k = c + ma - b.$$

This is part (ii) with c replaced by k, and translation does not change area. So $p = |c + ma - b|/\sqrt{1+m^2}$, and the area of the minor segment is

$$R^2 \arccos\left(\frac{|c+ma-b|}{R\sqrt{1+m^2}}\right) - \frac{|c+ma-b|\sqrt{R^2(1+m^2)-(c+ma-b)^2}}{1+m^2}.$$

Putting $m = 0$ recovers part (i), and putting $a = b = 0$ recovers part (ii).

Ask yourself Why can't we just write $c - b$ in part (i), without the modulus?

If $c < b$ then $(c-b)/R$ is negative, arccos returns an angle $\pi - \theta$ greater than $\frac{\pi}{2}$, and the second term changes sign. The expression becomes $R^2(\pi - \theta) + p\sqrt{R^2 - p^2}$, which is the area of the *major* segment, not the minor one.

Checks

- With $R = 2$, $d = 1$ we get $\theta = \frac{\pi}{3}$, and $(*)$ gives $\frac{4\pi}{3} - \sqrt{3} \approx 2.4567$. This is less than half the disc ($2\pi \approx 6.28$), as it should be.
- As $d \to R$ the area tends to 0, and at $d = 0$ it equals $\frac{1}{2}\pi R^2$.
- We checked all four formulas against direct numerical integration of chord lengths over hundreds of random circles and lines. The agreement was within 2×10^{-6}.

Common slips

- Leaving out the modulus. A signed distance gives the major segment whenever the centre is on the "wrong" side of the line.

- Using the vertical gap $|c|$ instead of the perpendicular distance $|c|/\sqrt{1+m^2}$ in parts (ii) and (iii).
- Using $\frac{1}{2}R^2\theta$ for the sector, forgetting that the angle at the centre is 2θ, not θ.

Beyond STEP

The question rests on the fact that area is unchanged by *isometries*, the translations, rotations and reflections of the plane. In a first-year course this becomes the statement that the Jacobian determinant of an isometry has modulus 1, so the change-of-variables formula for double integrals leaves areas unchanged. Classifying isometries, and the quantities they preserve, is a good route into geometry and group theory. Strang's *Introduction to Linear Algebra* covers rotation matrices and determinants as scale factors for area.

WORKED SOLUTION

STEP II 2003, Question 5

Pure • Vector Geometry of Lines

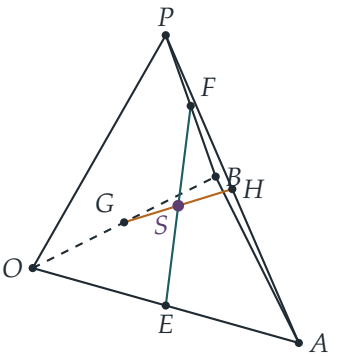
STEP II 2003, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q5

The key idea

Write down the four midpoints, put each line in the form "point + parameter × direction", and equate. Two lines in three dimensions usually miss each other, so the work lies in showing that the three component equations in two unknown parameters are consistent. The **k**-components settle things quickly. Geometrically, O, A, B, P form a tetrahedron, and EF and GH each join the midpoints of a pair of opposite edges. That suggests they meet at the "average" of the four vertices.



Solution

The lines EF and GH meet

Write $\mathbf{p} = l\mathbf{i} + m\mathbf{j} + n\mathbf{k}$. The midpoints have position vectors

$$\mathbf{e} = \tfrac{1}{2}a\mathbf{i}, \qquad \mathbf{f} = \tfrac{1}{2}\big(l\mathbf{i} + (b+m)\mathbf{j} + n\mathbf{k}\big),$$
$$\mathbf{g} = \tfrac{1}{2}b\mathbf{j}, \qquad \mathbf{h} = \tfrac{1}{2}\big((a+l)\mathbf{i} + m\mathbf{j} + n\mathbf{k}\big).$$

A general point of EF is $\mathbf{e} + s(\mathbf{f} - \mathbf{e})$, and a general point of GH is $\mathbf{g} + t(\mathbf{h} - \mathbf{g})$. Doubling both to clear the halves,

$$2\big(\mathbf{e} + s(\mathbf{f} - \mathbf{e})\big) = \big(a + s(l - a)\big)\mathbf{i} + s(b + m)\mathbf{j} + sn\,\mathbf{k},$$
$$2\big(\mathbf{g} + t(\mathbf{h} - \mathbf{g})\big) = t(a + l)\mathbf{i} + \big(b + t(m - b)\big)\mathbf{j} + tn\,\mathbf{k}.$$

The lines meet if and only if there are values of s and t that make these equal, so we need all three of

$$\begin{aligned} \mathbf{i}: &\quad a + s(l - a) = t(a + l), \\ \mathbf{j}: &\quad s(b + m) = b + t(m - b), \\ \mathbf{k}: &\quad sn = tn. \end{aligned}$$

Ask yourself Three equations in two unknowns: why should they have a solution at all?

In general they don't, and that's why two lines in space usually miss each other. Here we solve two of the equations and check that the third holds too. Since $n \neq 0$, the $\mathbf{k}$-equation gives $s = t$. The $\mathbf{i}$-equation then becomes $a + sl - sa = sa + sl$, so $a = 2sa$, and since $a \neq 0$ we get $s = \frac{1}{2}$. Finally, with $s = t = \frac{1}{2}$ the $\mathbf{j}$-equation reads

$$\tfrac{1}{2}(b + m) = b + \tfrac{1}{2}(m - b) = \tfrac{1}{2}(b + m),$$

which is true. So all three equations hold when $s = t = \frac{1}{2}$, and the lines EF and GH intersect, as required.

The point of intersection is the midpoint of EF (and also of GH):

$$\mathbf{s} = \tfrac{1}{4}\big((a + l)\mathbf{i} + (b + m)\mathbf{j} + n\mathbf{k}\big) = \tfrac{1}{4}\big(\mathbf{0} + a\mathbf{i} + b\mathbf{j} + \mathbf{p}\big),$$

the average of the position vectors of O, A, B and P.

The point T

Since S is the midpoint of DT, we have $\mathbf{s} = \frac{1}{2}(\mathbf{d} + \mathbf{t})$, so $\mathbf{t} = 2\mathbf{s} - \mathbf{d}$:

$$\mathbf{t} = \tfrac{1}{2}(a + l)\,\mathbf{i} + \tfrac{1}{2}(b + m)\,\mathbf{j} + \big(\tfrac{1}{2}n - d\big)\mathbf{k}.$$

The plane OAB contains O and the vectors $a\mathbf{i}$ and $b\mathbf{j}$. Since a and b are non-zero, these point along the x- and y-axes, so the plane OAB is the plane $z = 0$. So T lies in it exactly when its $\mathbf{k}$-component vanishes, that is, $\frac{1}{2}n - d = 0$. Hence

$$d = \tfrac{1}{2}n.$$

This is non-zero because $n \neq 0$, so it is consistent with the condition $d \neq 0$.

Checks

- Take $a = 2$, $b = 4$, $\mathbf{p} = (2,0,4)$. Then $E = (1,0,0)$, $F = (1,2,2)$, $G = (0,2,0)$, $H = (2,0,2)$. The midpoints of EF and GH are both $(1,1,1)$, which is $\frac{1}{4}\big((0,0,0) + (2,0,0) + (0,4,0) + (2,0,4)\big)$.
- With $d = 2 = \frac{1}{2}n$, $T = 2(1,1,1) - (0,0,2) = (2,2,0)$, which lies in $z = 0$.
- A computer check confirmed, in exact rational arithmetic for 20 000 random choices of a, b, l, m, n, d, that the lines meet at $s = t = \frac{1}{2}$ at the point above, and that T lies in the plane OAB (scalar triple product zero) exactly when $d = \frac{1}{2}n$.

Common slips

- Equating the two lines with the *same* parameter from the start. That assumes the answer $s = t$, which here comes from the **k**-equation.
- Solving two of the component equations and stopping. Showing that two lines in space intersect needs the third equation checked as well.
- Forgetting to say where $a \neq 0$ and $n \neq 0$ are used: they are what let us divide to find s and t.

Beyond STEP

The three lines joining midpoints of opposite edges of any tetrahedron (the *bimedians*) all pass through its centroid, the average of the four vertices, and each is bisected there. This is the three-dimensional cousin of the medians of a triangle meeting at the centroid, and it is a first taste of thinking about points as weighted averages, the idea behind barycentric and affine coordinates. Strang's *Introduction to Linear Algebra* develops the vector and linear-equation viewpoint used here, including when a system with more equations than unknowns is consistent.

WORKED SOLUTION

STEP II 2003, Question 6

Pure • Iterated Functions and Integration

STEP II 2003, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q6

The key idea

Applying f once more means "subtract 1, then reflect anything negative back above the axis". Do that to a graph whose values on some stretch lie between 0 and 1, and that stretch is simply turned upside down inside the strip $0 \le y \le 1$, while the long straight arms gain one extra tooth. So the graphs become zigzags (triangle waves), and once you see that, both integrals in the final part are just sums over identical periods.

Solution

The sketches

Two facts do all the work.

Recurrence. Since $g_{n+1} = f \circ g_n$, we have $g_{n+1}(x) = |g_n(x) - 1|$: shift the graph of g_n down by 1 and reflect the parts below the x-axis.

Symmetry. Since $f(2-x) = |1-x| = f(x)$, we get $g_n(2-x) = g_{n-1}\big(f(2-x)\big) = g_{n-1}\big(f(x)\big) = g_n(x)$ for $n \ge 2$, and directly for $n = 1$. So every g_n is symmetric about the line $x = 1$, and it is enough to understand $x \ge 1$.

Start with $g_1(x) = |x-1|$, a V with vertex $(1, 0)$.

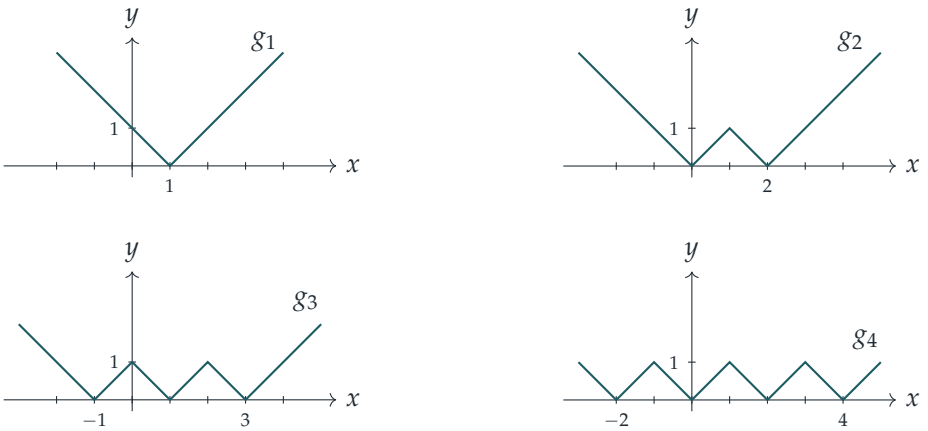
For g_2: on $[0, 2]$ we have $0 \le g_1 \le 1$, so $g_2 = 1 - |x-1|$, a tent rising to 1 at $x = 1$ and falling to 0 at $x = 0$ and $x = 2$. For $x \ge 2$, $g_2 = x - 2$; by symmetry $g_2 = -x$ for $x \le 0$. So g_2 is a W shape.

Each further step does the same thing: the zigzag part (with values in $[0, 1]$) is turned upside down, and the straight arm $x - n$ becomes $|x - n - 1|$, which adds one more tooth on $[n, n+1]$ (and, by symmetry, on $[1-n, 2-n]$). Writing $d_{\text{even}}(x)$ and $d_{\text{odd}}(x)$

for the distance from x to the nearest even and nearest odd integer, this gives

$$g_n(x) = \begin{cases} 2-n-x, & x \le 2-n, \\ d_{\text{even}}(x) \text{ if } n \text{ is even, } d_{\text{odd}}(x) \text{ if } n \text{ is odd}, & 2-n \le x \le n, \\ x-n, & x \ge n. \end{cases}$$

This follows by induction from the recurrence: $d_{\text{even}} + d_{\text{odd}} = 1$ for every real x (between consecutive integers the two distances are t and $1-t$), so "1 minus the zigzag" swaps the parity; and on $[n, n+1]$ the new piece $n+1-x$ is exactly the distance to $n+1$, the nearest integer of the new parity. The left-hand side follows by symmetry about $x = 1$.



In words: g_3 has zeros at $-1, 1, 3$ and peaks of height 1 at 0 and 2, with arms $y = -1 - x$ and $y = x - 3$; g_4 has zeros at $-2, 0, 2, 4$ and peaks of height 1 at $-1, 1, 3$, with arms $y = -2 - x$ and $y = x - 4$. All four graphs are symmetric about $x = 1$.

The integral

Let n be even. Since $n \ge 2$, the interval $[0, n]$ lies inside $[2-n, n]$, so on it $g_n(x) = d_{\text{even}}(x)$.

? **Ask yourself** Why is it enough to integrate over $[0, 2]$?

Both functions have period 2: $d_{\text{even}}(x+2) = d_{\text{even}}(x)$ because shifting by 2 maps even integers to even integers, and $\left|\sin \frac{\pi(x+2)}{2}\right| = \left|-\sin \frac{\pi x}{2}\right| = h(x)$. As n is even, $[0, n]$ is exactly $\frac{n}{2}$ whole periods $[0, 2], [2, 4], \ldots, [n-2, n]$, so each integral over $[0, n]$ is $\frac{n}{2}$ times the integral over $[0, 2]$.

On $[0, 2]$, g_n is the tent $1 - |x - 1|$: a triangle with base 2 and height 1, so

$$\int_0^2 g_n(x)\,\mathrm{d}x = \tfrac{1}{2} \cdot 2 \cdot 1 = 1.$$

On $[0,2]$, $\frac{\pi x}{2} \in [0,\pi]$, so $\sin\frac{\pi x}{2} \geq 0$ and the modulus can be dropped:

$$\int_0^2 h(x)\,\mathrm{d}x = \int_0^2 \sin\frac{\pi x}{2}\,\mathrm{d}x = \Big[-\frac{2}{\pi}\cos\frac{\pi x}{2}\Big]_0^2 = \frac{2}{\pi}+\frac{2}{\pi} = \frac{4}{\pi}.$$

Therefore

$$\int_0^n \big(h(x)-g_n(x)\big)\,\mathrm{d}x = \frac{n}{2}\Big(\frac{4}{\pi}-1\Big) = \frac{2n}{\pi}-\frac{n}{2},$$

as required.

Checks

- Pick a value and iterate by hand: $g_4(-1)$: $-1 \to 2 \to 1 \to 0 \to 1$, so $g_4(-1) = 1$, a peak, as in the sketch.
- The answer is positive, since $4/\pi \approx 1.27 > 1$: the sine arch bulges above the straight-sided tent with the same base and height, which matches the picture of the two graphs.
- Numerically, Simpson's rule for $n = 2$ gives 0.273240, agreeing with $\frac{4}{\pi} - 1 \approx 0.273240$; the same agreement holds for $n = 4, 6, 8, 10$.

Common slips

- Forgetting the left-hand side: for $x < 0$ the graphs are not straight lines once $n \geq 3$. The symmetry $g_n(2-x) = g_n(x)$ is the quickest way to get them right.
- Integrating $\sin\frac{\pi x}{2}$ over $[0,n]$ without the modulus, which gives 0 for even n instead of $\frac{2n}{\pi}$.

Beyond STEP

Iterating a simple map and watching what the graphs do is the starting point of discrete dynamical systems. Here the zigzags settle into a triangle wave, and the closely related "tent map" is a classic example of chaos. The comparison of h with the triangle wave also points to Fourier series, where a triangle wave is written as a sum of sines and cosines, and $|\sin|$ is one of the first examples. Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction to Fourier series.

WORKED SOLUTION

STEP II 2003, Question 7

Pure • Improper Integrals and Series

STEP II 2003, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q7

The key idea

An integrand of the form $\ln x \times (\text{power of } x)$ is made for integration by parts: differentiate the logarithm and it disappears. That handles the first integral. The rest of the question turns the exact value $2/(n^2\mathrm{e})$ round: it lets us *write* $1/n^2$ as an integral. Then we move all the lower limits to the same place, which can only make each integral bigger, and add up a geometric series inside the integral.

Solution

The integral

For $a > 1$ and $n > 0$ let

$$I_n(a) = \int_a^\infty \frac{\ln x}{x^{n+1}}\,\mathrm{d}x.$$

Integrate by parts with $u = \ln x$ and $\mathrm{d}v = x^{-n-1}\,\mathrm{d}x$, so $v = -\frac{1}{n}x^{-n}$ (this is where we use $n \neq 0$). Over $[a, X]$,

$$\begin{aligned}\int_a^X \frac{\ln x}{x^{n+1}}\,\mathrm{d}x &= \left[-\frac{\ln x}{nx^n}\right]_a^X + \frac{1}{n}\int_a^X x^{-n-1}\,\mathrm{d}x \\ &= \frac{\ln a}{na^n} - \frac{\ln X}{nX^n} + \frac{1}{n^2}\left(\frac{1}{a^n} - \frac{1}{X^n}\right).\end{aligned}$$

Now let $X \to \infty$. Since $n > 0$, $X^{-n} \to 0$. For the other term, put $Y = X^n$, so that $\ln X = \frac{1}{n}\ln Y$ and

$$\frac{\ln X}{X^n} = \frac{1}{n}\cdot\frac{\ln Y}{Y} \to 0,$$

because $Y \to \infty$ as $X \to \infty$ and we are allowed to assume $(\ln Y)/Y \to 0$. (We can't just quote $(\ln X)/X \to 0$ here: when $0 < n < 1$, X^n grows more slowly than X.) So the integral converges and

$$I_n(a) = \frac{\ln a}{na^n} + \frac{1}{n^2 a^n}.$$

At $a = \mathrm{e}^{1/n}$ we have $a^n = \mathrm{e}$ and $\ln a = 1/n$, so

$$I_n\left(\mathrm{e}^{1/n}\right) = \frac{1}{n^2\mathrm{e}} + \frac{1}{n^2\mathrm{e}} = \frac{2}{n^2\mathrm{e}},$$

which is the required result.

Comparing the two integrals

If $1 < a < b$, then

$$I_n(a) - I_n(b) = \int_a^b \frac{\ln x}{x^{n+1}}\,\mathrm{d}x.$$

For $x > 1$ we have $\ln x > 0$ and $x^{n+1} > 0$, so the integrand is strictly positive on $[a, b]$. It is also continuous, and $b > a$, so this integral is strictly positive. Hence $I_n(b) < I_n(a)$: moving the lower limit to the right throws away a piece of positive area.

The inequality

By the first part, for every positive integer n,

$$\frac{1}{n^2} = \frac{\mathrm{e}}{2}\, I_n\left(\mathrm{e}^{1/n}\right).$$

Ask yourself The sum has a different lower limit $\mathrm{e}^{1/n}$ in each term, but the right-hand side has a single lower limit $\mathrm{e}^{1/N}$. How do we get from one to the other?

Take $1 \le n \le N$. Then $1/N \le 1/n$, so $1 < \mathrm{e}^{1/N} \le \mathrm{e}^{1/n}$. When $n < N$ the inequality is strict, and the second part (with $a = \mathrm{e}^{1/N}$, $b = \mathrm{e}^{1/n}$) gives

$$I_n\left(\mathrm{e}^{1/n}\right) < I_n\left(\mathrm{e}^{1/N}\right);$$

when $n = N$ the two sides are equal. Adding over $n = 1, \ldots, N$, the sum is strict because $N > 1$ means at least one term (for instance $n = 1$) has $n < N$:

$$\sum_{n=1}^{N} \frac{1}{n^2} = \frac{\mathrm{e}}{2} \sum_{n=1}^{N} I_n\left(\mathrm{e}^{1/n}\right) < \frac{\mathrm{e}}{2} \sum_{n=1}^{N} I_n\left(\mathrm{e}^{1/N}\right).$$

This is the point where we need $N > 1$. With $N = 1$ the two sides would be equal.

It remains to add up the integrals. A finite sum of convergent integrals over the same range is the integral of the sum, so

$$\sum_{n=1}^{N} I_n\left(\mathrm{e}^{1/N}\right) = \int_{\mathrm{e}^{1/N}}^{\infty} \left(\sum_{n=1}^{N} x^{-(n+1)} \right) \ln x\,\mathrm{d}x.$$

The bracket is a geometric series with first term x^{-2}, ratio $x^{-1} \neq 1$ (as $x > 1$) and N terms:

$$\sum_{n=1}^{N} x^{-(n+1)} = \frac{x^{-2}\left(1 - x^{-N}\right)}{1 - x^{-1}} = \frac{1 - x^{-N}}{x^2 - x},$$

after multiplying top and bottom by x^2. Therefore

$$\sum_{n=1}^{N} \frac{1}{n^2} < \frac{\mathrm{e}}{2} \int_{\mathrm{e}^{1/N}}^{\infty} \left(\frac{1 - x^{-N}}{x^2 - x} \right) \ln x \, \mathrm{d}x,$$

as required.

Checks

- Differentiate the formula for $I_n(a)$ with respect to a: you should get $-\ln a / a^{n+1}$, the negative of the integrand, as the fundamental theorem of calculus demands.
- Numerically, with the substitution $x = \mathrm{e}^u$ the integral in the final part becomes $\int_{1/N}^{\infty} u(1 - \mathrm{e}^{-Nu})/(\mathrm{e}^u - 1)\,\mathrm{d}u$. For $N = 2$ the right-hand side is about 1.4865, against $1 + \frac{1}{4} = 1.25$. For $N = 10$ it is about 2.0579, against 1.5498.
- As $N \to \infty$ the right-hand side tends to $\frac{\mathrm{e}}{2} \int_1^{\infty} \frac{\ln x}{x^2 - x}\,\mathrm{d}x$, and the substitution $x = 1/t$ turns this into $\frac{\mathrm{e}}{2} \int_0^1 \frac{-\ln t}{1-t}\,\mathrm{d}t = \frac{\mathrm{e}}{2} \cdot \frac{\pi^2}{6} \approx 2.236$. This is comfortably above $\pi^2/6 \approx 1.645$, as it must be.

Common slips

- Using $(\ln X)/X \to 0$ directly to kill $(\ln X)/X^n$. That works for $n \geq 1$, but the result is claimed for all $n > 0$.
- Moving the lower limits the wrong way. Since $\mathrm{e}^{1/N}$ is the *smallest* of the $\mathrm{e}^{1/n}$, every integral gets bigger.
- Claiming strict inequality without saying why. The $n = N$ term gives equality, so strictness comes from the terms with $n < N$, which exist because $N > 1$.

Beyond STEP

Writing a number as an integral and then summing under the integral sign is a standard move in analysis. Taken to the limit it gives $\sum 1/n^2 = \int_0^1 \frac{-\ln t}{1-t}\,\mathrm{d}t$, one route to $\pi^2/6$. Letting $N \to \infty$ swaps an infinite sum with an improper integral. That swap needs justification, through the monotone or dominated convergence theorems, which you meet in a first analysis or measure theory course. Abbott's *Understanding Analysis* is a friendly place to see why such interchanges can fail and when they are safe.

WORKED SOLUTION

STEP II 2003, Question 8

Pure • Separable Differential Equations

STEP II 2003, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q8

The key idea

The equation is first order and separable: $\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}$ equals a rational function of t. Dividing the quadratic by $t+1$ splits that function into a polynomial plus a simple fraction, each of which integrates at once. The stationary points then come straight from the differential equation itself, with no need to differentiate the solution: $\frac{\mathrm{d}y}{\mathrm{d}t}=0$ exactly where the quadratic t^2-3t+2 vanishes.

Solution

Finding y

Polynomial division gives

$$\frac{t^2-3t+2}{t+1}=t-4+\frac{6}{t+1},$$

since $(t+1)(t-4)+6=t^2-3t-4+6=t^2-3t+2$.

The solution must start at $y=A>0$. If y were ever zero, then by uniqueness of solutions to this linear equation it would be the zero solution, which contradicts $y(0)=A$. So $y>0$ throughout (on $t>-1$, where the coefficient is defined), and we may divide by y:

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}t}=-k\left(t-4+\frac{6}{t+1}\right).$$

Integrating from 0 to t, and using $\ln 1=0$,

$$\ln\frac{y}{A}=-k\left(\frac{t^2}{2}-4t+6\ln(t+1)\right),$$

so

$$\boxed{y=A\,(t+1)^{-6k}\,\mathrm{e}^{-k(t^2/2-4t)}=A\,(t+1)^{-6k}\,\mathrm{e}^{k(8t-t^2)/2}}.$$

The stationary values

From the differential equation,

$$\frac{\mathrm{d}y}{\mathrm{d}t} = -k\,\frac{(t-1)(t-2)}{t+1}\,y.$$

We assume $k \neq 0$ (otherwise $y \equiv A$ and every point is stationary). Since $y > 0$ and $t+1 > 0$, the derivative vanishes only at $t = 1$ and $t = 2$, and it changes sign at each of them because each is a simple root. So there are exactly two stationary values:

$$y(1) = A \cdot 2^{-6k}\,\mathrm{e}^{-k(1/2-4)} = A \cdot 2^{-6k}\,\mathrm{e}^{7k/2},$$
$$y(2) = A \cdot 3^{-6k}\,\mathrm{e}^{-k(2-8)} = A \cdot 3^{-6k}\,\mathrm{e}^{6k}.$$

Their ratio is

$$\frac{y(1)}{y(2)} = \frac{3^{6k}}{2^{6k}}\,\mathrm{e}^{7k/2-6k} = \left(\frac{3}{2}\right)^{6k}\mathrm{e}^{-5k/2},$$

as required.

Ask yourself Which of the two stationary values is the larger?

Take logarithms:

$$\ln\frac{y(1)}{y(2)} = k\left(6\ln\tfrac{3}{2} - \tfrac{5}{2}\right),$$

and $6\ln 1.5 \approx 2.433 < 2.5$, so the bracket is negative (about -0.067). Hence $y(1) < y(2)$ when $k > 0$ and $y(1) > y(2)$ when $k < 0$. The difference is small, so the wiggle in the graph is gentle. Similarly

$$\ln\frac{y(1)}{A} = k\left(\tfrac{7}{2} - 6\ln 2\right) \approx -0.659k, \qquad \ln\frac{y(2)}{A} = k(6 - 6\ln 3) \approx -0.592k,$$

so both stationary values lie below A when $k > 0$ and above A when $k < 0$.

Behaviour as $t \to +\infty$

Write

$$\ln\frac{y}{A} = -k\left(\frac{t^2}{2} - 4t + 6\ln(t+1)\right).$$

As $t \to \infty$ the bracket tends to $+\infty$, dominated by $\frac{1}{2}t^2$ (the ln term grows far more slowly). So:

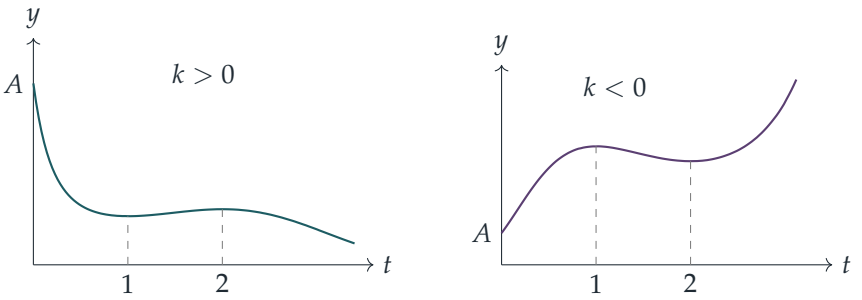
- if $k > 0$, then $\ln(y/A) \to -\infty$, so $y \to 0$ (very rapidly, like $\mathrm{e}^{-kt^2/2}$);
- if $k < 0$, then $\ln(y/A) \to +\infty$, so $y \to \infty$ (like $\mathrm{e}^{|k|t^2/2}$).

Sketches

The sign of $\frac{\mathrm{d}y}{\mathrm{d}t}$ is the sign of $-k(t-1)(t-2)$ for $t > 0$.

- $k > 0$: y decreases from A on $(0,1)$ to a local minimum at $t = 1$, rises to a local maximum at $t = 2$ (still below A), then decreases towards 0.
- $k < 0$: y increases from A to a local maximum at $t = 1$, falls to a local minimum at $t = 2$ (still above A), then increases without bound.

The graphs below are drawn accurately with $A = 1$ and $k = \pm 2$.



Checks

- At $t = 0$ the formula gives $A \cdot 1 \cdot \mathrm{e}^0 = A$. Differentiating $\ln y = \ln A - 6k\ln(t+1) - k(t^2/2 - 4t)$ gives $\frac{y'}{y} = -\frac{6k}{t+1} - k(t-4)$, which recombines to the original equation.
- Numerically, integrating the differential equation step by step (Runge–Kutta) with $A = 1.5$ and $k = \pm 2$ reproduces the formula at $t = 3$ to eight significant figures. With $k = 2$ the ratio $y(1)/y(2)$ is 0.8742, matching $(1.5)^{12}\mathrm{e}^{-5}$.

Common slips

- Getting the division wrong: the remainder is 6, not 2. Multiply back by $t+1$ to check.
- Inverting the ratio: the target is $y(1)/y(2)$, and the signs in the exponent make it easy to get $y(2)/y(1)$ by mistake.
- Sketching both turning points on the wrong side of A: the earlier calculation places both below A for $k > 0$ and both above A for $k < 0$.

Beyond STEP

The equation is linear, $y' + p(t)y = 0$, and its solution $A\exp\big(-\int_0^t p\big)$ is the simplest case of the integrating-factor method. It also illustrates a general principle you will meet in a first ODE course: you can often read off qualitative features (stationary points, sign of the solution, growth or decay) directly from the equation without solving it. Boyce and DiPrima, *Elementary Differential Equations*, develops both ideas, including the uniqueness theorem we used to show that y never vanishes.

WORKED SOLUTION

STEP II 2003, Question 9

Mechanics • Limiting Friction Equilibrium

STEP II 2003, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q9

The key idea

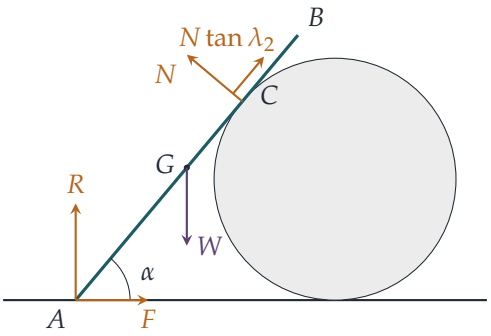
Because the coefficients of friction are given as $\tan\lambda_1$ and $\tan\lambda_2$, the natural tool is the *total reaction* (normal reaction plus friction combined into one force). When friction is limiting, the total reaction makes the angle of friction λ with the normal. That leaves the rod under just three forces whose directions we know, and resolving horizontally and vertically gives two equations in two unknown magnitudes. The only real work is deciding which way each friction force points.

Solution

The model and the directions of friction

We treat the rod as a rigid uniform rod and the cylinder as fixed. The rod touches the cylinder along a tangent, so the normal reaction at C is perpendicular to the rod. Take the cylinder to lie on the side of A in the positive horizontal direction. The forces on the rod are then:

- its weight W, acting vertically down at the midpoint G;
- at A: a normal reaction R vertically up and a friction force F along the floor;
- at C: a normal reaction N perpendicular to the rod, pointing up and back towards A, together with a friction force along the rod.



Ask yourself Which way does friction act at A and at C? Could the rod be on the point of slipping the other way?

The rod has only one way to move while keeping both contacts: as A slides along the floor, the angle α changes and the rod slides over the cylinder. If A moves away from the cylinder, the rod drops (so α decreases) and the rod's point at C slides *down* along the rod relative to the cylinder. If A moves towards the cylinder, everything is reversed.

In the second case friction at A would point away from the cylinder and friction at C would point down the rod. Then all three horizontal forces (friction at A, the horizontal part of N, and the horizontal part of friction at C) would point away from the cylinder, and horizontal equilibrium would be impossible. So in limiting equilibrium the rod is about to slide down: friction on the rod at A acts *towards* the cylinder, and friction at C acts *up* the rod, as drawn.

Total reactions

Let S be the total reaction at A and T the total reaction at C. Since friction is limiting at both contacts:

- S makes angle λ_1 with the upward vertical, tilted towards the cylinder, because $F = R\tan\lambda_1$.
- The normal at C makes angle α with the upward vertical, tilted back towards A, because the rod makes angle α with the horizontal. Friction up the rod turns T by λ_2 back towards the vertical. So T makes angle $\alpha - \lambda_2$ with the upward vertical, measured on the side away from the cylinder.

Resolving for the rod,

$$\begin{aligned} \text{horizontally:} \quad & S\sin\lambda_1 = T\sin(\alpha - \lambda_2), \\ \text{vertically:} \quad & S\cos\lambda_1 + T\cos(\alpha - \lambda_2) = W. \end{aligned}$$

To eliminate T, we multiply the first equation by $\cos(\alpha - \lambda_2)$, the second by $\sin(\alpha - \lambda_2)$, and subtract the first from the second:

$$\begin{aligned} S\left[\cos\lambda_1 \sin(\alpha - \lambda_2) + \sin\lambda_1 \cos(\alpha - \lambda_2)\right] &= W\sin(\alpha - \lambda_2), \\ S\sin(\alpha + \lambda_1 - \lambda_2) &= W\sin(\alpha - \lambda_2), \end{aligned}$$

using the addition formula for sin. We are told that $\alpha \neq \lambda_2 - \lambda_1$, so $\sin(\alpha + \lambda_1 - \lambda_2) \neq 0$ (all the angles are acute, so $\alpha + \lambda_1 - \lambda_2$ lies strictly between $-\frac{\pi}{2}$ and π). Hence we may divide:

$$S = \frac{W\sin(\alpha - \lambda_2)}{\sin(\alpha + \lambda_1 - \lambda_2)}.$$

The frictional force at A is the horizontal component of S, namely $F = S\sin\lambda_1$, so

$$F = \frac{W\sin\lambda_1\ \sin(\alpha - \lambda_2)}{\sin(\alpha + \lambda_1 - \lambda_2)},$$

as required. In the same way, $T = W\sin\lambda_1 / \sin(\alpha + \lambda_1 - \lambda_2)$.

Ask yourself We never took moments. Isn't that suspicious?

Not here. Resolving in two directions is enough to find the two unknown magnitudes. Taking moments about A (where S has no moment, and nor does the friction at C, which acts along the rod) gives $N \cdot AC = W \cdot AG\cos\alpha$ with $N = T\cos\lambda_2$. This only tells us *where* C must be for limiting equilibrium:

$$\frac{AC}{AB} = \frac{\cos\alpha\ \sin(\alpha + \lambda_1 - \lambda_2)}{2\sin\lambda_1\cos\lambda_2}.$$

The given condition $AC > CB$ isn't needed for the friction force. Physically we also need $S > 0$ and $T > 0$, which holds exactly when $\alpha > \lambda_2$: a rod resting on the cylinder at a shallower angle than λ_2 can't be on the point of sliding down it.

Checks

- Smooth floor, $\lambda_1 = 0$: the formula gives $F = 0$, as it must.
- $\alpha = \lambda_2$: then T is vertical, it alone holds up the rod, and $F = 0$. This is the borderline angle at which the rod can just rest on the cylinder.
- Numerically, with $\lambda_1 = 0.3$, $\lambda_2 = 0.5$, $\alpha = 1.0$ (radians), solving the component equations for R and N directly gives $F \approx 0.1975W$, which matches the formula. In this case moments about A then put C at about 0.75 of the way along the rod, consistent with $AC > CB$.

Common slips

- Taking the reaction at C to be vertical, or perpendicular to the cylinder's "top". The reaction is normal to the common tangent, which is the rod itself.

- Choosing the friction directions without checking them. The wrong pair makes horizontal equilibrium impossible.
- Tilting the total reaction at C the wrong way, which gives $\alpha + \lambda_2$ in place of $\alpha - \lambda_2$.

Beyond STEP

Taking the total reaction at angle λ to the normal is the idea behind the *cone of friction*: a contact can hold as long as the total reaction stays inside a cone of half-angle λ around the normal. With three forces, the lines of action must also pass through one point, which turns many statics problems into geometry. Problems with several rough contacts, where the order in which they slip matters, are a good introduction to rigid-body statics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and rigid-body equilibrium carefully.

WORKED SOLUTION

STEP II 2003, Question 10

Mechanics • Constrained Motion with Friction

STEP II 2003, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q10

The key idea

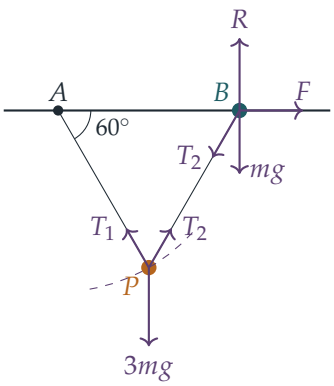
At the start, $AP = PB = AB = \ell$, so APB is an equilateral triangle. Because AP stays taut, P moves on a circle about A. Everything begins at rest, so there is no centripetal term, and P's initial acceleration must be along the tangent to that circle. This fixes its direction. Then P is always at the apex of an isosceles triangle on AB, so P's horizontal position is always half of B's. For the final part, write down Newton's second law for B and for P. Resolving for P along its direction of acceleration (perpendicular to AP) means the tension in AP never appears.

Solution

We model B and P as particles and the string as light and inextensible. Both halves of the string stay taut. Take A as the origin, with the wire along the x-axis and B initially at $(\ell, 0)$. Let θ be the angle PAB. Since $AP = PB = \ell$,

$$B = (2\ell\cos\theta,\, 0), \qquad P = (\ell\cos\theta,\, -\ell\sin\theta),$$

and at the start $\theta = \frac{\pi}{3}$.



The direction of P's acceleration

Since $AP = \ell$ is fixed, P moves on a circle of radius ℓ centred at A (dashed in the diagram). The acceleration of a point on a circle has a component $\ell\dot{\theta}^2$ towards the centre and a component $\ell\ddot{\theta}$ along the tangent. At the start $\dot{\theta} = 0$, because the system is released from rest. So the initial acceleration of P points along the tangent, which is perpendicular to AP.

AP makes an angle of 60° with the horizontal, so the perpendicular to AP makes an angle of 30° with the horizontal. If the acceleration has magnitude a, then

$$a_1 = a\cos 30^\circ = \tfrac{\sqrt{3}}{2}a, \qquad a_2 = a\sin 30^\circ = \tfrac{1}{2}a,$$

and so $a_1 = \sqrt{3}\,a_2$, as required.

In terms of θ: differentiating P's position and setting $\dot{\theta} = 0$ gives an initial acceleration of $\ell\ddot{\theta}(-\sin\theta, -\cos\theta)$. With $\ddot{\theta} > 0$, P moves down and towards A. Its magnitude is $a = \ell\ddot{\theta}$.

The acceleration of B

The x-coordinate of B is always exactly twice that of P, because P lies on the perpendicular bisector of AB. Differentiating twice shows that B's acceleration is always twice the horizontal acceleration of P. So the initial acceleration of B has magnitude $2a_1$, directed towards A, as required.

> **Ask yourself** Why can we say B moves towards A, and which way does friction act?

As P falls, θ increases, so $x_B = 2\ell\cos\theta$ decreases. So B slides towards A, and friction on B acts away from A (in the positive x-direction). Here we are assuming that B really does move. The question confirms this by giving the friction as $\frac{\sqrt{3}}{6}R$, which is its value during sliding.

The magnitude of the initial acceleration

Let T_1 and T_2 be the tensions in AP and PB. Let R be the normal reaction on B and $F = \frac{\sqrt{3}}{6}R$ the friction. Write a for the magnitude of P's initial acceleration, so $a_1 = \frac{\sqrt{3}}{2}a$, and B's acceleration is $2a_1 = \sqrt{3}\,a$ towards A.

Bead B. The string BP pulls B at 60° below the horizontal, towards A. B has no vertical acceleration, so resolving vertically gives

$$R = mg + \tfrac{\sqrt{3}}{2}T_2.$$

Resolving horizontally, taking the direction towards A as positive:

$$m(\sqrt{3}\,a) = \tfrac{1}{2}T_2 - \tfrac{\sqrt{3}}{6}\left(mg + \tfrac{\sqrt{3}}{2}T_2\right) = \tfrac{1}{4}T_2 - \tfrac{\sqrt{3}}{6}mg,$$

so

$$T_2 = 4\sqrt{3}\,ma + \tfrac{2\sqrt{3}}{3}mg. \tag{1}$$

Particle P. Resolve along the direction of P's acceleration, that is, down and towards A at 30° below the horizontal. This direction is perpendicular to AP, so T_1 contributes nothing. The weight $3mg$ makes an angle of 60° with this direction, so it contributes $3mg\cos 60^\circ = \frac{3}{2}mg$. The tension T_2 acts along PB, up and away from A at 60° above the horizontal. That makes an angle of 150° with the direction of acceleration, so it contributes $T_2\cos 150^\circ = -\frac{\sqrt{3}}{2}T_2$. Hence

$$3ma = \tfrac{3}{2}mg - \tfrac{\sqrt{3}}{2}T_2. \tag{2}$$

Substituting (1) into (2):

$$3ma = \tfrac{3}{2}mg - \tfrac{\sqrt{3}}{2}\left(4\sqrt{3}\,ma + \tfrac{2\sqrt{3}}{3}mg\right) = \tfrac{3}{2}mg - 6ma - mg,$$
$$9ma = \tfrac{1}{2}mg,$$

so the magnitude of the initial acceleration of P is

$$a = \frac{g}{18},$$

as required.

The answer is consistent: (1) gives $T_2 = \frac{8\sqrt{3}}{9}mg > 0$, so the string PB is taut. Then $R = \frac{7}{3}mg > 0$. Resolving horizontally for P gives $T_1 = T_2 + 6ma_1 = \frac{19\sqrt{3}}{18}mg > 0$, so AP is taut too.

Checks

- Vertical equation for P, which we did not use: $\frac{\sqrt{3}}{2}(T_1 + T_2) - 3mg = \frac{35}{12}mg - 3mg = -\frac{1}{12}mg$. This equals $-3ma_2 = -3m\cdot\frac{g}{36}$, as it should.
- Numerically, we differentiated the positions of B and P to get $a_1/a_2 \approx 1.7320508$ and $a_B/a_1 = 2$. We then solved the three force equations for T_1, T_2 and $\ddot{\theta}$ with $g = 9.81$. This gave $|a| = 0.545 = g/18$, with $T_2/mg \approx 1.5396$ and $R/mg \approx 2.3333$.
- Without friction, (1) would be $T_2 = 2\sqrt{3}\,ma$ and (2) would give $6ma = \frac{3}{2}mg$, so $a = g/4$. Friction should slow things down, and $g/18 < g/4$.

Common slips

- Including a centripetal term. At the instant of release $\dot{\theta} = 0$, so the acceleration is purely tangential.
- Forgetting that T_2 pulls B downwards, which increases R and therefore the friction. Taking $R = mg$ gives the wrong answer.
- Resolving for P horizontally and vertically. This works, but it brings in T_1 and needs one more equation. Resolving perpendicular to AP removes it.

Beyond STEP

This is a constrained system with one degree of freedom: everything is fixed by the single angle θ. Writing positions in terms of θ and differentiating is the first step of Lagrangian mechanics. In Lagrangian mechanics, constraint forces such as T_1 are removed systematically rather than by a clever choice of direction. Friction does not fit neatly into that framework, which is one reason questions like this are good practice in Newtonian methods. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers constrained motion and non-uniform circular motion carefully.

WORKED SOLUTION

STEP II 2003, Question 11

Mechanics • Projectile Motion

STEP II 2003, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

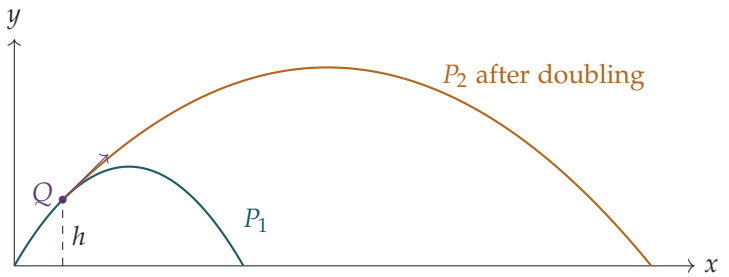
step.maths.org/questions/03-s2-q11

The key idea

Work with the horizontal and vertical components of velocity separately. The direction of motion makes 45° with the horizontal exactly when the two components are equal, and that gives the first time directly. When the speed is doubled the direction doesn't change, so both components double, and the rest of the flight is a new projectile problem that starts at a known height.

Solution

We model each particle as a point projectile moving under gravity alone, with no air resistance. Write $s = \sin\alpha$ and $c = \cos\alpha$. Since $\alpha > 45^\circ$ we have $s > c > 0$.



The sketch uses $\alpha = 60^\circ$. Both particles follow the teal path as far as Q, where the direction first makes 45° with the horizontal. After that P_2 follows the amber path.

The flight time of P_1

The vertical displacement at time t is $y = Vs\,t - \frac{1}{2}gt^2 = t\left(Vs - \frac{1}{2}gt\right)$. This is zero again when

$$T_1 = \frac{2V\sin\alpha}{g}.$$

The first 45° moment

At time t the velocity components are

$$\dot{x} = Vc, \qquad \dot{y} = Vs - gt.$$

The horizontal component never changes. At launch the direction is steeper than 45°, because $\dot{y} = Vs > Vc = \dot{x}$. As $\dot{y}$ decreases, the angle first falls to 45° when $\dot{y} = \dot{x}$, with the particle still rising:

$$Vs - gt_1 = Vc \quad \Longrightarrow \quad t_1 = \frac{V(s-c)}{g}.$$

(The direction makes 45° again on the way down, when $\dot{y} = -Vc$, but that is later.) Since $\frac{1}{2}T_1 = Vs/g$,

$$t_1 = \frac{Vs}{g}\left(1 - \frac{c}{s}\right) = \tfrac{1}{2}(1 - \cot\alpha)\, T_1,$$

as required.

The flight of P_2

Up to time t_1, P_2 moves exactly like P_1. At t_1 both velocity components equal Vc, and the height is

$$h = Vs\,t_1 - \tfrac{1}{2}gt_1^2 = \frac{V^2}{g}\left[s(s-c) - \tfrac{1}{2}(s-c)^2\right]$$
$$= \frac{V^2}{2g}(s-c)\left[2s - (s-c)\right] = \frac{V^2}{2g}\,(s^2 - c^2).$$

This is positive because $s > c$.

Ask yourself What does "the speed is doubled" do to each component of the velocity?

The speed changes but the direction doesn't, so the whole velocity vector is multiplied by 2. Both components become $2Vc$: the particle sets off from height h at 45° above the horizontal with speed $2\sqrt{2}\,Vc$.

Let τ be the time from the doubling until P_2 lands. Measuring vertical displacement upwards from Q, landing means a displacement of $-h$:

$$2Vc\,\tau - \tfrac{1}{2}g\tau^2 = -\frac{V^2}{2g}(s^2 - c^2).$$

Multiply by $-2/g$ and write $\tau = (V/g)u$, where u is dimensionless:

$$u^2 - 4cu - (s^2 - c^2) = 0 \quad \Longrightarrow \quad u = 2c \pm \sqrt{4c^2 + s^2 - c^2} = 2c \pm \sqrt{s^2 + 3c^2}.$$

The product of the roots is $-(s^2 - c^2) < 0$, so one root is negative. We need $\tau > 0$, so we take the plus sign:

$$\tau = \frac{V}{g}\left(2c + \sqrt{s^2 + 3c^2}\right).$$

Therefore

$$T_2 = t_1 + \tau = \frac{V}{g}\left(s - c + 2c + \sqrt{s^2 + 3c^2}\right) = \frac{V}{g}\left(s + c + \sqrt{s^2 + 3c^2}\right).$$

Dividing by $\frac{1}{2}T_1 = Vs/g$ and using $\sqrt{s^2 + 3c^2}/s = \sqrt{1 + 3c^2/s^2}$ (valid because $s > 0$):

$$\frac{2T_2}{T_1} = \frac{s + c + \sqrt{s^2 + 3c^2}}{s} = 1 + \cot\alpha + \sqrt{1 + 3\cot^2\alpha},$$

which is the required result.

Checks

- As $\alpha \to 90^\circ$, $\cot\alpha \to 0$ and the ratio tends to 2. This makes sense: the 45° moment is now the top of the flight, where the speed is zero, so doubling it does nothing and $T_2 = T_1$.
- At $\alpha = 45^\circ$ (the boundary case) the doubling happens at launch, so P_2 is just projected at $2V$, and $T_2 = 2T_1$. The formula gives $1 + 1 + 2 = 4$, which agrees.
- At $\alpha = 60^\circ$, $t_1/T_1 = \frac{1}{2}(1 - 1/\sqrt{3}) \approx 0.2113$ and $2T_2/T_1 = 1 + 1/\sqrt{3} + \sqrt{2} \approx 2.9916$. A numerical step-by-step simulation of P_2's flight for 2000 random choices of α and V matched the formula for T_2.

Common slips

- Doubling only the vertical component, or only the horizontal one. The direction is unchanged, so both double.
- Restarting P_2's motion from ground level instead of from height h.
- Taking the negative root for τ without checking its sign.

Beyond STEP

An instantaneous change of velocity is an *impulse*: a very large force acting for a very short time. Impulses are how first-year mechanics treats collisions, rocket burns and kicks. Splitting a trajectory into pieces joined by impulses is exactly how orbital manoeuvres are planned, where a short engine burn moves a spacecraft from one conic orbit onto another. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops impulse and momentum carefully and has many problems of this kind.

WORKED SOLUTION

STEP II 2003, Question 12

Probability and Statistics • Branching Processes

STEP II 2003, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q12

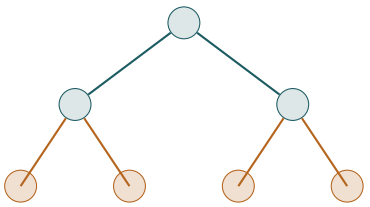
This question has a full guided treatment in *Beyond STEP*, Section 12.15, *Extinction or Explosion*: https://beyondstep.keithmcnulty.org/ch12/branching.html

The key idea

This is a branching process: each generation is built from the previous one, with every particle splitting independently. The one fact that does all the work is that if m particles each produce a $\mathrm{B}(2, p)$ number of offspring, independently, the total is $\mathrm{B}(2m, p)$. So we condition on the size of the previous generation, both for the probabilities and for the expectation.

Solution

We assume, as the model intends, that different particles disintegrate independently of each other and of the past. Write $X_0 = 1$ for the single particle alive during $0 < t < T$. At $t = nT$ each of the X_{n-1} particles alive is replaced by its own offspring, so X_n is the total number of offspring of generation $n - 1$.



$X_0 = 1$

$X_1 = 2$

$X_2 \leq 4$

The picture shows the case $X_1 = 2$: the two particles then have up to four offspring between them, and each of the four possible offspring is born with probability p, independently.

The joint probability

Since $X_1 \sim \mathrm{B}(2, p)$, we have $\mathrm{P}(X_1 = 2) = p^2$. Given $X_1 = 2$, each of the two particles produces a $\mathrm{B}(2, p)$ number of offspring, independently, so X_2 is the number of successes in 4 independent trials with success probability p; that is, $X_2 \mid X_1 = 2 \sim \mathrm{B}(4, p)$. Hence

$$\mathrm{P}(X_1 = 2 \text{ and } X_2 = 2) = \mathrm{P}(X_1 = 2)\,\mathrm{P}(X_2 = 2 \mid X_1 = 2) = p^2 \cdot \binom{4}{2} p^2 q^2 = 6p^4 q^2,$$

as required.

The conditional probability

To use the definition of conditional probability we need $\mathrm{P}(X_2 = 2)$, which we find by conditioning on X_1:

- if $X_1 = 0$ the population has died out, so $X_2 = 0$;
- if $X_1 = 1$ (probability $2pq$), then $X_2 \sim \mathrm{B}(2, p)$ and $\mathrm{P}(X_2 = 2 \mid X_1 = 1) = p^2$;
- if $X_1 = 2$, the contribution is the $6p^4q^2$ just found.

So

$$\mathrm{P}(X_2 = 2) = 2pq \cdot p^2 + 6p^4q^2 = 2p^3q\,(1 + 3pq),$$

and therefore, for $0 < p < 1$,

$$\mathrm{P}(X_1 = 2 \mid X_2 = 2) = \frac{6p^4q^2}{2p^3q(1 + 3pq)} = \frac{3pq}{1 + 3pq}.$$

Setting this equal to $\frac{9}{25}$ gives $75pq = 9 + 27pq$, so $48pq = 9$ and $pq = \frac{3}{16}$. With $q = 1 - p$ this is

$$p^2 - p + \tfrac{3}{16} = 0 \quad \Longleftrightarrow \quad \left(p - \tfrac{1}{4}\right)\left(p - \tfrac{3}{4}\right) = 0,$$

so $p = \frac{1}{4}$ or $p = \frac{3}{4}$. (The cases $p = 0$ and $p = 1$ are excluded: for $p = 0$ the event $X_2 = 2$ is impossible, and for $p = 1$ the conditional probability is 0.)

? **Ask yourself** Why do both values of p give the same answer?

The conditional probability depends on p only through $pq = p(1 - p)$, which is unchanged when p is replaced by $1 - p$. So the roots must come in a pair $p, 1 - p$, here $\frac{1}{4}$ and $\frac{3}{4}$.

The expectation

Suppose $X_{n-1} = m$. Each of these m particles produces on average $\mathrm{E}(X) = 2p$ offspring, so the expected size of the next generation is $2pm$:

$$\mathrm{E}(X_n \mid X_{n-1} = m) = 2pm.$$

Averaging over the possible values of m (the law of total expectation),

$$\mathrm{E}(X_n) = \sum_m \mathrm{P}(X_{n-1} = m)\, 2pm = 2p\,\mathrm{E}(X_{n-1}).$$

In words: every particle alive in one interval contributes $2p$ particles, on average, to the next. Since $\mathrm{E}(X_1) = 2p$ (or $\mathrm{E}(X_0) = 1$), the recurrence gives a geometric sequence:

$$\mathrm{E}(X_n) = (2p)^n.$$

The limits

For $p = \frac{1}{4}$ we get $\mathrm{E}(X_n) = \left(\frac{1}{2}\right)^n \to 0$ as $n \to \infty$, since $0 < \frac{1}{2} < 1$.

For $p = \frac{3}{4}$ we get $\mathrm{E}(X_n) = \left(\frac{3}{2}\right)^n \to +\infty$, since $\frac{3}{2} > 1$.

So the two values of p that look identical in the conditional probability lead to completely different long-term behaviour: on average the population dies away when $p = \frac{1}{4}$ and explodes when $p = \frac{3}{4}$.

Checks

- With $p = \frac{3}{4}$: $6p^4q^2 = \frac{243}{2048} \approx 0.1187$, and a simulation of 10^6 two-generation populations gave 0.1186.
- At $p = \frac{1}{4}$, $pq = \frac{3}{16}$, so $3pq = \frac{9}{16}$ and $\dfrac{9/16}{25/16} = \frac{9}{25}$, as it should be.
- Computing the exact distributions of $X_1, \ldots, X_5$ (each generation $\mathrm{B}(2m, p)$ given the last) for several p gives means exactly $(2p)^n$.

Common slips

- Forgetting the route $X_1 = 1$, $X_2 = 2$ when finding $\mathrm{P}(X_2 = 2)$; the route through $X_1 = 0$ correctly contributes nothing.
- Taking $X_2 \mid X_1 = 2$ to be $2 \times \mathrm{B}(2, p)$ (twice one particle's offspring) rather than a sum of two independent copies, which is $\mathrm{B}(4, p)$.

Beyond STEP

This is a Galton–Watson branching process, first studied to explain why surnames die out. The mean number of offspring $\mu = 2p$ decides everything: if $\mu \le 1$ the population dies out with probability 1, while if $\mu > 1$ it survives for ever with positive probability. Even for $p = \frac{3}{4}$, the population still dies out with probability $\frac{1}{9}$, the smallest root of $s = (q + ps)^2$, which comes from the probability generating function. Grimmett and Welsh, *Probability: An Introduction*, treat branching processes with generating functions in a way that follows on directly from this question.

WORKED SOLUTION

STEP II 2003, Question 13

Probability and Statistics • Truncated Poisson Distribution

STEP II 2003, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q13

The key idea

This is a Poisson distribution with the value 0 removed and the remaining probabilities scaled up so they still sum to 1. Every sum we need is therefore a Poisson sum with its $k = 0$ term missing, and for the mean and variance the $k = 0$ term contributes nothing anyway. So we can borrow the familiar Poisson facts $\sum k\,p_k = \lambda$ and $\sum k^2 p_k = \lambda + \lambda^2$ and simply multiply by A.

Solution

The constant A

Write $p_k = \dfrac{\lambda^k \mathrm{e}^{-\lambda}}{k!}$ for the ordinary $\mathrm{Po}(\lambda)$ probabilities. Since $\sum_{k\geq 0} \lambda^k/k! = \mathrm{e}^{\lambda}$,

$$\sum_{k=1}^{\infty} p_k = \mathrm{e}^{-\lambda}\left(\mathrm{e}^{\lambda} - 1\right) = 1 - \mathrm{e}^{-\lambda}.$$

The probabilities $\mathrm{P}(X = k) = Ap_k$ must sum to 1, so $A(1 - \mathrm{e}^{-\lambda}) = 1$, that is,

$$A = (1 - \mathrm{e}^{-\lambda})^{-1},$$

as required. Note that $0 < 1 - \mathrm{e}^{-\lambda} < 1$ for $\lambda > 0$, so $A > 1$: the leftover probability p_0 has been shared out among $k \geq 1$.

The mean

The $k = 0$ term of $\sum k\,p_k$ is zero, so

$$\sum_{k=1}^{\infty} k\,p_k = \sum_{k=0}^{\infty} k\,p_k = \lambda,$$

the mean of $\mathrm{Po}(\lambda)$. (Directly: $\sum_{k\geq 1} k\lambda^k \mathrm{e}^{-\lambda}/k! = \lambda \sum_{k\geq 1} \lambda^{k-1}\mathrm{e}^{-\lambda}/(k-1)! = \lambda$.) Hence

$$\mu = \mathrm{E}(X) = A\lambda = \frac{\lambda}{1-\mathrm{e}^{-\lambda}}.$$

The variance

In the same way, the $k = 0$ term of $\sum k^2 p_k$ is zero, and for $\mathrm{Po}(\lambda)$ we have $\mathrm{E}(Y^2) = \mathrm{Var}(Y) + (\mathrm{E}\,Y)^2 = \lambda + \lambda^2$. If you'd rather not quote that, write $k^2 = k(k-1) + k$:

$$\sum_{k\geq 2} k(k-1)\frac{\lambda^k \mathrm{e}^{-\lambda}}{k!} = \lambda^2 \sum_{k\geq 2} \frac{\lambda^{k-2}\mathrm{e}^{-\lambda}}{(k-2)!} = \lambda^2, \qquad \text{so} \qquad \sum_{k\geq 1} k^2 p_k = \lambda^2 + \lambda.$$

Therefore

$$\begin{aligned} \mathrm{E}(X^2) &= A\lambda(1+\lambda) = \mu(1+\lambda), \\ \mathrm{Var}(X) &= \mathrm{E}(X^2) - \mu^2 = \mu(1+\lambda) - \mu^2 = \mu(1-\mu+\lambda), \end{aligned}$$

as required.

The inequalities

> **Ask yourself** Where could a strict inequality $\mu < 1 + \lambda$ come from?

From the variance. X takes every value $1, 2, 3, \ldots$ with positive probability, so it is not constant, and a non-constant random variable has strictly positive variance (as $\mathrm{Var}(X) = \mathrm{E}\big((X-\mu)^2\big)$ and $(X-\mu)^2 > 0$ with positive probability). So

$$\mu(1-\mu+\lambda) > 0.$$

Since $\mu > 0$ (it is the mean of a positive random variable), we may divide by μ to get $1 - \mu + \lambda > 0$, i.e. $\mu < 1 + \lambda$.

For the lower bound, $\mu = A\lambda$ with $A > 1$ and $\lambda > 0$, so $\mu > \lambda$. Together,

$$\lambda < \mu < 1 + \lambda.$$

Normal approximation with $\lambda = 100$

Now e^{-100} is astronomically small (about 4×10^{-44}), so to any sensible accuracy

$$\mu = \frac{100}{1-\mathrm{e}^{-100}} \approx 100, \qquad \mathrm{Var}(X) = \mu(101-\mu) \approx 100 \times 1 = 100.$$

So we approximate X by $Y \sim \mathrm{N}(100, 10^2)$. Since X is integer-valued, we use a continuity correction: the event $X = 100$ corresponds to $99.5 < Y < 100.5$. Standardising with $Z = (Y - 100)/10$,

$$\begin{aligned} \mathrm{P}(X = 100) &\approx \mathrm{P}(-0.05 < Z < 0.05) \\ &= 2\Phi(0.05) - 1 \\ &\approx 2(0.51994) - 1 = 0.0399, \end{aligned}$$

so $\mathrm{P}(X = 100) \approx 0.04$ to 2 decimal places.

Checks

- As $\lambda \to 0$, $\mu = \lambda/(1 - \mathrm{e}^{-\lambda}) \to 1$ and the variance $\to 0$: almost all the probability sits at $k = 1$, as it should. This matches the inequality $\lambda < \mu < 1 + \lambda$ squeezing μ towards 1.
- Summing the series numerically for many random λ reproduces A, μ and $\mu(1 - \mu + \lambda)$ to machine accuracy.
- For $\lambda = 100$ the exact probability is $\dfrac{100^{100}\mathrm{e}^{-100}}{100!\,(1 - \mathrm{e}^{-100})} \approx 0.03986$, against the normal approximation 0.03988. Both round to 0.04.

Common slips

- Forgetting the continuity correction and computing $\mathrm{P}(Y = 100) = 0$, or using a density value without justification.
- Proving only $\mu \leq 1 + \lambda$: the strict inequality needs the observation that X is not constant.
- Dividing by μ without noting that $\mu > 0$.

Beyond STEP

X is a *zero-truncated Poisson* distribution, which statisticians use when a count of zero can't be observed (for example, the number of occupants of cars passing a checkpoint: no car has zero occupants). The normal approximation works because a $\mathrm{Po}(100)$ variable is the sum of 100 independent $\mathrm{Po}(1)$ variables, so the central limit theorem applies; the truncation is invisible at this scale. Grimmett and Welsh, *Probability: An Introduction*, covers generating functions and the central limit theorem behind both halves of the question.

WORKED SOLUTION

STEP II 2003, Question 14

Probability and Statistics • Hypothesis Testing Errors

STEP II 2003, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q14

The key idea

The number of throws needed to get the first 6 has a geometric distribution, and its tail has a beautifully simple form: the event "more than M throws" is just "the first M throws all fail". So both error probabilities are pure Mth powers, and eliminating M between them is a matter of taking logarithms.

Solution

The two error probabilities

Model the throws as independent, each giving a 6 with the same probability p. Then X, the number of throws up to and including the first 6, satisfies

$$X > M \iff \text{none of the first } M \text{ throws is a 6}, \qquad \text{so} \qquad \mathrm{P}(X > M) = (1-p)^M,$$

using independence of the throws. (There is no need to sum the geometric series $\sum p(1-p)^{k-1}$: thinking about the event directly is quicker.)

The test accepts H_1 exactly when $X > M$. So α, the probability of accepting H_1 when in fact $p = B$, is

$$\alpha = (1-B)^M,$$

and β, the probability of accepting H_0 (that is, $X \leq M$) when in fact $p = A$, is

$$\beta = 1 - \mathrm{P}(X > M \mid p = A) = 1 - (1-A)^M.$$

Eliminating M

We want to write $(1-A)^M$ as a power of $\alpha = (1-B)^M$. Since $0 < B < 1$, $\ln(1-B)$ is defined and non-zero, so taking logarithms of α gives

$$M = \frac{\ln \alpha}{\ln(1-B)}.$$

Then

$$(1-A)^M = \mathrm{e}^{M\ln(1-A)} = \exp\left(\frac{\ln(1-A)}{\ln(1-B)}\ln\alpha\right) = \alpha^K, \qquad \text{where} \quad K = \frac{\ln(1-A)}{\ln(1-B)}.$$

Hence

$$\beta = 1-\alpha^K, \qquad K = \frac{\ln(1-A)}{\ln(1-B)},$$

and K depends only on A and B, not on M, as required.

Ask yourself What can we say about the size of K?

Since $0 < A < B < 1$ we have $1 > 1-A > 1-B > 0$, so both logarithms are negative and $\ln(1-B) < \ln(1-A) < 0$. Dividing, K is positive and

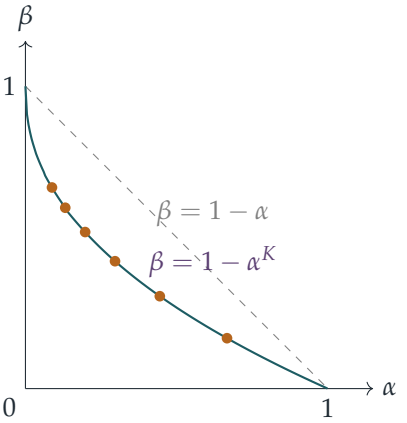
$$0 < K < 1.$$

This is what fixes the shape of the graph.

The graph of β against α

As M runs through $1, 2, 3, \ldots$, $\alpha = (1-B)^M$ takes values in $(0,1)$, decreasing towards 0; strictly the graph is a sequence of points, all lying on the curve $\beta = 1-\alpha^K$, $0 \le \alpha \le 1$. For that curve:

- it passes through $(0,1)$ and $(1,0)$;
- $\dfrac{\mathrm{d}\beta}{\mathrm{d}\alpha} = -K\alpha^{K-1} < 0$, so β decreases as α increases: making one error less likely makes the other more likely;
- since $K - 1 < 0$, the gradient tends to $-\infty$ as $\alpha \to 0^+$ (the curve leaves $(0,1)$ vertically) and equals $-K$, which is between -1 and 0, at $\alpha = 1$;
- $\dfrac{\mathrm{d}^2\beta}{\mathrm{d}\alpha^2} = K(1-K)\alpha^{K-2} > 0$, so the curve is convex;
- for $0 < \alpha < 1$ and $0 < K < 1$ we have $\alpha^K > \alpha$, so $\beta < 1-\alpha$: the curve lies below the straight line joining $(0,1)$ to $(1,0)$.



The curve is drawn for $A = \frac{1}{6}$, $B = \frac{1}{3}$ (so $K \approx 0.45$), with the points for $M = 1, \ldots, 6$ marked; increasing M moves the point up and to the left.

Checks

- Take $A = \frac{1}{6}$, $B = \frac{1}{3}$, $M = 4$. Directly, $\alpha = (\frac{2}{3})^4 \approx 0.1975$ and $\beta = 1 - (\frac{5}{6})^4 \approx 0.5177$. With $K = \ln \frac{5}{6} / \ln \frac{2}{3} \approx 0.4497$, $1 - \alpha^K \approx 0.5177$ too. A simulation of 2×10^5 runs of the test gave 0.198 and 0.518.
- Extreme cases behave sensibly: if A is close to B then $K \approx 1$ and $\beta \approx 1 - \alpha$, so the test is little better than ignoring the throws and deciding at random; if A is tiny, $K \approx 0$ and β can be made small for any reasonable α.

Common slips

- Mixing up which hypothesis is which: here H_0 is the *larger* value B, so a long wait for a 6 is evidence against H_0.
- Getting K upside down, $\ln(1 - B) / \ln(1 - A)$, which is greater than 1; the check $\beta < 1 - \alpha$ (any sensible test beats guessing) catches this.

Beyond STEP

The curve of β against α is the trade-off curve between the two types of error, closely related to the ROC curve used throughout statistics and machine learning. The Neyman–Pearson lemma shows that, for two simple hypotheses like these, a test based on the likelihood ratio gives the smallest possible β for each α; here the likelihood ratio is monotonic in X, so the test "reject H_0 when X is large" is in fact optimal. For more on these ideas, see Grimmett and Welsh, *Probability: An Introduction*, as a foundation, then any first course in mathematical statistics.

WORKED SOLUTION

STEP III 2003, Question 1

Pure • Inverse Hyperbolic Integration

STEP III 2003, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q1

The key idea

The first two results are careful chain-rule calculations. The useful thing they produce is a template: the derivative of each inverse function has the shape $\dfrac{\text{constant}}{(\text{linear})\sqrt{\text{linear}}}$, and that is exactly the shape of both integrands. So for each integral we choose a and b to match the template. The bracket outside the root tells us which formula to use. If it is $x+b$, the larger constant, we use arcsin; if it is $x+a$, the smaller one, we use arcosh.

Solution

The two derivatives

Let $y = \dfrac{x+a}{x+b}$. By the quotient rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(x+b)-(x+a)}{(x+b)^2} = \frac{b-a}{(x+b)^2}.$$

Since $x+a>0$ and $x+b>0$ we have $y>0$, and since $b>a$ the numerator is smaller than the denominator, so $0<y<1$ and $\arcsin y$ is differentiable there. Next,

$$1-y^2 = \frac{(x+b)^2-(x+a)^2}{(x+b)^2} = \frac{(b-a)(2x+a+b)}{(x+b)^2},$$

using the difference of two squares. Since $x+b>0$, the square root of $(x+b)^2$ is $x+b$ itself, so

$$\sqrt{1-y^2} = \frac{\sqrt{b-a}\,\sqrt{a+b+2x}}{x+b}.$$

By the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}\arcsin\left(\frac{x+a}{x+b}\right) = \frac{1}{\sqrt{1-y^2}}\cdot\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x+b}{\sqrt{b-a}\,\sqrt{a+b+2x}}\cdot\frac{b-a}{(x+b)^2}$$
$$= \frac{\sqrt{b-a}}{(x+b)\sqrt{a+b+2x}},$$

since $(b-a)/\sqrt{b-a} = \sqrt{b-a}$. This is the required result.

For the arcosh, let $z = \dfrac{x+b}{x+a}$. Here $z > 1$, because $x+b > x+a > 0$. Then

$$\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{(x+a)-(x+b)}{(x+a)^2} = -\frac{b-a}{(x+a)^2}, \qquad z^2-1 = \frac{(b-a)(2x+a+b)}{(x+a)^2}.$$

Since $x+a>0$, $\sqrt{z^2-1} = \sqrt{b-a}\,\sqrt{a+b+2x}\,/(x+a)$, and so

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcosh}\left(\frac{x+b}{x+a}\right) = \frac{x+a}{\sqrt{b-a}\,\sqrt{a+b+2x}}\cdot\left(-\frac{b-a}{(x+a)^2}\right) = -\frac{\sqrt{b-a}}{(x+a)\sqrt{a+b+2x}}.$$

The two results look almost alike. The differences are the sign and which bracket sits outside the root.

Ask yourself Which of the hypotheses did the arcsin calculation actually use?

We needed $x+b>0$ to simplify $\sqrt{(x+b)^2}$, and we needed $1-y^2>0$. From the factorised form, $1-y^2>0$ holds exactly when $a+b+2x>0$, given $b>a$. We never used $x+a>0$ itself. So the arcsin formula holds whenever $x+b>0$ and $a+b+2x>0$, even if $x+a$ is negative, in which case y lies in $(-1,0]$. This matters in (ii).

Part (i)

The bracket outside the root is $x+1$, so we try to match it with $x+a$ in the arcosh formula. Write $\sqrt{x+3} = \sqrt{2x+6}\,/\sqrt{2}$, so we need $a+b=6$ as well as $a=1$. That gives $b=5$, so $b-a=4$. With these values the hypotheses $x+a>0$, $x+b>0$ and $b>a$ all hold for $x>-1$, and

$$\frac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcosh}\left(\frac{x+5}{x+1}\right) = -\frac{2}{(x+1)\sqrt{2x+6}} = -\frac{\sqrt{2}}{(x+1)\sqrt{x+3}}.$$

Dividing by $-\sqrt{2}$,

$$\int \frac{1}{(x+1)\sqrt{x+3}}\,\mathrm{d}x = -\frac{1}{\sqrt{2}}\operatorname{arcosh}\left(\frac{x+5}{x+1}\right) + C.$$

Part (ii)

Now the bracket outside is $x+3$, so we match it with $x+b$ in the arcsin formula: $b=3$. Inside the root we need $a+b+2x = 2(x+1)$, so $a+b=2$ and $a=-1$. Then $b-a=4$

and

$$\frac{\mathrm{d}}{\mathrm{d}x}\arcsin\left(\frac{x-1}{x+3}\right)=\frac{2}{(x+3)\sqrt{2x+2}}=\frac{\sqrt{2}}{(x+3)\sqrt{x+1}}.$$

Ask yourself With $a=-1$, the hypothesis $x+a>0$ requires $x>1$. Does our answer still hold on the whole range $x>-1$?

Yes, it does. As we saw above, the arcsin formula needs only $x+b>0$ and $a+b+2x>0$. Here these say $x+3>0$ and $2x+2>0$, and both hold for every $x>-1$. On this range $(x-1)/(x+3)$ goes from -1 up towards 1, so the arcsin is defined. Hence, for $x>-1$,

$$\int\frac{1}{(x+3)\sqrt{x+1}}\,\mathrm{d}x=\frac{1}{\sqrt{2}}\arcsin\left(\frac{x-1}{x+3}\right)+C.$$

Otherwise: a substitution

Both integrals also give way to $u=\sqrt{x+c}$, where $x+c$ is the expression under the root, since then $\mathrm{d}x=2u\,\mathrm{d}u$. In (ii), with $u=\sqrt{x+1}$ the integral becomes $\int\frac{2}{u^2+2}\,\mathrm{d}u=\sqrt{2}\arctan\left(\sqrt{(x+1)/2}\right)+C$. This differs from our answer only by the constant $-\pi/(2\sqrt{2})$. In (i), with $u=\sqrt{x+3}$ we get $\int\frac{2}{u^2-2}\,\mathrm{d}u=\frac{1}{\sqrt{2}}\ln\left|\frac{\sqrt{x+3}-\sqrt{2}}{\sqrt{x+3}+\sqrt{2}}\right|+C$. This is exactly the arcosh answer written in logarithms.

Checks

- Differentiate the answers back. For example, at $x=0$ the derivative of $\frac{1}{\sqrt{2}}\arcsin\frac{x-1}{x+3}$ should be $\frac{1}{3}$, and the formula above gives $\frac{\sqrt{2}}{\sqrt{2}\cdot3\cdot1}=\frac{1}{3}$.
- Numerically, $\int_0^1\frac{\mathrm{d}x}{(x+1)\sqrt{x+3}}\approx0.374543$ and $\int_0^1\frac{\mathrm{d}x}{(x+3)\sqrt{x+1}}\approx0.240301$, by Simpson's rule. Both values agree with the antiderivatives to better than 10^{-12}. The agreement also holds on intervals reaching down to $x=-0.9$, inside the range $-1<x<1$ where $x+a<0$ in (ii).
- Signs: the integrand in (i) is positive, so its integral should be increasing. Our answer is, because $\frac{x+5}{x+1}$ decreases as x increases, so minus its arcosh increases.

Common slips

- Matching the wrong bracket to the wrong formula: in (i) the $x+1$ outside the root is the *smaller* constant, so it has to be $x+a$, which means arcosh, not arcsin.
- Forgetting the factor $\sqrt{2}$ that comes from writing $\sqrt{x+3}$ as $\sqrt{2x+6}/\sqrt{2}$.
- Dropping the minus sign in the arcosh derivative, which comes from $\mathrm{d}z/\mathrm{d}x<0$.

Beyond STEP

The two answers to (ii), one with arcsin and one with arctan, differ by a constant. That is guaranteed because two functions with the same derivative on an *interval* differ by a constant, a consequence of the mean value theorem. On a set with two separate pieces the constants could be different. The logarithm in (i) is no accident either: inverse hyperbolic functions are logarithms in disguise, and that is why integrals of $1/\sqrt{\text{quadratic}}$ give either inverse trigonometric functions or logarithms. Abbott's *Understanding Analysis* proves the mean value theorem carefully, and Siklos's *Advanced Problems in Mathematics* has more integrals of this kind.

WORKED SOLUTION

STEP III 2003, Question 2

Pure • Binomial Series

STEP III 2003, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q2

The key idea

The central binomial coefficient $\binom{2r}{r}$ hides inside the binomial series for negative half-integer powers. Once the opening identity splits $(2r)!$ into its odd and even factors, the coefficient of p^r in $(1-p)^{-1/2}$ turns out to be exactly $\binom{2r}{r}/4^r$. Each sum is then a binomial series in disguise, and the only job is to pick p (and, in (ii), the right power).

Solution

The identity

Split the factors of $(2r)!$ into odd and even ones:

$$(2r)! = \big[1 \times 3 \times \cdots \times (2r-1)\big] \times \big[2 \times 4 \times \cdots \times 2r\big].$$

Each of the r even factors is 2 times one of $1, 2, \ldots, r$, so the second bracket is $2^r\, r!$. Hence

$$\binom{2r}{r} = \frac{(2r)!}{r!\,r!} = \frac{1 \times 3 \times \cdots \times (2r-1) \times 2^r\, r!}{r!\,r!} = \frac{1 \times 3 \times \cdots \times (2r-1)}{r!} \times 2^r,$$

as required.

Part (i)

For $|p| < 1$, the binomial series gives

$$\begin{aligned}(1-p)^{-\frac{1}{2}} &= 1 + \left(-\tfrac{1}{2}\right)(-p) + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}(-p)^2 \\ &\quad + \frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)}{3!}(-p)^3 + \cdots \\ &= 1 + \tfrac{1}{2}p + \tfrac{3}{8}p^2 + \tfrac{5}{16}p^3 + \cdots.\end{aligned}$$

The general term follows the same pattern. In the coefficient of p^r there are r minus signs from the factors $-\frac{1}{2}, -\frac{3}{2}, \ldots$ and r more from $(-p)^r$, so they cancel, leaving

$$\frac{\frac{1}{2} \times \frac{3}{2} \times \cdots \times \frac{2r-1}{2}}{r!}\, p^r = \frac{1 \times 3 \times \cdots \times (2r-1)}{2^r\, r!}\, p^r.$$

Ask yourself How does this connect to $\binom{2r}{r}$?

By the identity, $\dfrac{1 \times 3 \times \cdots \times (2r-1)}{r!} = \dfrac{\binom{2r}{r}}{2^r}$. So the coefficient of p^r is $\binom{2r}{r}/4^r$, which also holds for $r = 0$ since both sides are 1. Therefore

$$(1-p)^{-\frac{1}{2}} = \sum_{r=0}^{\infty} \binom{2r}{r} \left(\frac{p}{4}\right)^r \qquad (|p| < 1).$$

The sum we want has $(p/4)^r = 1/8^r$, so we take $p = \frac{1}{2}$, which is inside the interval of convergence. Then

$$\sum_{r=0}^{\infty} \frac{\binom{2r}{r}}{8^r} = \left(1 - \tfrac{1}{2}\right)^{-\frac{1}{2}} = \left(\tfrac{1}{2}\right)^{-\frac{1}{2}} = \sqrt{2}.$$

Part (ii)

The extra factor $2r+1$ suggests going one power further, to $(1-p)^{-3/2}$. For $|p| < 1$, the coefficient of p^r in its binomial series is, with the signs cancelling as before,

$$\begin{aligned}\frac{\frac{3}{2} \times \frac{5}{2} \times \cdots \times \frac{2r+1}{2}}{r!} &= \frac{3 \times 5 \times \cdots \times (2r+1)}{2^r\, r!} \\ &= (2r+1)\, \frac{1 \times 3 \times \cdots \times (2r-1)}{2^r\, r!} = \frac{(2r+1)\binom{2r}{r}}{4^r},\end{aligned}$$

since $3 \times 5 \times \cdots \times (2r+1) = (2r+1) \times \left[1 \times 3 \times \cdots \times (2r-1)\right]$, and then using the identity exactly as in (i). (For $r = 0$ both ends equal 1.) So

$$(1-p)^{-\frac{3}{2}} = \sum_{r=0}^{\infty} (2r+1) \binom{2r}{r} \left(\frac{p}{4}\right)^r \qquad (|p| < 1).$$

Now we need $p/4 = \frac{1}{5}$, that is $p = \frac{4}{5}$, again with $|p| < 1$. Hence

$$\sum_{r=0}^{\infty} \frac{(2r+1)\binom{2r}{r}}{5^r} = \left(1 - \tfrac{4}{5}\right)^{-\frac{3}{2}} = \left(\tfrac{1}{5}\right)^{-\frac{3}{2}} = 5^{\frac{3}{2}} = \left(\sqrt{5}\right)^3,$$

as required.

Alternatively, write $f(x) = \sum \binom{2r}{r} x^r = (1-4x)^{-1/2}$ for $|x| < \frac{1}{4}$. Differentiating the power series term by term inside its interval of convergence, $\sum r\binom{2r}{r} x^r = x f'(x) = 2x(1-4x)^{-3/2}$, so

$$\sum (2r+1) \binom{2r}{r} x^r = 4x(1-4x)^{-\frac{3}{2}} + (1-4x)^{-\frac{1}{2}} = (1-4x)^{-\frac{3}{2}},$$

and $x = \frac{1}{5}$ gives the same answer.

Checks

- The first few terms of (i) are $1+\frac{2}{8}+\frac{6}{64}+\frac{20}{512}\approx 1.371$, already close to $\sqrt{2}\approx 1.414$; summing 200 terms gives $1.41421356\ldots$.
- In (ii) the series converges more slowly (ratio of terms tends to $\frac{4}{5}$), but 400 terms give $11.18033988\ldots$, matching $5\sqrt{5}\approx 11.1803$.
- The coefficients $\binom{2r}{r}/4^r$ for $r=0,1,2,3$ are $1, \frac{1}{2}, \frac{3}{8}, \frac{5}{16}$, agreeing with the four terms found directly.

Common slips

- Losing track of signs: the r negative factors in the coefficient and the $(-1)^r$ from $(-p)^r$ always cancel, so every coefficient is positive.
- Forgetting to check that the chosen p satisfies $|p|<1$; without it, the series need not converge to the function.
- In (ii), using $(1-p)^{-1/2}$ again with a different p: the factor $2r+1$ cannot come from that series alone.

Beyond STEP

The function $\sum\binom{2r}{r}x^r=(1-4x)^{-1/2}$ is the generating function of the central binomial coefficients, a standard tool in combinatorics; for example, squaring it gives $\sum_r 4^r x^r$, which proves $\sum_k \binom{2k}{k}\binom{2r-2k}{r-k}=4^r$. Justifying term-by-term differentiation, and knowing exactly where a power series converges, are first-year analysis topics. See Abbott, *Understanding Analysis*, for power series, and Siklos, *Advanced Problems in Mathematics*, for more STEP practice with series.

WORKED SOLUTION

STEP III 2003, Question 3

Pure • Curve Sketching

STEP III 2003, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q3

The key idea

Write $u = (1+x)^m$ and $v = (1-x)^m$. Then $\mathrm{f} = (u-v)/(u+v)$, and the quotient rule collapses beautifully because $u'v - uv'$ has a common factor $(1-x^2)^{m-1}$. That derivative is never negative, which tells us almost everything about the shape of $y = \mathrm{f}(x)$. For the reciprocal curve there is a hidden symmetry: replacing x by $1/x$ turns f into $1/\mathrm{f}$.

Solution

The denominator is never zero

Suppose $(1+x)^m + (1-x)^m = 0$ for some real x.

If m is even, both terms are non-negative, so both must be zero. That needs $x = -1$ and $x = 1$ at the same time, which is impossible.

If m is odd, then $(1+x)^m = -(1-x)^m = (x-1)^m$. The map $t \mapsto t^m$ is strictly increasing on $\mathbb{R}$ when m is odd, so it is one-to-one, and $1 + x = x - 1$, that is, $1 = -1$. This is a contradiction.

So $(1+x)^m + (1-x)^m \neq 0$ for every real x, and f is defined and differentiable on all of $\mathbb{R}$.

The derivative

With $u = (1+x)^m$ and $v = (1-x)^m$ we have $u' = m(1+x)^{m-1}$ and $v' = -m(1-x)^{m-1}$. By the quotient rule,

$$\mathrm{f}'(x) = \frac{(u'-v')(u+v) - (u-v)(u'+v')}{(u+v)^2} = \frac{2(u'v - uv')}{(u+v)^2}.$$

The numerator simplifies:

$$\begin{aligned} u'v - uv' &= m(1+x)^{m-1}(1-x)^m + m(1+x)^m(1-x)^{m-1} \\ &= m(1+x)^{m-1}(1-x)^{m-1}\big[(1-x)+(1+x)\big] \\ &= 2m(1-x^2)^{m-1}. \end{aligned}$$

Therefore

$$\mathrm{f}'(x) = \frac{4m\,(1-x^2)^{m-1}}{\big[(1+x)^m + (1-x)^m\big]^2}.$$

The case $m = 5$: the curve $y = \mathrm{f}(x)$

Expanding with the binomial theorem, the even powers cancel in the numerator and the odd powers cancel in the denominator:

$$\mathrm{f}(x) = \frac{2(5x + 10x^3 + x^5)}{2(1 + 10x^2 + 5x^4)} = \frac{x(x^4 + 10x^2 + 5)}{5x^4 + 10x^2 + 1}, \qquad \mathrm{f}'(x) = \frac{20(1-x^2)^4}{\big[(1+x)^5 + (1-x)^5\big]^2}.$$

Now we read off the features.

- *Symmetry.* $\mathrm{f}(-x) = -\mathrm{f}(x)$, so the curve has rotational symmetry about the origin. It passes through $(0,0)$ with gradient $\mathrm{f}'(0) = 20/2^2 = 5$.
- *Monotonic.* $\mathrm{f}'(x) \geq 0$ for all x, with equality only at $x = \pm 1$. So f is increasing everywhere. At $x = \pm 1$ it has stationary points, and since $(1-x^2)^4$ does not change sign these are points of inflection, not turning points. They are at $(1,1)$ and $(-1,-1)$: the $(1-x)^5$ or $(1+x)^5$ term vanishes, leaving $\mathrm{f} = \pm 1$.
- *Large $|x|$.* $\mathrm{f}(x) \approx x/5$. More precisely,

  $$\mathrm{f}(x) - \frac{x}{5} = \frac{x(40x^2 + 24)}{5(5x^4 + 10x^2 + 1)},$$

  which tends to 0 as $|x| \to \infty$. So $y = x/5$ is an oblique asymptote. The curve lies above it for $x > 0$ and below it for $x < 0$.

Ask yourself Where is $\mathrm{f}(x)$ bigger than 1 in size, and where is it smaller?

Since f is increasing with $\mathrm{f}(\pm 1) = \pm 1$, we have $|\mathrm{f}(x)| < 1$ for $|x| < 1$ and $|\mathrm{f}(x)| > 1$ for $|x| > 1$. This matters for the reciprocal. (For $|x| < 1$, putting $x = \tanh t$ gives $(1+x)/(1-x) = \mathrm{e}^{2t}$, and dividing top and bottom of f by $(1-x)^5$ shows $\mathrm{f}(\tanh t) = \tanh 5t$. That is another way to see the S-shape between -1 and 1.)

The curve $y = 1/\mathrm{f}(x)$

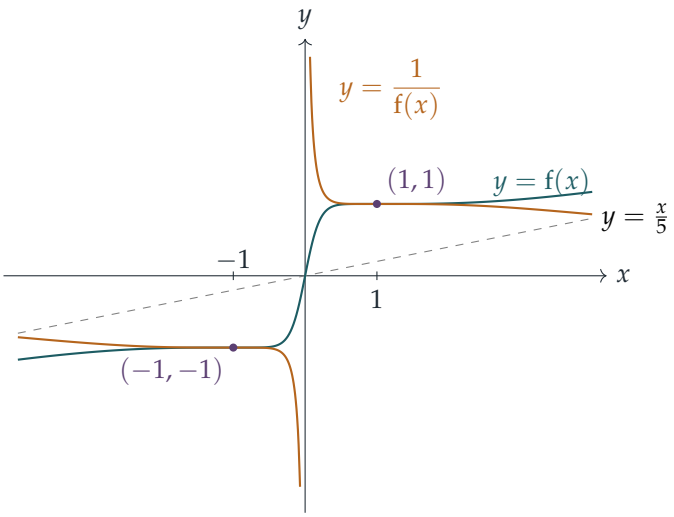
Replacing x by $1/x$ and multiplying top and bottom by x^5 gives

$$\mathrm{f}(1/x) = \frac{(x+1)^5 - (x-1)^5}{(x+1)^5 + (x-1)^5} = \frac{(1+x)^5 + (1-x)^5}{(1+x)^5 - (1-x)^5} = \frac{1}{\mathrm{f}(x)},$$

using $(x-1)^5 = -(1-x)^5$. So $y = 1/\mathrm{f}(x)$ is the curve $y = \mathrm{f}(1/x)$, and its features follow from those of f:

- It is odd, and undefined at $x = 0$, the only zero of f. Near 0, $1/\mathrm{f}(x) \approx 1/(5x)$, so the y-axis is a vertical asymptote, with $y \to +\infty$ as $x \to 0^+$ and $y \to -\infty$ as $x \to 0^-$.
- Its derivative is $-\mathrm{f}'(x)/\mathrm{f}(x)^2 \le 0$, so it is decreasing on $x > 0$ and on $x < 0$. It has stationary points of inflection at $(1,1)$ and $(-1,-1)$.
- As $|x| \to \infty$, $1/\mathrm{f}(x) \approx 5/x \to 0$, so the x-axis is an asymptote. The curve lies above it for $x > 0$ and below it for $x < 0$.

The two curves meet only where $\mathrm{f}(x)^2 = 1$, that is, at $(1,1)$ and $(-1,-1)$. Both curves are horizontal there, so they touch rather than cross.



Checks

- Take $m = 1$: the formula gives $\mathrm{f}'(x) = 4/2^2 = 1$, and indeed $\mathrm{f}(x) = 2x/2 = x$.
- Numerically, $\mathrm{f}(1000)/1000 \approx 0.2000016$, consistent with the asymptote $y = x/5$. A central-difference derivative agrees with the formula for f′ at random points for $m = 1, \ldots, 8$.
- $\mathrm{f}(\tanh t) = \tanh 5t$ and $\mathrm{f}(1/x) = 1/\mathrm{f}(x)$ both check out to 10^{-9} at random points.

Common slips

- Calling $x = \pm 1$ maxima or minima. The derivative touches zero there but does not change sign, so these are stationary points of inflection.
- Missing the oblique asymptote: f is a quintic over a quartic, so it grows like $x/5$ rather than levelling off.

- Drawing $y = 1/\mathrm{f}(x)$ crossing $y = \mathrm{f}(x)$ at $(\pm 1, \pm 1)$. Both curves are flat there, so they touch.

Beyond STEP

The substitution $x = \tanh t$ shows that f is the formula for $\tanh 5t$ in terms of $\tanh t$, the hyperbolic version of the multiple-angle formula for $\tan 5\theta$. Seen this way, f is a rational map that conjugates to multiplication by 5. Maps like this, and how they behave under iteration, are a starting point of complex dynamics. Needham's *Visual Complex Analysis* is a beautiful place to see how such rational functions act on the whole complex plane.

WORKED SOLUTION

STEP III 2003, Question 4

Pure • Tangents to Parametric Curves

STEP III 2003, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q4

The key idea

A straight line meets this curve where a *cubic* in the parameter vanishes. A tangent gives a double root at t, so the sum of the roots tells us the third root T at once. The formula for T then looks just like the double-angle formula for tan, which suggests writing $t = \tan\theta$: the map $t \mapsto T$ becomes a simple linear map on angles, and iterating it is easy.

Solution

The formula for T

Since $y = t + t^3$ is strictly increasing in t, different parameters give different points, so points of the curve and values of the parameter correspond one-to-one.

Differentiating, $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 2t$ and $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 1 + 3t^2$, so for $t \neq 0$ the tangent at t has gradient

$$m = \frac{1 + 3t^2}{2t}.$$

(The tangent isn't vertical, because $t \neq 0$.) Its equation is $y - (t + t^3) = m(x - t^2)$. The point with parameter s lies on it exactly when

$$s + s^3 - (t + t^3) = m(s^2 - t^2), \qquad \text{that is} \qquad f(s) \equiv s^3 - ms^2 + s - \left(t + t^3 - mt^2\right) = 0.$$

Clearly $f(t) = 0$, and $f'(t) = 3t^2 - 2mt + 1 = 0$ by the choice of m. So $s = t$ is (at least) a double root of this cubic, and the roots are $t,\ t,\ T$ for some real T. The sum of the roots of $s^3 - ms^2 + \ldots$ is m, so

$$2t + T = \frac{1 + 3t^2}{2t}$$

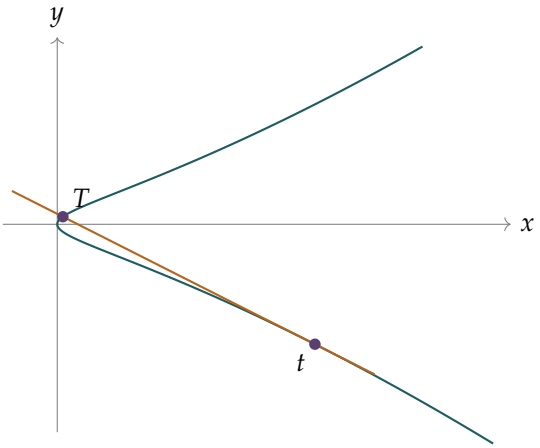
$$T = \frac{1 + 3t^2 - 4t^2}{2t} = \frac{1 - t^2}{2t}.$$

This root T is the only other place the line can meet the curve.

The tangent meets the curve *again* only if $T \neq t$. Now

$$T = t \iff 1 - t^2 = 2t^2 \iff 3t^2 = 1,$$

so we need $3t^2 \neq 1$. (When $3t^2 = 1$ the root t is triple and the tangent touches the curve only once: these are the points of inflection.) This proves both results.



The sequence with $t_0 = \tan\frac{7}{18}\pi$

Write $t = \tan\theta$. By the double-angle formula,

$$T = \frac{1 - \tan^2\theta}{2\tan\theta} = \frac{1}{\tan 2\theta} = \cot 2\theta = \tan\Big(\frac{\pi}{2} - 2\theta\Big).$$

So if $t_i = \tan\theta_i$ then we may take $\theta_{i+1} = \frac{\pi}{2} - 2\theta_i$. Since tan has period π and is one-to-one on any interval of length π (missing the odd multiples of $\frac{\pi}{2}$), two parameters are equal exactly when their angles differ by a multiple of π.

Ask yourself Before iterating, what must we check about each t_i so that the next point exists?

The construction needs $t_i \neq 0$ and $3t_i^2 \neq 1$ at every step, that is, θ_i is not a multiple of $\frac{\pi}{2}$ and not $\pm\frac{\pi}{6}$ plus a multiple of π. In multiples of $\frac{\pi}{18}$, we must avoid $0, \pm 3, \pm 9$ modulo 18.

Starting from $\theta_0 = \frac{7\pi}{18}$:

$$\begin{aligned}
\theta_1 &= \frac{9\pi}{18} - \frac{14\pi}{18} = -\frac{5\pi}{18}, \\
\theta_2 &= \frac{9\pi}{18} + \frac{10\pi}{18} = \frac{19\pi}{18} \equiv \frac{\pi}{18} \pmod{\pi}, \\
\theta_3 &= \frac{9\pi}{18} - \frac{2\pi}{18} = \frac{7\pi}{18}.
\end{aligned}$$

None of $7, -5, 1$ is forbidden, so every step is valid. Hence $t_3 = \tan\frac{7\pi}{18} = t_0$, so $P_3 = P_0$. But $\theta_1 - \theta_0 = -\frac{12\pi}{18}$ is not a multiple of π, so $t_1 = \tan(-\frac{5\pi}{18}) < 0 < t_0$ and $P_1 \neq P_0$. The three points have parameters $\tan 70^\circ$, $-\tan 50^\circ$ and $\tan 10^\circ$.

A second starting value

Write $g(\theta) = \frac{\pi}{2} - 2\theta$. Iterating,

$$g^2(\theta) = -\frac{\pi}{2} + 4\theta, \qquad g^3(\theta) = \frac{3\pi}{2} - 8\theta.$$

So $P_3 = P_0$ exactly when $\frac{3\pi}{2} - 8\theta \equiv \theta \pmod{\pi}$, i.e. $9\theta \equiv \frac{3\pi}{2} \pmod{\pi}$, which gives

$$\theta = \frac{\pi}{6} + \frac{k\pi}{9} = \frac{(3+2k)\pi}{18}, \qquad k \in \mathbb{Z}.$$

Similarly $P_1 = P_0$ exactly when $3\theta \equiv \frac{\pi}{2} \pmod{\pi}$, i.e. $\theta = \frac{(3+6j)\pi}{18}$. These are exactly the cases $3 + 2k \equiv 3, 9, 15 \pmod{18}$: they give $t = \pm\frac{1}{\sqrt{3}}$ or no finite t at all, and are excluded anyway.

What remains, modulo π, is $\theta \in \{\frac{\pi}{18}, \frac{5\pi}{18}, \frac{7\pi}{18}, \frac{11\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}\}$. For $t_0 > 0$ we need $0 < \theta < \frac{\pi}{2}$, giving $\tan\frac{\pi}{18}$, $\tan\frac{5\pi}{18}$ and the given $\tan\frac{7\pi}{18}$. The value $\tan\frac{\pi}{18}$ is just P_2 of the cycle above. The value that gives a genuinely new cycle is

$$t_0 = \tan\frac{5\pi}{18} \ (= \tan 50^\circ),$$

with $\theta_0 = \frac{5\pi}{18}$, $\theta_1 = -\frac{\pi}{18}$, $\theta_2 = \frac{11\pi}{18}$, $\theta_3 = \frac{9\pi}{18} - \frac{22\pi}{18} = -\frac{13\pi}{18} \equiv \frac{5\pi}{18}$. None of $5, -1, 11$ is forbidden, and $t_1 = -\tan 10^\circ \neq t_0$. Either $\tan\frac{5\pi}{18}$ or $\tan\frac{\pi}{18}$ answers the question.

? **Ask yourself** Why is it legitimate to work with angles "modulo π"?

Because the only thing we use is $t = \tan\theta$, and $\tan(\theta + \pi) = \tan\theta$. Adding π to θ_i changes $g(\theta_i)$ by 2π, which doesn't change t_{i+1}. So the whole sequence of parameters depends only on θ_0 modulo π.

Checks

- With $t = -1$ we get $T = 0$: the tangent at $(1, -2)$ has gradient -2, and it does pass through the origin.
- Numerically, $t_0 = \tan 70^\circ \approx 2.747477$ gives $t_1 \approx -1.191754$, $t_2 \approx 0.176327$ and $t_3 \approx 2.747477$. Likewise $\tan 50^\circ \approx 1.191754 \to -0.176327 \to -2.747477 \to 1.191754$.
- Solving the cubic for the other intersection numerically, for random t, agrees with $T = (1 - t^2)/(2t)$.

Common slips

- Finding the gradient of the chord and equating, then forgetting to cancel the factor $T - t$. Vieta's formula for the sum of the roots avoids this altogether.
- Iterating the map on t directly with decimals, which hides the exact cycle. The substitution $t = \tan\theta$ is what makes $P_3 = P_0$ exact.
- Giving $\tan\frac{\pi}{6}$ as the second value: there $T = t$, so there is no next point at all.

Beyond STEP

The chord-and-tangent construction is the heart of the theory of elliptic curves, where the same "third intersection" rule turns the points of a cubic curve into a group. Here the curve $y^2 = x(1+x)^2$ is a singular cubic (it has an isolated point at $(-1, 0)$), and the substitution $t = \tan\theta$ shows that the tangent map is really angle-doubling in disguise. That's why the cycles come from dividing a circle into nine equal parts. Silverman and Tate, *Rational Points on Elliptic Curves*, develops this beautifully at undergraduate level.

WORKED SOLUTION

STEP III 2003, Question 5

Pure • Roots of Cubic Equations

STEP III 2003, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q5

The key idea

Count roots by looking at the graph. A quadratic with positive leading coefficient crosses the x-axis twice exactly when its minimum lies below the axis. A cubic with positive leading coefficient crosses three times exactly when its maximum is above the axis and its minimum below. To make all the roots positive, look at which side of the y-axis the turning points sit, and at the sign of the curve where it meets the y-axis. The last equation is a disguised case of the shifted cubic: complete the cube to bring it to that form.

Solution

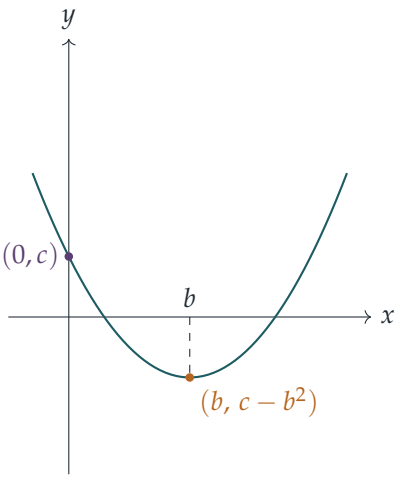
The quadratic

Completing the square,

$$y = x^2 - 2bx + c = (x-b)^2 + c - b^2,$$

so the turning point (a minimum) is at $(b,\ c-b^2)$.

When there are two distinct real roots, the curve is a $\cup$-shaped parabola, symmetric about $x = b$, whose vertex lies below the x-axis. It meets the y-axis at $(0, c)$.



Two distinct real roots. The parabola has positive leading coefficient, so $y \to +\infty$ as $x \to \pm\infty$, and its smallest value is $c - b^2$. If $c - b^2 < 0$, the curve dips below the axis and comes back up on each side of $x = b$, so it crosses twice. If $c - b^2 = 0$ it just touches the axis (a repeated root), and if $c - b^2 > 0$ it never meets it. So the condition is

$$b^2 > c.$$

Two distinct positive roots. We claim the conditions are

$$b^2 > c, \qquad b > 0, \qquad c > 0.$$

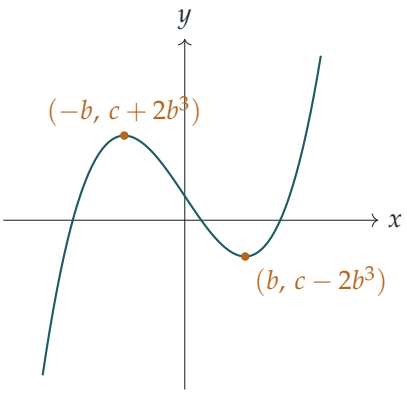
Necessary: the roots α, β satisfy $\alpha + \beta = 2b$ and $\alpha\beta = c$, and if both are positive then both of these are positive. **Sufficient:** if $b^2 > c$ there are two roots, one on each side of $x = b$. Since $b > 0$, the right-hand root is positive. The left-hand root lies in the interval where the curve is decreasing; the curve is positive at $x = 0$ (height $c > 0$) and negative at $x = b > 0$, so this root lies strictly between 0 and b, and is positive too.

The cubic

For $y = x^3 - 3b^2x + c$ we have $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 3b^2 = 3(x - b)(x + b)$, which is zero at $x = \pm b$. Substituting,

$$x = -b:\ y = -b^3 + 3b^3 + c = c + 2b^3, \qquad x = b:\ y = b^3 - 3b^3 + c = c - 2b^3.$$

Since $b > 0$, the turning points are a maximum at $(-b,\ c + 2b^3)$ and a minimum at $(b,\ c - 2b^3)$: the derivative is positive for $x < -b$, negative between, and positive for $x > b$.



The curve rises from $-\infty$ to the maximum, falls to the minimum, then rises to $+\infty$, and is strictly monotonic on each of these three pieces. So it has one root on each piece, giving three distinct roots, exactly when the maximum is above the axis and the minimum below:

$$c+2b^3>0 \text{ and } c-2b^3<0, \qquad \text{that is,} \qquad -2b^3<c<2b^3.$$

If either turning value is zero we get a repeated root, and if both have the same strict sign there is only one real root. So $|c|<2b^3$ is necessary and sufficient.

Three distinct positive roots of the shifted cubic

The curve $y=(x-a)^3-3b^2(x-a)+c$ is the previous curve translated a to the right (we keep $b>0$). So it has three distinct real roots if and only if $|c|<2b^3$, and then its maximum is at $x=a-b$, its minimum at $x=a+b$, and the roots $x_1<x_2<x_3$ satisfy

$$x_1<a-b<x_2<a+b<x_3.$$

All three are positive if and only if the smallest, x_1, is positive. Write

$$y(0)=(-a)^3-3b^2(-a)+c=c-a^3+3ab^2$$

for the height at which the curve crosses the y-axis. We claim that the conditions are

$$|c|<2b^3, \qquad a>b, \qquad c<a^3-3ab^2.$$

Ask yourself Why isn't "$y(0)<0$" enough on its own to put all three roots to the right of the origin?

Because y is also negative between x_2 and x_3. A negative value at $x=0$ could mean that the origin sits between the middle and largest roots. We must also know that 0 lies on the first rising piece of the curve, to the left of the maximum. That is the role of $a>b$.

Necessary. Suppose there are three distinct positive roots. Then $|c|<2b^3$ from the previous part. Also $0<x_1<a-b$, so $a>b$. Finally, y is strictly increasing on $(-\infty,a-b]$ and $y(x_1)=0$, so $y(0)<0$ because $0<x_1$. This gives $c<a^3-3ab^2$.

Sufficient. Suppose all three conditions hold. There are three distinct real roots, and $0 < a - b$, so 0 lies in $(-\infty, a-b]$, where y is strictly increasing. Since $y(0) < 0 = y(x_1)$, we get $0 < x_1$, and hence all three roots are positive.

So the equation has three distinct positive roots if and only if

$$a > b \quad \text{and} \quad -2b^3 < c < \min\big(2b^3,\ a^3 - 3ab^2\big).$$

(Since $a^3 - 3ab^2 + 2b^3 = (a-b)^2(a+2b) > 0$ when $a > b > 0$, these bounds on c are consistent.)

The final equation

Divide by 2 and complete the cube. The x^2 term in $(x-a)^3$ is $-3ax^2$, so we need $3a = \frac{9}{2}$, that is, $a = \frac{3}{2}$. Putting $x = t + \frac{3}{2}$,

$$\begin{aligned} x^3 - \tfrac{9}{2}x^2 + \tfrac{7}{2}x - \tfrac{1}{2} &= t^3 + \big(\tfrac{27}{4} - \tfrac{27}{2} + \tfrac{7}{2}\big)t + \big(\tfrac{27}{8} - \tfrac{81}{8} + \tfrac{21}{4} - \tfrac{1}{2}\big) \\ &= t^3 - \tfrac{13}{4}t - 2, \end{aligned}$$

where the t^2 terms cancel by the choice of a. So

$$2x^3 - 9x^2 + 7x - 1 = 0 \iff \big(x - \tfrac{3}{2}\big)^3 - 3b^2\big(x - \tfrac{3}{2}\big) + c = 0, \qquad a = \tfrac{3}{2},\ b^2 = \tfrac{13}{12},\ c = -2,$$

with $b = \sqrt{13/12} > 0$. Now we check the three conditions.

- $|c| < 2b^3$: this says $2 < 2b^3$, i.e. $b^2 > 1$, which holds because $b^2 = \frac{13}{12} > 1$.
- $a > b$: both are positive and $a^2 = \frac{9}{4} = \frac{27}{12} > \frac{13}{12} = b^2$.
- $c < a^3 - 3ab^2$: here $a^3 - 3ab^2 = a(a^2 - 3b^2) = \frac{3}{2}\big(\frac{9}{4} - \frac{13}{4}\big) = -\frac{3}{2}$, and $-2 < -\frac{3}{2}$.

All three hold, so $2x^3 - 9x^2 + 7x - 1 = 0$ has three distinct positive roots, as required.

Checks

- Sign changes confirm the final answer directly. With $p(x) = 2x^3 - 9x^2 + 7x - 1$ we have $p(0) = -1$, $p(\frac{1}{2}) = \frac{1}{2}$, $p(1) = -1$, $p(3) = -7$ and $p(4) = 11$. So there are roots in $(0, \frac{1}{2})$, $(\frac{1}{2}, 1)$ and $(3, 4)$. Numerically they are 0.185, 0.760 and 3.555.
- The value $y(0) = c - a^3 + 3ab^2 = -2 + \frac{3}{2} = -\frac{1}{2}$ matches the constant term of the monic cubic $x^3 - \frac{9}{2}x^2 + \frac{7}{2}x - \frac{1}{2}$, as it must.
- A random test of thousands of choices of (a, b, c) agreed with the stated conditions every time.

Common slips

- Using only $y(0) < 0$ (or only $a > b$) for positivity. Both are needed: $y(0) < 0$ also holds when the origin lies between the two larger roots.

- Forgetting that equality in $|c| \leq 2b^3$ gives a repeated root, so it is not "three distinct roots".
- Getting the shift the wrong way round: the curve in $x - a$ is the original curve moved a to the *right*.

Beyond STEP

Any cubic can be moved into the "depressed" form $t^3 + pt + q$ by a shift, as here. It then has three distinct real roots exactly when $4p^3 + 27q^2 < 0$; with $p = -3b^2$, $q = c$ this is the condition $c^2 < 4b^6$ found above. The quantity $-(4p^3 + 27q^2)$ is the *discriminant*, and it is the start of the story of how cubics were solved (Cardano's formula) and, eventually, of Galois theory. Allenby's *Rings, Fields and Groups* is a good next step towards that theory.

WORKED SOLUTION

STEP III 2003, Question 6

Pure • Telescoping Trigonometric Sums

STEP III 2003, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q6

The key idea

A sum of cosines whose angles go up in equal steps is awkward as it stands, but multiplying every term by $2\sin\frac{1}{2}\theta$ turns each cosine into a *difference* of two sines, and the sum then telescopes. What is left is a difference of two sines, which the factor formula turns into a product. Setting a product to zero is easy. The one thing to watch is that we multiplied by $\sin\frac{1}{2}\theta$, which can itself be zero.

Solution

The identity

Using $\sin(A+B)-\sin(A-B)=2\cos A\sin B$ with $A=r\theta$ and $B=\frac{1}{2}\theta$:

$$\sin\left(r+\tfrac{1}{2}\right)\theta-\sin\left(r-\tfrac{1}{2}\right)\theta=2\cos r\theta\,\sin\tfrac{1}{2}\theta,$$

which is the required result.

Summing the series

Write

$$S(\theta)=\sum_{r=a}^{b-1}\cos r\theta.$$

Multiplying by $2\sin\frac{1}{2}\theta$ and using the identity for each r from a to $b-1$,

$$\begin{aligned}2\sin\tfrac{1}{2}\theta\;S(\theta)&=\sum_{r=a}^{b-1}\left[\sin\left(r+\tfrac{1}{2}\right)\theta-\sin\left(r-\tfrac{1}{2}\right)\theta\right]\\&=\sin\left(b-\tfrac{1}{2}\right)\theta-\sin\left(a-\tfrac{1}{2}\right)\theta,\end{aligned}$$

because each $\sin\left(r+\frac{1}{2}\right)\theta$ cancels the $-\sin\left((r+1)-\frac{1}{2}\right)\theta$ from the next term, leaving only the top end of the last term and the bottom end of the first.

Now we use $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$, with $\frac{P+Q}{2} = \frac{1}{2}(a+b-1)\theta$ and $\frac{P-Q}{2} = \frac{1}{2}(b-a)\theta$:

$$\sin\tfrac{1}{2}\theta\ S(\theta) = \cos\tfrac{1}{2}(a+b-1)\theta\ \sin\tfrac{1}{2}(b-a)\theta. \qquad (*)$$

Solving $S(\theta) = 0$

? **Ask yourself** Can we just divide $(*)$ by $\sin\frac{1}{2}\theta$?

Only when $\sin\frac{1}{2}\theta \neq 0$, that is, when θ is not a multiple of 2π. So we deal with those values first. If $\theta = 2m\pi$ for an integer m, then every term $\cos r\theta$ equals 1, so $S(\theta) = b - a \geq 2$. These values are never solutions.

For every other θ we have $\sin\frac{1}{2}\theta \neq 0$, and by $(*)$,

$$S(\theta) = 0 \iff \cos\tfrac{1}{2}(a+b-1)\theta = 0 \ \text{ or } \ \sin\tfrac{1}{2}(b-a)\theta = 0.$$

- $\cos\frac{1}{2}(a+b-1)\theta = 0$ exactly when $\frac{1}{2}(a+b-1)\theta = \left(k+\frac{1}{2}\right)\pi$, that is,

$$\theta = \frac{(2k+1)\pi}{a+b-1}, \qquad k \in \mathbb{Z}.$$

  (Note $a+b-1 \geq 3$, so we are not dividing by zero.) None of these is a multiple of 2π: that would need $2k+1 = 2m(a+b-1)$, an odd number equal to an even one. So all of them are solutions.

- $\sin\frac{1}{2}(b-a)\theta = 0$ exactly when $\frac{1}{2}(b-a)\theta = k\pi$, that is,

$$\theta = \frac{2k\pi}{b-a}, \qquad k \in \mathbb{Z}.$$

  This is a multiple of 2π exactly when $b-a$ divides k, and we have already seen that those values fail. So we must exclude them.

? **Ask yourself** Why must the multiples of 2π be thrown out of the second family, when $(*)$ seems to be satisfied there?

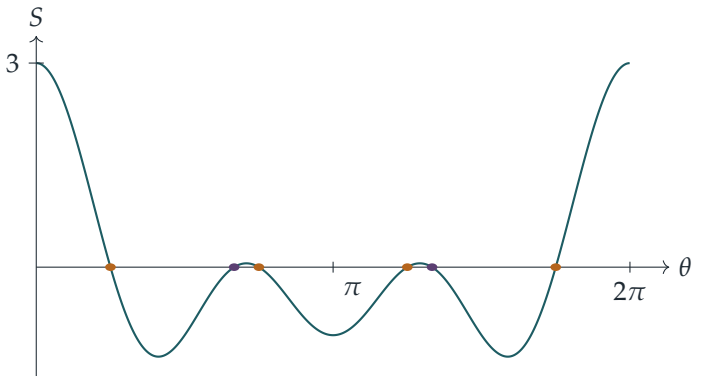
At $\theta = 2m\pi$ both sides of $(*)$ are zero whatever S is, so $(*)$ tells us nothing about S there. It was a false lead created by multiplying through by $\sin\frac{1}{2}\theta$. Since $b-a \geq 2$, the second family still contains genuine solutions, such as $\theta = 2\pi/(b-a)$.

Putting this together, the solutions are exactly

$$\boxed{\theta = \frac{(2k+1)\pi}{a+b-1}\ (k \in \mathbb{Z}) \qquad \text{or} \qquad \theta = \frac{2k\pi}{b-a}\ (k \in \mathbb{Z},\ b-a \nmid k).}$$

The two families can overlap. That does no harm, since a value in both is still just one solution.

For example, with $a = 1$ and $b = 4$ the equation is $\cos\theta + \cos 2\theta + \cos 3\theta = 0$. In $(0, 2\pi)$ its roots are $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ from the first family and $\frac{2\pi}{3}, \frac{4\pi}{3}$ from the second. Note that $\theta = 0$ and $\theta = 2\pi$, where $S = 3$, are left out.



The curve $S = \cos\theta + \cos 2\theta + \cos 3\theta$: the amber roots come from the cosine factor and the plum roots from the sine factor.

Checks

- With $a = 1$, $b = 4$: at $\theta = \frac{\pi}{4}$, $\frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} = 0$, and at $\theta = \frac{2\pi}{3}$, $-\frac{1}{2} - \frac{1}{2} + 1 = 0$.
- Numerically, $(*)$ holds at random θ for many pairs a, b. For every $1 \le a \le 6$, $a + 2 \le b \le 11$, a fine scan of S on $(0, 6\pi)$ finds exactly the roots given by the formula and no others. For instance, with $a = 1$, $b = 3$ the root $\theta = 13\pi/3 \approx 13.6136$ comes from the first family.

Common slips

- Including $\theta = 2m\pi$ by dividing $(*)$ by $\sin\frac{1}{2}\theta$ without checking; there $S = b - a \neq 0$.
- Getting the end terms of the telescoping sum wrong: they are $\sin\left(b - \frac{1}{2}\right)\theta$ and $\sin\left(a - \frac{1}{2}\right)\theta$, because the last cosine is $\cos(b-1)\theta$.
- Giving only solutions in one interval such as $[0, 2\pi)$, when the question asks for all of them.

Beyond STEP

The closed form for $\sum \cos r\theta$ is the real part of a geometric series in $e^{i\theta}$. With $a = -n$ and $b = n + 1$ it gives the *Dirichlet kernel* $\sin\left(n + \frac{1}{2}\right)\theta / \sin\frac{1}{2}\theta$, which is at the heart of the theory of Fourier series and of whether they converge. Needham's *Visual Complex Analysis* is a good place to see why trigonometric sums like this become simple once you work with $e^{i\theta}$.

WORKED SOLUTION

STEP III 2003, Question 7

Pure • Simson Line

STEP III 2003, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q7

The key idea

The foot of a perpendicular is found by walking from P along the normal to the line until we hit it, so write $N = P + \lambda(\text{normal})$ and solve for λ. For the second part, the condition on s and t is a product of two gradients equal to -1: it says that PA and PB are perpendicular. Turned into a circle equation, it is exactly what makes the collinearity check collapse.

Solution

The coordinates of N

The line AB has equation $\dfrac{x}{a} + \dfrac{y}{b} = 1$, that is

$$bx + ay = ab.$$

A normal vector to this line is (b, a), so N, the foot of the perpendicular from P, has the form

$$N = (s + \lambda b,\ t + \lambda a)$$

for some λ. Since N lies on AB,

$$b(s + \lambda b) + a(t + \lambda a) = ab \quad \Longrightarrow \quad \lambda = \frac{ab - bs - at}{a^2 + b^2}.$$

Hence

$$x = s + \lambda b = \frac{s(a^2 + b^2) + ab^2 - b^2 s - abt}{a^2 + b^2} = \frac{ab^2 - abt + a^2 s}{a^2 + b^2} = \frac{ab^2 - a(bt - as)}{a^2 + b^2},$$

$$y = t + \lambda a = \frac{t(a^2 + b^2) + a^2 b - abs - a^2 t}{a^2 + b^2} = \frac{a^2 b + b^2 t - abs}{a^2 + b^2} = \frac{a^2 b + b(bt - as)}{a^2 + b^2},$$

as required.

N lies on XY

First, what the condition says. For it to make sense we need $s \neq 0$ and $s \neq a$, and since the product is -1 rather than 0, also $t \neq 0$ and $t \neq b$. Multiplying out,

$$t(t-b) = -s(s-a) \quad \Longleftrightarrow \quad s^2 + t^2 = as + bt. \qquad (*)$$

Now $X = (s, 0)$ and $Y = (0, t)$. As $s \neq 0$ and $t \neq 0$, these are distinct points off the origin, and the line XY is

$$\frac{x}{s} + \frac{y}{t} = 1, \qquad \text{that is} \qquad tx + sy = st.$$

So we must show that $tx + sy = st$ for the coordinates of N.

Ask yourself Should we substitute and hope, or is there a tidy way to organise the algebra?

Write $k = bt - as$ to keep things short. Using the formulae above,

$$\begin{aligned}(a^2 + b^2)(tx + sy) &= t(ab^2 - ak) + s(a^2 b + bk) \\ &= ab^2 t + a^2 bs + k(bs - at).\end{aligned}$$

The last product expands neatly:

$$k(bs - at) = (bt - as)(bs - at) = b^2 st - abt^2 - abs^2 + a^2 st = (a^2 + b^2)st - ab(s^2 + t^2).$$

Therefore

$$(a^2 + b^2)(tx + sy) = (a^2 + b^2)st + ab\big(as + bt - s^2 - t^2\big).$$

This holds for every P. By $(*)$ the bracket is zero, so $tx + sy = st$ and N lies on the line XY.

Geometrical interpretation

The gradient of PB is $\dfrac{t-b}{s}$ and the gradient of PA is $\dfrac{t}{s-a}$, so the condition says that $PA \perp PB$: the angle APB is a right angle. Equivalently, $(*)$ is

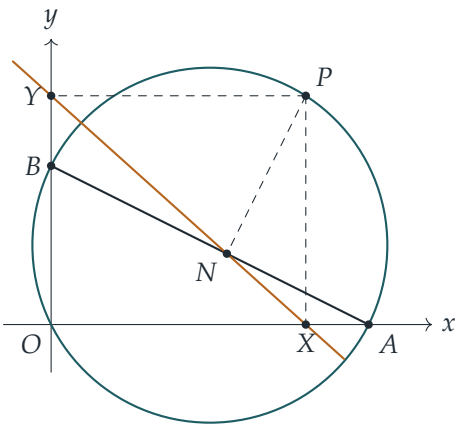
$$\left(s - \tfrac{a}{2}\right)^2 + \left(t - \tfrac{b}{2}\right)^2 = \frac{a^2 + b^2}{4},$$

so P lies on the circle with diameter AB. Since the angle AOB is also a right angle, this circle passes through O: it is the *circumcircle of triangle OAB*.

The points X, Y and N are the feet of the perpendiculars from P to the three sides OA, OB and AB of the triangle. So the result says:

If P lies on the circumcircle of a triangle, the feet of the perpendiculars from P to the three sides (extended if necessary) are collinear.

This is the *Simson line* theorem, here proved for a right-angled triangle placed with its right angle at the origin.



With $a = 4$, $b = 2$: P on the circle through O, A, B; the line XY (amber) passes through N.

Checks

- Put $P = O$, i.e. $s = t = 0$: the formula gives $N = \left(\frac{ab^2}{a^2+b^2}, \frac{a^2b}{a^2+b^2}\right)$, which lies on $bx + ay = ab$ and on the line $y = \frac{a}{b}x$ through O perpendicular to AB, as it should.
- With $a = 4$, $b = 2$ and $P = (2 + \sqrt{5}\cos 1,\ 1 + \sqrt{5}\sin 1) \approx (3.208, 2.882)$ on the circle, the formula gives $N \approx (2.214, 0.893)$, and $\frac{2.214}{3.208} + \frac{0.893}{2.882} \approx 0.690 + 0.310 = 1$, so N is on XY.
- The identity $(a^2 + b^2)(tx + sy - st) = ab(as + bt - s^2 - t^2)$ also gives the converse: for P off the axes, N is on XY *only* if P is on the circle.

Common slips

- Using (a, b) as the normal to AB. The direction of AB is $(-a, b)$, so the normal is (b, a).
- Expanding everything at once and drowning. Grouping $bt - as$ as one symbol, as the given answer suggests, keeps the algebra to a few lines.
- Stopping at "angle APB is a right angle". The interpretation the question is after is the Simson line: X, Y, N are feet of perpendiculars to the three sides of triangle OAB.

Beyond STEP

The Simson line holds for any triangle, and a synthetic proof uses cyclic quadrilaterals and equal angles in the same segment. Its converse is also true, and as P moves round the circumcircle the Simson lines envelope a deltoid (Steiner's hypocycloid). Coordinate

and complex-number methods make such results mechanical; Needham, *Visual Complex Analysis*, shows how complex numbers handle circle geometry elegantly, and Coxeter and Greitzer, *Geometry Revisited*, covers the Simson line and much more classical triangle geometry.

WORKED SOLUTION

STEP III 2003, Question 8

Pure • First-Order Differential Equations

STEP III 2003, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q8

The key idea

Part (i) is implicit differentiation, but it is really a worked example of a trick: a family of curves $(y+ax)^n(y+bx)=c$ has a gradient that is a ratio of two linear expressions in x and y. Part (ii) runs the trick backwards. We differentiate the general family, read off the gradient in terms of a, b and n, and choose them to match the given equation. Once they match, the expression is constant along every solution curve, and that constant relationship is the general solution.

Solution

Part (i)

Differentiate $(y+2x)^3(y-4x)=c$ with respect to x, using the product rule and the chain rule, and write y' for $\frac{\mathrm{d}y}{\mathrm{d}x}$:

$$3(y+2x)^2(y'+2)(y-4x)+(y+2x)^3(y'-4)=0.$$

At points with $y+2x\neq 0$ we can divide by $(y+2x)^2$:

$$3(y'+2)(y-4x)+(y+2x)(y'-4)=0.$$

Collecting the terms in y',

$$\begin{aligned} y'\big[3(y-4x)+(y+2x)\big] &= -6(y-4x)+4(y+2x),\\ y'(4y-10x) &= 32x-2y, \end{aligned}$$

and dividing top and bottom by 2,

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{16x-y}{2y-5x},$$

as required. (On the line $y=-2x$ the curve has $c=0$, and there the formula still gives the right gradient, -2.)

Part (ii)

Let $F = (y+ax)^n(y+bx)$, where y is a function of x. Exactly as in part (i),

$$\begin{aligned}\frac{\mathrm{d}F}{\mathrm{d}x} &= n(y+ax)^{n-1}(y'+a)(y+bx) + (y+ax)^n(y'+b)\\ &= (y+ax)^{n-1}\Big[y'\big((n+1)y+(nb+a)x\big) + (na+b)y + (n+1)ab\,x\Big].\end{aligned}$$

So F is constant along a curve precisely when, away from $y=-ax$,

$$y' = -\frac{(na+b)y + (n+1)ab\,x}{(n+1)y + (nb+a)x}. \qquad (*)$$

(Part (i) is the case $a=2$, $b=-4$, $n=3$, which gives $(32x-2y)/(4y-10x)$.)

Ask yourself How do we match $(*)$ with $\dfrac{10x-4y}{3x-y}$, when a fraction can be written in many ways?

A fraction is unchanged when the top and bottom are multiplied by the same number, so we can only match the two up to a common factor. Write the target as $-(4y-10x)/(y-3x)$ and multiply the top and bottom by $k=n+1$, so that the coefficients of y in the denominators agree. We then need

$$na+b = -4(n+1), \qquad (n+1)ab = 10(n+1), \qquad nb+a = -3(n+1).$$

The middle equation gives $ab=10$, since $n+1\neq 0$ (if $n=-1$ the denominator of $(*)$ has no y term). Adding the outer two equations gives $(n+1)(a+b) = -7(n+1)$, so $a+b=-7$. Hence a and b are the roots of $t^2+7t+10=0$, namely -2 and -5. Subtracting the outer equations gives

$$(n-1)(a-b) = -(n+1).$$

- With $a=-5$, $b=-2$: $-3(n-1) = -(n+1)$, so $n=2$.
- With $a=-2$, $b=-5$: $3(n-1) = -(n+1)$, so $n=\frac{1}{2}$.

Both choices describe the same curves, since $(y-2x)^{1/2}(y-5x) = c$ squares to $(y-5x)^2(y-2x) = c^2$. We take $n=2$, which avoids square roots.

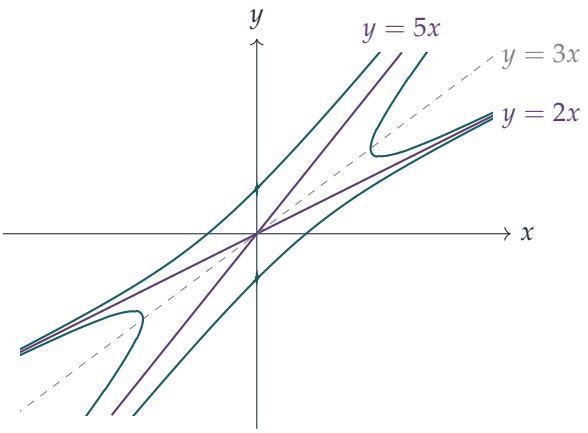
To be sure, check directly. With $G = (y-5x)^2(y-2x)$,

$$\begin{aligned}\frac{\mathrm{d}G}{\mathrm{d}x} &= (y-5x)\big[2(y'-5)(y-2x) + (y-5x)(y'-2)\big]\\ &= (y-5x)\big[3y'(y-3x) - 12y + 30x\big]\\ &= 3(y-5x)\big[y'(y-3x) - (4y-10x)\big].\end{aligned}$$

If y satisfies the differential equation, then $y'(y-3x) = 4y-10x$, so the bracket vanishes and $\mathrm{d}G/\mathrm{d}x = 0$. Hence G is constant on every solution curve, and the general solution is

$$\boxed{(y-5x)^2(y-2x) = C}$$

with C an arbitrary constant. The value $C = 0$ gives the straight lines $y = 5x$ and $y = 2x$. Both really are solutions: on $y = 2x$ the right-hand side is $2x/x = 2$, and on $y = 5x$ it is $(-10x)/(-2x) = 5$. The curves all have vertical tangents where they cross the line $y = 3x$, where the equation is not defined.



The curves $(y - 5x)^2(y - 2x) = \pm 1$ (teal) and the line solutions $C = 0$ (plum). The branches between the two lines have a vertical tangent where they meet the dashed line $y = 3x$.

An alternative for part (ii)

The right-hand side depends only on y/x, so the substitution $y = vx$ also works. It gives

$$x\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{(v-2)(v-5)}{3-v},$$

and separating the variables and using partial fractions leads to $\ln|x| = -\frac{1}{3}\ln|v-2| - \frac{2}{3}\ln|v-5| + \text{const}$, which rearranges to the same answer. The method in the question avoids the partial fractions.

Checks

- Part (i) contains a check of $(*)$: putting $a = 2$, $b = -4$, $n = 3$ in it reproduces the gradient found there.
- At the point $(1, 0)$ the equation gives gradient $10/3$. Differentiating $(y - 5x)^2(y - 2x) = C$ implicitly there gives $(-5)\big[2(y' - 5)(-2) + (-5)(y' - 2)\big] = 0$, so $-9y' + 30 = 0$ and $y' = 10/3$.
- Numerically, we stepped the differential equation with a Runge–Kutta method from fifty random starting points. The value of $(y - 5x)^2(y - 2x)$ stayed constant to about 10^{-13} along each path.

Common slips

- Matching the coefficients of $(*)$ with $10x - 4y$ and $3x - y$ directly, without the common factor, which gives inconsistent equations.
- Leaving the answer as $(y - 2x)^{1/2}(y - 5x) = c$ without noticing that it is the same family; it also loses the solutions with $y < 2x$ unless you are careful with signs.
- Forgetting that $C = 0$ is allowed, which gives the line solutions $y = 2x$ and $y = 5x$.

Beyond STEP

The equation here is *homogeneous*: the right-hand side depends only on y/x. Looking for a function $G(x, y)$ that is constant along solutions is the idea behind *exact equations* and *integrating factors*. In mechanics, the same idea is the search for conserved quantities such as energy. Both are covered early in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 2003, Question 9

Mechanics • Stability of Equilibrium

STEP III 2003, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q9

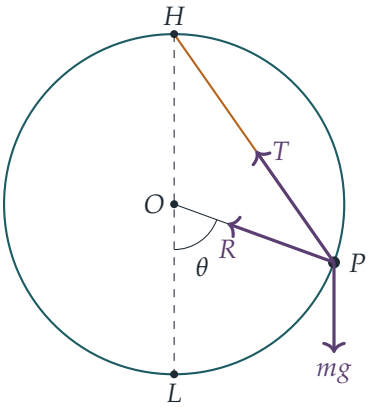
The key idea

The wire is smooth, so its reaction is always along the radius and does no work. Everything therefore happens in the *tangential* direction: resolve the tension and the weight along the tangent and you have a single equation for $\ddot{\theta}$ that gives both the equilibrium and the small oscillations. The geometry of the isosceles triangle OHP supplies all the angles.

Solution

Setting up

Model P as a particle on a smooth wire, so the reaction R from the wire acts along PO. The string is light and obeys Hooke's law. Take $0 < \theta < \pi$, so P is on one side of the vertical diameter LH.



Since $OH = OP = a$, triangle OHP is isosceles with $\angle HOP = \pi - \theta$, so its base angles

are

$$\angle OHP = \angle OPH = \tfrac{1}{2}\theta, \qquad HP = 2a\cos\tfrac{1}{2}\theta.$$

The string is taut when $HP > a$, that is, when $\cos\frac{1}{2}\theta > \frac{1}{2}$, and then Hooke's law gives

$$T = \frac{\alpha mg}{a}\left(2a\cos\tfrac{1}{2}\theta - a\right) = \alpha mg\left(2\cos\tfrac{1}{2}\theta - 1\right).$$

Now resolve along the tangent at P, in the direction of increasing θ (towards H). The string makes angle $\frac{1}{2}\theta$ with PO, so it makes angle $\frac{1}{2}\pi - \frac{1}{2}\theta$ with this tangent, and its tangential component is $T\sin\frac{1}{2}\theta$. The weight has tangential component $-mg\sin\theta$, and R has none. With tangential acceleration $a\ddot{\theta}$,

$$\begin{aligned} ma\ddot{\theta} &= T\sin\tfrac{1}{2}\theta - mg\sin\theta \\ &= mg\left[\alpha\left(2\cos\tfrac{1}{2}\theta - 1\right)\sin\tfrac{1}{2}\theta - 2\sin\tfrac{1}{2}\theta\cos\tfrac{1}{2}\theta\right], \end{aligned}$$

so, while the string is taut,

$$\ddot{\theta} = \frac{g}{a}\sin\tfrac{1}{2}\theta\left[2(\alpha-1)\cos\tfrac{1}{2}\theta - \alpha\right]. \qquad (*)$$

(The same equation comes from differentiating the energy $\frac{1}{2}ma^2\dot{\theta}^2 - mga\cos\theta + \frac{1}{2}\alpha mga(2\cos\frac{1}{2}\theta - 1)^2$.)

The equilibrium

In equilibrium $\ddot{\theta} = 0$. For $0 < \theta < \pi$ we have $\sin\frac{1}{2}\theta \neq 0$, so we need

$$\cos\tfrac{1}{2}\theta = \frac{\alpha}{2(\alpha-1)},$$

which is allowed since $\alpha > 1$. For this to give $0 < \theta < \pi$ we need $0 < \cos\frac{1}{2}\theta < 1$. The value is positive, and

$$\frac{\alpha}{2(\alpha-1)} < 1 \iff \alpha < 2\alpha - 2 \iff \alpha > 2.$$

We also need the string to be taut there, as $(*)$ assumes: $\dfrac{\alpha}{2(\alpha-1)} > \dfrac{1}{2} \iff \alpha > \alpha - 1$, which is always true. So if $\alpha > 2$ there is an equilibrium position with $0 < \theta < \pi$, as required.

? **Ask yourself** What if the string were slack at the equilibrium?

Then only gravity would have a tangential component, and $-mg\sin\theta = 0$ has no solution with $0 < \theta < \pi$. That's why it matters that the equilibrium we found has $\cos\frac{1}{2}\theta > \frac{1}{2}$ automatically.

Small oscillations when $\alpha = 2 + \sqrt{2}$

Now

$$\cos\tfrac{1}{2}\theta = \frac{2+\sqrt{2}}{2(1+\sqrt{2})} = \frac{\sqrt{2}(\sqrt{2}+1)}{2(\sqrt{2}+1)} = \frac{1}{\sqrt{2}},$$

so the equilibrium is at $\frac{1}{2}\theta = \frac{1}{4}\pi$, that is, $\theta = \frac{1}{2}\pi$, with P level with O. Put $\theta = \frac{1}{2}\pi + \phi$, so $\ddot{\theta} = \ddot{\phi}$. Write $(*)$ as $\ddot{\theta} = \frac{g}{a} f(\theta)$, where $f(\frac{1}{2}\pi) = 0$. For small ϕ,

$$f(\tfrac{1}{2}\pi + \phi) \approx f'(\tfrac{1}{2}\pi)\,\phi.$$

Because the square bracket in $(*)$ vanishes at $\theta = \frac{1}{2}\pi$, the product rule leaves only one term:

$$f'(\tfrac{1}{2}\pi) = \sin\tfrac{1}{4}\pi \cdot \left[-2(\alpha-1)\cdot\tfrac{1}{2}\sin\tfrac{1}{4}\pi\right] = -(\alpha-1)\sin^2\tfrac{1}{4}\pi = -\frac{\alpha-1}{2} = -\frac{\sqrt{2}+1}{2}.$$

Therefore

$$\ddot{\phi} \approx -\frac{g(\sqrt{2}+1)}{2a}\,\phi$$

for small ϕ, as required.

Final part

The coefficient of ϕ is negative, so a small displacement makes the particle oscillate about $\theta = \frac{1}{2}\pi$ in (approximately) simple harmonic motion: the equilibrium is stable. The angular frequency is $\omega = \sqrt{g(\sqrt{2}+1)/(2a)}$, so the period is

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{2a}{(\sqrt{2}+1)g}} = 2\pi\sqrt{\frac{2(\sqrt{2}-1)a}{g}},$$

and the amplitude stays small, because energy is conserved (the wire is smooth and the string is elastic, so nothing dissipates energy).

Checks

- At $\alpha = 2$ the equilibrium formula gives $\cos\frac{1}{2}\theta = 1$, i.e. $\theta = 0$: the equilibrium merges with the lowest point, which fits the condition $\alpha > 2$.
- As $\alpha \to \infty$, $\cos\frac{1}{2}\theta \to \frac{1}{2}$, so $\theta \to \frac{2}{3}\pi$ and $HP \to a$: a very stiff string barely stretches.
- Stepping the full equation $(*)$ numerically from $\phi = 0.01$ with $a = 1.3\,\mathrm{m}$ and $g = 9.81\,\mathrm{m\,s^{-2}}$ gives a period of about $2.0819\,\mathrm{s}$, matching $2\pi\sqrt{2(\sqrt{2}-1)a/g} \approx 2.0818\,\mathrm{s}$.

Common slips

- Using $HP = 2a\sin\frac{1}{2}\theta$: that's the chord LP, not HP. The angle at O in triangle OHP is $\pi - \theta$.

- Forgetting to check that the string is taut at the equilibrium, since a slack string has no tension.
- Resolving radially instead of tangentially: the radial equation brings in the unknown reaction R and tells you nothing about $\ddot{\theta}$.

Beyond STEP

Linearising about an equilibrium, $\ddot{\phi} \approx f'(\theta_0)\,\phi$, is the standard first step in studying stability of any dynamical system: the sign of $f'(\theta_0)$ (equivalently of $V''(\theta_0)$ for the potential energy) decides between oscillation and runaway. In a first-year course this grows into the phase-plane analysis of nonlinear ODEs. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats small oscillations about stable equilibria via potential energy very clearly.

WORKED SOLUTION

STEP III 2003, Question 10

Mechanics • Variable Acceleration

STEP III 2003, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q10

The key idea

The acceleration $kx\dot{x}$ is an exact derivative with respect to time: it is $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}kx^2\right)$. So we can integrate once straight away and get a first-order equation $\dot{x} = \frac{1}{2}kx^2 + C$, which separates. The sign of the constant C decides everything: a positive constant gives an arctangent (and a solution that blows up), a negative one gives partial fractions and a logarithm (and a solution that settles down).

Solution

We treat the particle as moving on the x-axis with $\ddot{x} = kx\dot{x}$. Since $\frac{\mathrm{d}}{\mathrm{d}t}(x^2) = 2x\dot{x}$, the equation of motion is

$$\frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{1}{2}kx^2\right), \qquad \text{so} \qquad \dot{x} = \tfrac{1}{2}kx^2 + C$$

for a constant C. At $t = 0$ we have $x = d$ and $\dot{x} = U$, so $C = U - \frac{1}{2}kd^2$, and

$$\dot{x} = \tfrac{1}{2}k\left(x^2 - d^2\right) + U. \qquad (*)$$

Part (i)

With $U = kd^2$, equation $(*)$ becomes

$$\dot{x} = \tfrac{1}{2}k\left(x^2 + d^2\right).$$

The right-hand side is always positive, so we may separate variables:

$$\int_d^x \frac{\mathrm{d}u}{u^2 + d^2} = \int_0^t \tfrac{1}{2}k\,\mathrm{d}s \quad \Longrightarrow \quad \frac{1}{d}\left[\arctan\frac{x}{d} - \arctan 1\right] = \tfrac{1}{2}kt.$$

Since $\arctan 1 = \frac{\pi}{4}$, this gives $\arctan(x/d) = \frac{1}{2}dkt + \frac{\pi}{4}$, so

$$x = d\tan\left(\frac{dkt}{2} + \frac{\pi}{4}\right).$$

This is valid as long as the argument of tan stays below $\frac{\pi}{2}$ (the arctangent takes values in $(-\frac{\pi}{2}, \frac{\pi}{2})$). As t increases from 0, the argument increases from $\frac{\pi}{4}$, and it reaches $\frac{\pi}{2}$ when

$$\frac{dkt}{2} = \frac{\pi}{4}, \qquad \text{that is,} \qquad t = \frac{\pi}{2dk}.$$

Since $\tan\theta \to +\infty$ as $\theta \to \frac{\pi}{2}$ from below, and $d > 0$, we get $x \to \infty$ as $t \to \frac{\pi}{2dk}$, as required. The particle reaches infinity in a finite time.

Part (ii)

Now $U < 0$, so $C = U - \frac{1}{2}kd^2 < 0$. Write $C = -\frac{1}{2}ka^2$ with $a > 0$, that is,

$$a = \sqrt{d^2 - \frac{2U}{k}}, \qquad \text{so that} \qquad \dot{x} = \tfrac{1}{2}k\left(x^2 - a^2\right).$$

Because $U < 0$ and $k > 0$, we have $a > d > 0$.

Ask yourself Which partial-fraction logarithm should we use: $\ln\frac{x-a}{x+a}$ or $\ln\frac{a-x}{a+x}$?

It depends on where x lives. The constant solutions $x = \pm a$ are where $\dot{x} = 0$. Initially $x = d$ lies strictly between $-a$ and a, and a solution starting there can't cross the constant solutions $x = \pm a$ (two solutions of a first-order equation like this can't meet), so $-a < x < a$ for all time. In that range $\dot{x} < 0$, so x decreases, and the quantities $a - x$ and $a + x$ are both positive.

Using $\dfrac{1}{x^2 - a^2} = \dfrac{1}{2a}\left(\dfrac{1}{x-a} - \dfrac{1}{x+a}\right)$ and separating variables,

$$\frac{1}{2a}\left[\ln\frac{a-u}{a+u}\right]_d^x = \tfrac{1}{2}kt \quad \Longrightarrow \quad \frac{a-x}{a+x} = A\,\mathrm{e}^{akt}, \qquad \text{where } A = \frac{a-d}{a+d}.$$

Note $0 < A < 1$ because $a > d > 0$. Solving for x,

$$x = a\,\frac{1 - A\mathrm{e}^{akt}}{1 + A\mathrm{e}^{akt}}, \qquad a = \sqrt{d^2 - \frac{2U}{k}}, \quad A = \frac{a-d}{a+d}.$$

This is defined for every $t \geq 0$ (the denominator is positive). Equivalently, since $\ln A = -2\,\mathrm{artanh}(d/a)$, it can be written $x = a\tanh\left(\mathrm{artanh}(d/a) - \frac{1}{2}akt\right)$.

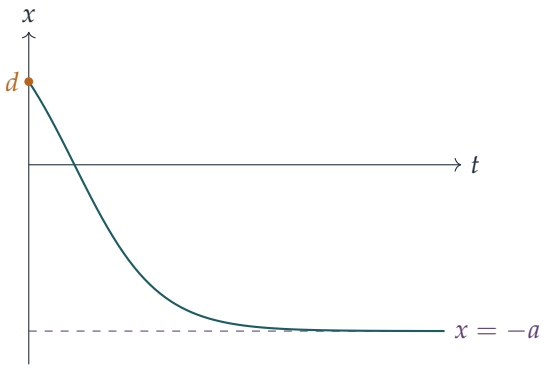
As $t \to \infty$, $A\mathrm{e}^{akt} \to \infty$ because $A > 0$ and $ak > 0$. Dividing top and bottom by $A\mathrm{e}^{akt}$,

$$x = a\,\frac{A^{-1}\mathrm{e}^{-akt} - 1}{A^{-1}\mathrm{e}^{-akt} + 1} \longrightarrow -a.$$

So x tends to the limit

$$-a = -\sqrt{d^2 - \frac{2U}{k}}.$$

(The limit also involves k; it can't be written using d and U alone, since changing k with d and U fixed changes it.)



The graph (drawn for $d = 1$, $a = 2$, $k = 1$) shows the particle starting at d with negative velocity, passing through the origin and approaching $-a$ ever more slowly.

Checks

- In part (i), $t = 0$ gives $x = d\tan\frac{\pi}{4} = d$, and differentiating gives $\dot{x} = \frac{1}{2}kd^2\sec^2\frac{\pi}{4} = kd^2 = U$.
- In part (ii), $t = 0$ gives $x = a\frac{1-A}{1+A} = d$. As $U \to 0^-$, $a \to d$ and $A \to 0$, so x stays at d: correct, since $x = d$, $\dot{x} = 0$ makes the acceleration zero.
- Numerically, stepping $\ddot{x} = kx\dot{x}$ with a Runge–Kutta method for random k, d and U matched both formulae to better than 10^{-6}. For example, with $k = 1$, $d = 1$, $U = -1.5$ we have $a = 2$, and the numerical x settles at -2.000000.

Common slips

- Using $\ln\frac{x-a}{x+a}$ without absolute values: for $|x| < a$ this is the log of a negative number.
- Taking the positive square root for the limit: x decreases from d, so the limit is $-a$, not $+a$.
- Writing the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$ and getting $v\frac{\mathrm{d}v}{\mathrm{d}x} = kxv$, then dividing by v without noting $v \neq 0$. It works here, but integrating $\frac{\mathrm{d}}{\mathrm{d}t}$ directly avoids the issue.

Beyond STEP

Part (i) is a small example of *finite-time blow-up*: an equation $\dot{x} = f(x)$ with f growing faster than linearly (here quadratically) can send the solution to infinity in finite time, so solutions need not exist for all t. Part (ii) shows the other behaviour: the constant solutions $x = \pm a$ are equilibria, $-a$ is stable and $+a$ unstable, and the phase-line picture predicts the limit before any integration. Both ideas, together with the uniqueness theorem that stops solutions crossing, are covered in Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 2003, Question 11

Mechanics • Interception Kinematics

STEP III 2003, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q11

The key idea

Forget the motion for a moment and think about the single point where the collision would happen. At that instant A, B and the collision point form a triangle whose sides are d, the distance P has travelled, and the distance Q has travelled, with the angle at B fixed by the bearing. The cosine rule then gives one equation, and writing both travelled distances in terms of the time turns it into a quadratic. Whether a collision is possible is then a question about whether that quadratic has a positive root.

Solution

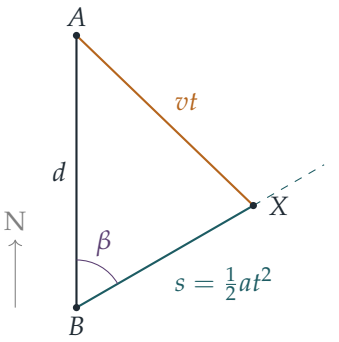
We model P and Q as particles moving in the horizontal plane. Bearings are measured clockwise from north, and $v > 0$.

The collision equation

Suppose Q strikes P at time $t > 0$, at the point X. By then P has started from rest with constant acceleration a along a fixed line, so

$$BX = s = \tfrac{1}{2}at^2,$$

and Q has moved in a straight line at constant speed, so $AX = vt$.



The direction from B to A is due north and the direction from B to X has bearing β, so the angle ABX is β (or $360^\circ - \beta$ if $\beta > 180^\circ$, which has the same cosine). The cosine rule in triangle ABX gives

$$v^2t^2 = d^2 + s^2 - 2ds\cos\beta.$$

(If $\beta = 0$ or 180° the "triangle" is flat, but the equation still holds: it becomes $(vt)^2 = (d \mp s)^2$.) Since $t^2 = 2s/a$, the left-hand side is $2v^2s/a$, and rearranging,

$$s^2 - 2\left(d\cos\beta + \frac{v^2}{a}\right)s + d^2 = 0. \qquad (*)$$

Conversely, if $(*)$ has a root $s > 0$, let $t = \sqrt{2s/a}$ and let X be the position of P at that time. Then $(*)$ says exactly that $AX = vt$, and $AX > 0$. So if Q is projected from A towards X, it reaches X at time t, just as P does, and strikes it. Hence

$$\text{a suitable direction exists} \iff (*) \text{ has a positive root } s.$$

The necessary condition

Write $k = d\cos\beta + v^2/a$, so that $(*)$ is $s^2 - 2ks + d^2 = 0$.

> **Ask yourself** The quadratic certainly needs real roots, but is that enough? What else must we know about them?

The roots must also be *positive*, because s is a distance travelled. The product of the roots is $d^2 > 0$, so the two roots have the same sign, and their sum is $2k$. So there is a positive root exactly when the discriminant is non-negative and $k > 0$:

$$k^2 \geq d^2 \quad \text{and} \quad k > 0 \iff k \geq d.$$

(If $k > 0$ then $k^2 \geq d^2$ is the same as $k \geq d$, as $d > 0$; and $k \geq d$ forces $k > 0$.)

So if Q can strike P, then

$$d\cos\beta + \frac{v^2}{a} \geq d, \qquad \text{that is,} \qquad v^2 \geq ad(1 - \cos\beta),$$

which is the required result.

Striking before P has moved a distance d

Now suppose $v^2 > ad(1-\cos\beta)$, so that $k > d > 0$. The discriminant $4(k^2 - d^2)$ is then strictly positive, and $(*)$ has two distinct roots $s_1 < s_2$. Both are positive, since their sum $2k$ and their product d^2 are positive. As they are distinct with

$$s_1 s_2 = d^2,$$

they can't both be at least d or both at most d, so

$$s_1 < d < s_2.$$

Explicitly, $s_1 = k - \sqrt{k^2 - d^2}$.

By the converse above, projecting Q towards the point X_1 at distance s_1 from B along P's path makes Q strike P at time $t_1 = \sqrt{2s_1/a}$, and at that moment P has moved only $s_1 < d$. So Q can be aimed to strike P before P has moved a distance d, as required.

Checks

- Take $\beta = 0$: P runs straight towards A, so they must meet whatever the value of v. The condition gives $v^2 \geq 0$, which always holds. Then $(*)$ becomes $s^2 - 2(d + v^2/a)s + d^2 = 0$, and the smaller root is below d: they meet before P reaches A, as common sense says.
- Take $a = d = 1$, $\beta = 90^\circ$ (so P runs due east), and $v^2 = 2 > 1 = ad(1-\cos\beta)$. Then $(*)$ is $s^2 - 4s + 1 = 0$, so $s_1 = 2 - \sqrt{3} \approx 0.268 < 1$. At $t_1 = \sqrt{2s_1}$ we have $s_1^2 + 1 = 2t_1^2$, so the collision point really is at distance vt_1 from A. A numerical check over many random values of a, d, β and v confirmed both the hit at $s_1 < d$ and, when $v^2 < ad(1-\cos\beta)$, that P always stays further than vt from A.
- In the borderline case $v^2 = ad(1-\cos\beta)$ we get $k = d$ and a double root $s = d$: the only possible hit is exactly when P has travelled d, which is why the strict inequality is needed in the final part.

Common slips

- Demanding only that the discriminant be non-negative. That gives $|k| \geq d$, which also allows $k \leq -d$, where both roots are negative and no collision happens.
- Using the wrong angle in the cosine rule. The angle at B is between BA (due north) and BX (bearing β), so it is β itself, not $180^\circ - \beta$.
- Only proving the necessary condition in the final part. You must also show that the aim towards X_1 actually works, which is the converse argument.

Beyond STEP

This is a pursuit-and-interception problem. Here the target's path is known in advance and the interceptor moves in a straight line, so the question reduces to a single equation in the meeting time. Missile guidance and "pursuit curves", where the chaser keeps turning towards the target, lead to differential equations instead. The habit of fixing the meeting point first and asking who can reach it in time is a standard tool in first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops the kinematics of relative motion carefully.

WORKED SOLUTION

STEP III 2003, Question 12

Probability and Statistics • Exponential Waiting Times

STEP III 2003, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q12

The key idea

If the number of interruptions in any interval is Poisson, then "the next interruption is more than t minutes away" is the same event as "there are *no* interruptions in the next t minutes", and that turns a discrete distribution into a continuous one: the waiting time is exponential. After that, the successive gaps between interruptions are independent, so the search for a long gap is a sequence of independent trials, each a "success" (a gap of half an hour or more) or a "failure". Geometric distributions and conditional expectation do the rest.

Solution

The distribution of T

Interruptions occur on average every ten minutes, so the rate is $\frac{1}{10}$ per minute and the number of interruptions in a period of t minutes is Poisson with mean $\frac{t}{10}$.

Start the clock at the interruption that has just occurred. The time T to the next one is at least t exactly when there are no interruptions in the next t minutes, so

$$\mathrm{P}(T \geq t) = \mathrm{P}(\text{no interruptions in time } t) = \mathrm{e}^{-t/10}.$$

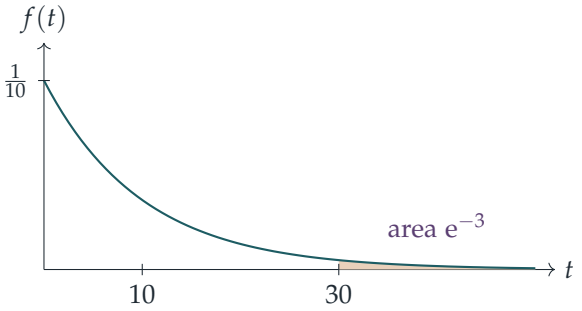
Hence, for $t \geq 0$, the probability of less than t minutes of work is

$$\mathrm{P}(T < t) = 1 - \mathrm{e}^{-t/10}.$$

This is the cumulative distribution function $F(t)$ of T (with $F(t) = 0$ for $t < 0$). Differentiating,

$$f(t) = F'(t) = \tfrac{1}{10}\,\mathrm{e}^{-t/10} \quad (t \geq 0), \qquad f(t) = 0 \quad (t < 0).$$

So T is exponential with mean 10 minutes.



The expected number of interruptions

Because the process has no memory (the Poisson counts in non-overlapping intervals are independent), each time an interruption occurs the time to the next one starts afresh with the same distribution as T, independently of everything before. So the successive gaps $T_1, T_2, \ldots$ are independent, each distributed like T.

Call a gap *long* if it is at least 30 minutes. Each gap is long with probability

$$p = \mathrm{P}(T \geq 30) = \mathrm{e}^{-30/10} = \mathrm{e}^{-3},$$

independently of the others. Let N be the number of interruptions before the first long gap (not counting the one that has just happened). Every short gap ends in an interruption and the first long gap is my uninterrupted half hour, so N is the number of short gaps before the first long one. Hence

$$\mathrm{P}(N = n) = (1-p)^n p, \qquad n = 0, 1, 2, \ldots$$

? **Ask yourself** How do we find the mean of this geometric distribution, which starts at 0 rather than 1?

Differentiate the geometric series $\sum_{n\geq 0} x^n = \frac{1}{1-x}$ to get $\sum_{n\geq 1} n x^{n-1} = \frac{1}{(1-x)^2}$ for $|x| < 1$. With $x = 1 - p$,

$$\mathrm{E}(N) = \sum_{n=1}^{\infty} n(1-p)^n p = p(1-p) \cdot \frac{1}{p^2} = \frac{1-p}{p}.$$

(Alternatively, $N + 1$ is the number of trials up to and including the first success, which has mean $\frac{1}{p}$.) With $p = \mathrm{e}^{-3}$,

$$\mathrm{E}(N) = \frac{1 - \mathrm{e}^{-3}}{\mathrm{e}^{-3}} = \mathrm{e}^3 - 1,$$

as required.

The expected length of a short gap

We want $\mathrm{E}(T \mid T < 30)$. The conditional density of T given $T < 30$ is $f(t)/\mathrm{P}(T < 30)$ on $0 \le t < 30$, so

$$\mathrm{E}(T \mid T < 30) = \frac{1}{1 - \mathrm{e}^{-3}} \int_0^{30} \frac{t}{10}\, \mathrm{e}^{-t/10}\, \mathrm{d}t.$$

Integrating by parts with $u = t$ and $\mathrm{d}v = \frac{1}{10}\mathrm{e}^{-t/10}\,\mathrm{d}t$, so $v = -\mathrm{e}^{-t/10}$,

$$\begin{aligned}\int_0^{30} \frac{t}{10}\, \mathrm{e}^{-t/10}\, \mathrm{d}t &= \Big[-t\, \mathrm{e}^{-t/10} \Big]_0^{30} + \int_0^{30} \mathrm{e}^{-t/10}\, \mathrm{d}t \\ &= -30\, \mathrm{e}^{-3} + 10\big(1 - \mathrm{e}^{-3}\big) \\ &= 10 - 40\, \mathrm{e}^{-3}.\end{aligned}$$

Therefore, multiplying top and bottom by e^3,

$$\mathrm{E}(T \mid T < 30) = \frac{10 - 40\, \mathrm{e}^{-3}}{1 - \mathrm{e}^{-3}} = \frac{10\, \mathrm{e}^3 - 40}{\mathrm{e}^3 - 1} \text{ minutes} \approx 8.43 \text{ minutes}.$$

It makes sense that this is a little less than 10: we have thrown away the long gaps.

The expected wait

The wait before the uninterrupted half hour begins is the sum of the N short gaps. Given $N = n$, these are n independent gaps each conditioned to be shorter than 30 minutes, so the expected wait given $N = n$ is $n\,\mathrm{E}(T \mid T < 30)$. Averaging over N,

$$\mathrm{E}(\text{wait}) = \mathrm{E}(N)\, \mathrm{E}(T \mid T < 30) = (\mathrm{e}^3 - 1) \cdot \frac{10\, \mathrm{e}^3 - 40}{\mathrm{e}^3 - 1} = 10\, \mathrm{e}^3 - 40 \text{ minutes},$$

which is about 161 minutes, or 2 hours 41 minutes. (If you want the time until the paper is *finished*, add the half hour itself: $10\, \mathrm{e}^3 - 10 \approx 191$ minutes.)

Checks

- $\int_0^{\infty} \frac{1}{10}\mathrm{e}^{-t/10}\, \mathrm{d}t = 1$, so f is a genuine density, and its mean is 10, matching "on average every ten minutes".
- Unconditionally, $\mathrm{E}(T) = 10$ should split as $\mathrm{P}(T < 30)\, \mathrm{E}(T \mid T < 30) + \mathrm{P}(T \ge 30)\, \mathrm{E}(T \mid T \ge 30)$. By memorylessness $\mathrm{E}(T \mid T \ge 30) = 40$, and indeed $(10 - 40\mathrm{e}^{-3}) + 40\mathrm{e}^{-3} = 10$.
- A simulation of 200 000 runs of the process gave an average of 19.1 interruptions (against $\mathrm{e}^3 - 1 \approx 19.09$) and an average wait of 161.1 minutes (against $10\mathrm{e}^3 - 40 \approx 160.86$).

Common slips

- Using $\mathrm{E}(N) = \frac{1}{p} = \mathrm{e}^3$: that counts the long gap as well, but it ends with no interruption before the half hour.

- Multiplying $\mathrm{E}(N)$ by the unconditional mean 10 to get the wait. The gaps that make up the wait are known to be short, so their mean is $\mathrm{E}(T \mid T < 30)$, not 10.

Beyond STEP

The interruptions form a *Poisson process*, and the result that the gaps are independent exponentials is one of its defining properties; the final step, $\mathrm{E}(\text{sum of } N \text{ terms}) = \mathrm{E}(N) \times \mathrm{E}(\text{term})$, is a conditioning argument closely related to *Wald's identity*. Together they lead into renewal theory and queueing models, studied in first- and second-year probability courses. Grimmett and Welsh, *Probability: An Introduction*, covers the Poisson process and conditional expectation at the right level.

WORKED SOLUTION

STEP III 2003, Question 13

Probability and Statistics • Absorbing Markov Chains

STEP III 2003, Question 13

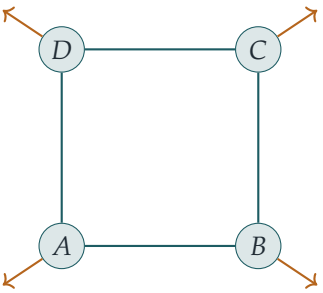
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q13

The key idea

This is a "first-step analysis": condition on what the rabbit does next, and the probability from one chamber is written in terms of the probabilities from the chambers it can reach. That gives four linear equations, and the symmetry of the square (reflecting in the diagonal AC swaps B and D) cuts the work down. For the final part we use the same trick on a modified warren in which entering C counts as failure, then apply the definition of conditional probability.

Solution



tunnels to the surface

Throughout, each move is chosen with probability $\frac{1}{3}$ independently of the rabbit's past (this is how we read "equally likely" for a rabbit "finding itself" in any chamber). Write E_A for the event that the rabbit eventually leaves through the tunnel from A.

Part (i)

Suppose the rabbit is in A, and condition on its next move.

- With probability $\frac{1}{3}$ it goes up the tunnel at A at once, and E_A has happened.

- With probability $\frac{1}{3}$ it runs to B. It is now in B, and because its future moves don't depend on how it got there, the probability that it eventually leaves from A is p_B.
- With probability $\frac{1}{3}$ it runs to D, and similarly the probability is then p_D.

By the law of total probability,

$$p_A = \tfrac{1}{3}\cdot 1 + \tfrac{1}{3}p_B + \tfrac{1}{3}p_D,$$

as required.

Part (ii)

The same argument from the other chambers gives three more equations. The only difference is that leaving through the tunnel at B, C or D means E_A has *failed*, so that branch contributes $\frac{1}{3}\cdot 0$:

$$\begin{aligned} p_B &= \tfrac{1}{3}p_A + \tfrac{1}{3}p_C, \\ p_C &= \tfrac{1}{3}p_B + \tfrac{1}{3}p_D, \\ p_D &= \tfrac{1}{3}p_A + \tfrac{1}{3}p_C. \end{aligned}$$

The equations for p_B and p_D have the same right-hand side, so $p_B = p_D$; call this common value b. (This is the reflection symmetry in AC.) Then

$$p_A = \tfrac{1}{3} + \tfrac{2}{3}b, \qquad p_C = \tfrac{2}{3}b,$$

and substituting into the equation for p_B,

$$b = \tfrac{1}{3}\left(\tfrac{1}{3} + \tfrac{2}{3}b + \tfrac{2}{3}b\right) = \tfrac{1}{9} + \tfrac{4}{9}b \quad\Longrightarrow\quad \tfrac{5}{9}b = \tfrac{1}{9} \quad\Longrightarrow\quad b = \tfrac{1}{5}.$$

Hence

$$p_A = \tfrac{1}{3} + \tfrac{2}{15} = \frac{7}{15}, \qquad \text{and also} \qquad p_B = p_D = \tfrac{1}{5},\ \ p_C = \tfrac{2}{15}.$$

> **Ask yourself** How do we know the rabbit leaves at all, so that these equations really pin down the probabilities?

At every step the rabbit leaves with probability $\frac{1}{3}$, so the chance it is still underground after n steps is $(\frac{2}{3})^n \to 0$: it leaves with probability 1. The four linear equations also have only one solution (we solved them without any choice), so the values above are the probabilities.

Part (iii)

Let N be the event that the rabbit never visits C. We want

$$\mathrm{P}(N \mid E_A) = \frac{\mathrm{P}(N \cap E_A)}{\mathrm{P}(E_A)},$$

with the rabbit starting in A, so the denominator is $p_A = \frac{7}{15}$.

For the numerator, let q_A, q_B, q_D be the probabilities of leaving through the tunnel at A *without ever entering* C, given that the rabbit is currently in A, B, D respectively (and has not yet visited C). First-step analysis works exactly as before, except that a move into C now counts as failure:

$$q_A = \tfrac{1}{3} + \tfrac{1}{3}q_B + \tfrac{1}{3}q_D,$$
$$q_B = \tfrac{1}{3}q_A + \tfrac{1}{3}\cdot 0 + \tfrac{1}{3}\cdot 0,$$

since from B the rabbit either runs back to A, runs into C (failure) or leaves through the wrong tunnel (failure). By symmetry $q_D = q_B = \frac{1}{3}q_A$, so

$$q_A = \tfrac{1}{3} + \tfrac{2}{9}q_A \quad\Longrightarrow\quad \tfrac{7}{9}q_A = \tfrac{1}{3} \quad\Longrightarrow\quad q_A = \tfrac{3}{7}.$$

Therefore

$$\mathrm{P}(N \mid E_A) = \frac{3/7}{7/15} = \frac{45}{49}.$$

Checks

- The four values p_A, p_B, p_C, p_D are the probabilities of leaving from A starting in each chamber. By symmetry they are also the probabilities, starting from A, of leaving from A, B, C, D respectively, so they must add to 1: indeed $\frac{7}{15} + \frac{3}{15} + \frac{2}{15} + \frac{3}{15} = 1$.
- $p_A > p_B = p_D > p_C$, as it should be: the further the rabbit starts from A, the less likely it is to leave there.
- The answer to (iii) is close to 1: a rabbit that leaves from A usually did so after only a short trip. A simulation of 4×10^5 rabbits gave 0.469 for p_A and 0.918 for the conditional probability, against $\frac{7}{15} \approx 0.467$ and $\frac{45}{49} \approx 0.918$.

Common slips

- Including the $\frac{1}{3}$ "leave now" term in the equations for p_B, p_C, p_D: leaving from any tunnel other than A's contributes 0.
- Giving $q_A = \frac{3}{7}$ as the answer to (iii). That is $\mathrm{P}(N \cap E_A)$; the question asks for a conditional probability, so we still divide by p_A.

Beyond STEP

The rabbit is performing a *Markov chain*: its next move depends only on where it is now. The surface is an *absorbing state*, and p_A, p_B, p_C, p_D are *absorption probabilities*, which first-step analysis always turns into a system of linear equations. The same idea gives expected times to absorption, gambler's-ruin probabilities and much more. Grimmett and Welsh, *Probability: An Introduction*, has a good first treatment of Markov chains.

WORKED SOLUTION

STEP III 2003, Question 14

Probability and Statistics • Probability Generating Functions

STEP III 2003, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q14

The key idea

The probability generating function (pgf) of a sum of independent scores is the product of their pgfs. So a question about *which dice give the same distribution of totals* becomes a question about *how a polynomial can be split into two factors*. Factorise everything into irreducible pieces, and the possible relabellings can be read off by sharing those pieces out between the two dice.

Solution

First part

Let X be the score on a standard fair die. Each of $1, \ldots, 6$ has probability $\frac{1}{6}$, so

$$G_X(t) = \mathrm{E}\left(t^X\right) = \tfrac{1}{6}\left(t + t^2 + t^3 + t^4 + t^5 + t^6\right) = \tfrac{1}{6}\, t\left(1 + t + t^2 + t^3 + t^4 + t^5\right).$$

To factorise the bracket, pair up terms and then use a difference of cubes:

$$1 + t + t^2 + t^3 + t^4 + t^5 = (1+t) + t^2(1+t) + t^4(1+t) = (1+t)(1+t^2+t^4),$$
$$1 + t^2 + t^4 = (1+t^2)^2 - t^2 = (1+t+t^2)(1-t+t^2).$$

Hence

$$G_X(t) = \tfrac{1}{6}\, t\,(1+t)(1-t+t^2)(1+t+t^2).$$

If X_1 and X_2 are the scores on two dice rolled independently, then t^{X_1} and t^{X_2} are independent, so $\mathrm{E}\left(t^{X_1+X_2}\right) = \mathrm{E}\left(t^{X_1}\right)\mathrm{E}\left(t^{X_2}\right)$ and

$$G_{X_1+X_2}(t) = G_X(t)^2 = \frac{1}{36}\, t^2(1+t)^2(1-t+t^2)^2(1+t+t^2)^2,$$

which is the required form.

Second part

Let Y and Z be the scores on the dice labelled $1,2,2,3,3,4$ and $1,3,4,5,6,8$. Counting faces,

$$G_Y(t) = \tfrac{1}{6}\left(t + 2t^2 + 2t^3 + t^4\right),$$
$$G_Z(t) = \tfrac{1}{6}\left(t + t^3 + t^4 + t^5 + t^6 + t^8\right).$$

For Y: $1 + 2t + 2t^2 + t^3 = (1+t)(1+t+t^2)$, which you can check by expanding. So

$$G_Y(t) = \tfrac{1}{6}\, t\,(1+t)(1+t+t^2).$$

For Z, the useful products from the first part are $(1+t)(1-t+t^2) = 1+t^3$ and $(1+t+t^2)(1-t+t^2) = 1+t^2+t^4$. Then

$$(1+t^3)(1+t^2+t^4) = 1+t^2+t^3+t^4+t^5+t^7,$$

and multiplying by t gives exactly $t+t^3+t^4+t^5+t^6+t^8$. Therefore

$$G_Z(t) = \tfrac{1}{6}\, t\,(1+t)(1+t+t^2)(1-t+t^2)^2.$$

Rolling independently,

$$G_{Y+Z}(t) = G_Y(t)G_Z(t) = \frac{1}{36}\, t^2(1+t)^2(1-t+t^2)^2(1+t+t^2)^2 = G_{X_1+X_2}(t).$$

Two pgfs that are equal as polynomials have equal coefficients, and the coefficient of t^n is the probability that the total is n. So every total has the same probability with these dice as with two standard dice.

Final part

A standard four-faced die has pgf

$$\tfrac{1}{4}\left(t + t^2 + t^3 + t^4\right) = \tfrac{1}{4}\, t\,(1+t)(1+t^2),$$

so the target pgf for the total of two of them is

$$\frac{1}{16}\, t^2(1+t)^2(1+t^2)^2.$$

Suppose the two new dice have pgfs $\frac{1}{4}f(t)$ and $\frac{1}{4}g(t)$. The coefficient of t^n in f is the number of faces labelled n, so f and g are polynomials with non-negative integer coefficients and $f(1) = g(1) = 4$. Since every label is a positive integer, $f(0) = g(0) = 0$. We need

$$f(t)\, g(t) = t^2(1+t)^2(1+t^2)^2.$$

Ask yourself Why must f be built from the same factors t, $1+t$, $1+t^2$, and nothing else?

The factors t and $1+t$ are linear, and $1+t^2$ has no real roots, so all three are irreducible over the real numbers. Polynomials factorise uniquely into irreducibles (up to constant multiples), so any factor of the right-hand side is

$$f(t) = c\,t^k(1+t)^a(1+t^2)^b, \qquad g(t) = \tfrac{1}{c}\,t^{2-k}(1+t)^{2-a}(1+t^2)^{2-b},$$

with $0 \le k, a, b \le 2$ and a constant $c > 0$.

Now we use the conditions on f and g one at a time.

- *The constant c.* The lowest-degree coefficient of f is c, and that of g is $1/c$. Both are counts of faces, so both are positive integers, which forces $c = 1$.
- *The power of t.* $f(0) = 0$ needs $k \ge 1$, and $g(0) = 0$ needs $2 - k \ge 1$. So $k = 1$: each die has exactly one factor t.
- *Four faces.* $f(1) = 2^a \cdot 2^b = 4$ gives $a + b = 2$, and then $g(1) = 2^{2-a}2^{2-b} = 4$ automatically.

So (a, b) is $(2,0)$, $(1,1)$ or $(0,2)$, with g taking the complementary choice. Since swapping the two dice gives the same pair, there are two cases.

- $(a, b) = (1,1)$ for both: $f = g = t(1+t)(1+t^2) = t + t^2 + t^3 + t^4$, the standard pair $1,2,3,4$ and $1,2,3,4$.
- $(a, b) = (2,0)$ and $(0,2)$:

  $$f(t) = t(1+t)^2 = t + 2t^2 + t^3, \qquad g(t) = t(1+t^2)^2 = t + 2t^3 + t^5,$$

  which are the dice labelled $1,2,2,3$ and $1,3,3,5$.

Both polynomials in each case have non-negative integer coefficients summing to 4, so both really are dice, and by construction they give the right pgf.

Hence the only labellings are the standard pair, and the pair $\{1,2,2,3\}$, $\{1,3,3,5\}$.

Checks

For the non-standard four-faced pair, list the totals: the 16 equally likely sums are $2,4,4,6,\ 3,5,5,7,\ 3,5,5,7,\ 4,6,6,8$, giving totals $2,\ldots,8$ with frequencies $1,2,3,4,3,2,1$, exactly as for two standard dice. Putting $t = 1$ in any of the pgfs above gives 1, a quick test of each factorisation. A computer search over all pairs of four-faced dice with labels from 1 to 7 finds exactly the two pairs above, and the totals for the $1,2,2,3,3,4$ and $1,3,4,5,6,8$ dice match the standard distribution exactly (for instance $\mathrm{P}(\text{total} = 7) = \frac{6}{36}$ in both cases).

Common slips

- Forgetting the constant c: you need a reason why one die can't carry a factor 2 and the other a factor $\frac{1}{2}$. The lowest coefficient being a positive integer for both dice is that reason.

- Allowing $k = 0$ or $k = 2$. Then one die would have a face labelled 0, which isn't a positive integer.
- Leaving $(1 - t + t^2)$ factorised into complex linear factors. Over the reals it's irreducible, and that's the form we need for sharing the factors out.

Beyond STEP

The six-sided pair here are the *Sicherman dice*, and the same method shows they are the only non-standard pair of six-sided dice with positive labels. The argument rests on unique factorisation of polynomials, which you'll meet in a first course in rings and fields as the polynomial version of the fundamental theorem of arithmetic. Generating functions also turn many hard counting problems into algebra. For both ideas, see Allenby, *Rings, Fields and Groups*, and Grimmett and Welsh, *Probability: An Introduction*.

Beyond STEP

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